

A STATISTICAL STUDY ON PRICES OF CRUDE OIL, GOLD AND EXCHANGE RATE

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IN
STATISTICS**

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CERTIFICATE

This is to certify that the project titled “**A Statistical Study on Prices of Crude Oil, Gold and Exchange Rate**” that is being submitted by **Ms. Aswani A.** to the University of Kerala, Thiruvananthapuram, is a bonafide record of study carried out by Ms. Aswani A. , Reg. No. 98724615005, MSc. Statistics student in the Department of Statistics, University of Kerala, Kariavattom during 2024-2026 under my supervision and guidance, in partial fulfilment of the requirements for the MSc. Degree in Statistics from the University of Kerala.

July 2026

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DECLARATION

I, Aswani A., hereby declare that this project report entitled “**A Statistical Study on Prices of Crude Oil, Gold and Exchange Rate**” that is being submitted to University of Kerala is a bonafide record of work done by me under the supervision of Dr. A. Riyaz, Professor, Department of Statistics, University of Kerala, in partial fulfilment of the requirements for the award of the **Degree of Master of Science in Statistics** of the university of Kerala and further that no part thereof has been presented for any other degree.

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ABSTRACT

Crude oil, gold, and the exchange rate are among the most closely watched prices in the Indian economy, and their movements bear directly on inflation, household savings, and macroeconomic policy. This study models and forecasts three series — the monthly price of crude oil, the daily price of gold, and the daily USD/INR exchange rate — and examines the long-run relationships among them, using data extending to 2026.

For each series the price level is modelled within the Box–Jenkins (ARIMA/SARIMA) framework. The model is identified through formal stationarity testing, the autocorrelation and partial autocorrelation functions, and information criteria, and is validated out-of-sample on a held-out test period rather than judged on its in-sample fit. For gold the autoregressive integrated model is compared against Holt’s linear exponential smoothing, and for the exchange rate against naive and random-walk benchmarks, so that the model carried forward is the one that demonstrably forecasts best. The diagnostic testing reveals volatility clustering in the residuals which is confirmed by the ARCH-LM test and the conditional variance is modelled explicitly with a GARCH(1,1) specification under a Student-t distribution to accommodate the heavy tails typical of financial data. The interdependence among the three series is then studied through Johansen’s cointegration test, a Vector Error Correction Model, Granger causality, impulse response analysis, and forecast error variance decomposition. The study further compares these classical models with machine-learning approaches, which are reported separately.

The analysis shows that the univariate models describe each series well and forecast accurately in stable periods, while the gold and exchange-rate residuals display pronounced volatility clustering that the GARCH layer captures. A central qualitative finding is that a univariate model, knowing only the past of its own series, cannot anticipate an exogenous shock: the large forecast errors observed for crude oil from early 2026 measure the size of such a shock rather than a failure of the model, a point made clear by the accuracy of the immediately preceding forecasts. The multivariate analysis characterises the long-run equilibrium among the three series and the direction of influence between them.

Overall, the study offers an integrated, India-focused treatment that forecasts all three series individually, models their volatility where the data demand it, and analyses their interdependence within a single coherent framework using recent data.

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Chapter 1

Introduction

1.1 Background and Motivation

Few prices are watched as closely in India as those of crude oil, gold, and the rupee. Crude oil is the country's largest single import and feeds directly into transport costs, manufacturing, and the general price level; gold is simultaneously a cultural mainstay, a household savings instrument, and a recognised hedge against inflation and uncertainty; and the USD/INR exchange rate sits at the junction of the two, shaping the rupee cost of every imported barrel and every ounce of gold. Movements in these three series ripple through inflation, the trade balance, monetary policy, and the investment decisions of ordinary households, which makes both their forecasting and an understanding of how they move together questions of real practical consequence.

The classical approach to forecasting such series is the Box–Jenkins methodology, in which an autoregressive integrated moving average (ARIMA) model is identified, estimated, and validated through a disciplined sequence of diagnostic checks. These models are interpretable, statistically well grounded, and effective whenever a series carries genuine linear temporal structure. They rest, however, on assumptions, most importantly constant variance and the absence of structural change that financial and commodity series routinely violate. Daily price changes in particular tend to arrive in bursts of calm and turbulence, a feature known as volatility clustering, which a plain ARIMA model assumes. When formal testing detects this behaviour, the natural response is to model the variance explicitly through a generalised autoregressive conditional heteroscedasticity (GARCH) specification, treating the size of risk as a quantity worth forecasting in its own right.

Forecasting each series on its own, however, ignores a large body of evidence that gold, oil, and the exchange rate are economically linked. Studies set in the Indian market reach strikingly different conclusions about the strength and even the existence of these links, with results that depend on the sample period, the assets included, and the methods used. This motivates a second, multivariate strand of analysis built on cointegration and the Vector Error Correction Model (VECM), which

can capture a stable long-run relationship among series that individually wander, together with the short-run dynamics and the direction of influence between them.

This study brings the two strands together for the Indian case using recent data. It forecasts crude oil, gold, and the USD/INR exchange rate individually, models their volatility where the data warrant it, and examines their interdependence, before comparing the classical models against machine-learning alternatives. Throughout, the guiding principle is that modelling choices should be led by the evidence in the data rather than fixed in advance, and that every model should be judged on genuinely unseen observations.

1.2 Objectives

The specific objectives of this study are as follows.

- 1.To identify the most suitable univariate time series model for forecasting the monthly price of crude oil, evaluated through out-of-sample forecast accuracy.
- 2.To determine the best-performing forecasting model for the daily price of gold by comparing the ARIMA and exponential smoothing (ETS) approaches on out-of-sample data.
- 3.To identify the most appropriate univariate model for forecasting the daily USD/INR exchange rate, assessed against naive and random-walk benchmarks using out-of-sample performance.
- 4.To examine the long-run relationships and short-run interdependence among crude oil, gold, and the USD/INR exchange rate, and the direction of influence between them.
- 5.To compare the out-of-sample forecasting performance of the classical time series models with machine-learning models, and to assess whether the differences in accuracy are statistically significant.

A further methodological element runs through the first three objectives: wherever diagnostic testing reveals volatility clustering in a fitted model's residuals, a GARCH model is estimated to describe the time-varying risk. This is treated as a data-driven step rather than a separate objective, since it is the evidence in the residuals , not a prior decision that calls for it.

1.3 Scope and Limitations

The study covers three Indian series over a recent horizon: crude oil at monthly frequency from April 2000, and gold and the USD/INR exchange rate at daily frequency over roughly 2020 to 2026. The methods span the classical Box–Jenkins family and exponential smoothing for the price level, GARCH models for volatility, cointegration and the VECM for interdependence, and machine-learning models for the comparative study. Every point forecast is validated out-of-sample, and competing models are compared on common test data.

Several limitations should be stated plainly. First, a univariate model is informed only by the past of its own series and therefore cannot anticipate an exogenous shock, such as a sudden geopolitical disruption; large errors in such episodes reflect the surprise itself rather than a defect in the model. Second, gold is quoted in rupees per ten grams, a figure that already embeds the exchange rate, so part of any gold–rupee association is mechanical; the multivariate analysis converts gold to dollar terms to mitigate this, but the choice of units remains a caveat. Third, the sample spans the 2008 financial crisis and the 2020 pandemic, periods of structural change that can affect the test statistics, and these are acknowledged rather than explicitly modelled. Fourth, the crude-oil series is monthly and comparatively short, which is too small to train machine-learning models reliably, so the classical-versus-machine-learning comparison is confined to the two daily series. Finally, the GARCH component is used descriptively, and fitted on the training data, for crude oil and the exchange rate and gold, where the resulting prediction interval is itself validated against held-out days; this difference is deliberate and is explained where it arises.

1.4 Organization of the Report

The remainder of the report is organised as follows. Chapter 2 reviews the relevant literature on forecasting these series, on their interdependence, and on the classical-versus-machine-learning question, and identifies the gap the study addresses. Chapter 3 sets out the theoretical background, defining each statistical test and model used in the analysis. Chapter 4 describes the data and the methodology, including the cleaning, the train–test design, and the model configurations. Chapter 5 presents the univariate forecasting and volatility results for crude oil, gold, and the exchange rate. Chapter 6 reports the interdependence analysis among the three series. Chapter 7

presents the comparison between the classical and machine-learning models. Chapter 8 summarises the findings, states the limitations, and suggests directions for further work.

Chapter 2

Literature Review

The literature relevant to this study falls into three broad strands. The first concerns the forecasting of individual commodity and financial price series of gold, crude oil, and the exchange rate by using classical time series models and, more recently, machine learning. The second examines how these series move together, that is, the long-run and short-run interdependence among gold, oil, the exchange rate, and the equity market. The third, which cuts across the first two, is the methodological question of whether modern machine learning models actually forecast these series better than the classical models that have long been the standard. This chapter reviews the work in each strand, with particular attention to studies set in the Indian market, and closes by identifying the gap that the present study addresses.

2.1 Forecasting of Individual Commodity and Financial Series

The most direct precedent for the gold component of this study is the work of Maity and Majumder (2026), who forecast monthly gold prices in India over 2004 to 2025 using the Box–Jenkins methodology. After differencing and the usual identification steps, they settled on an ARIMA(1,1,1) model, which showed strong statistical validity and low forecasting errors, and projected a steady rise in prices from December 2024 to April 2025. They attribute much of gold’s behaviour to festival and wedding demand and to its role as a hedge against inflation, and suggest that accuracy could be improved by bringing qualitative information and external variables into the ARIMA framework. The present treatment of gold follows the same Box–Jenkins route (Box, Jenkins, Reinsel and Ljung, 2015), which remains the standard procedure for univariate price forecasting.

For crude oil, Anastasiadis and Siskos (2023) provide a useful methodological template. Working with monthly WTI prices from January 1980 to May 2022 drawn from the EIA database, they modelled the conditional mean with an ARMA process and the conditional variance with the GARCH family, comparing ARMA-GARCH, ARMA-EGARCH and ARMA-FIGARCH at orders (1,2) and (1,20). Selecting on the

Akaike and Schwarz criteria and evaluating forecasts with RMSE, MAE and MAPE, they identified ARMA-EGARCH(1,20) as the most accurate specification. The study matters here less for the particular winning model than for the workflow it illustrates: when the residuals of a fitted mean model show ARCH effects, the variance is modelled explicitly with a GARCH-type specification (Engle, 1982; Bollerslev, 1986). This is precisely the data-driven step taken in this study only where diagnostic testing warrants it.

Rathi (2026) offers a different kind of contribution. Rather than a forecasting model, the paper is a descriptive analysis of gold price trends over 2024 and 2025, tracing a pronounced upward movement to global economic instability, inflation, geopolitical tension and currency fluctuations. It does not forecast, but it usefully situates the recent behaviour of the series and the macroeconomic drivers behind it.

2.2 Interdependence among Gold, Crude Oil, Exchange Rate and Equity Markets

A large body of Indian work studies these assets jointly, and the findings are notably inconsistent, which is itself instructive. Seyyedi (2017) examined gold prices, oil prices and the rupee–dollar exchange rate from January 2004 to April 2015 using Johansen’s cointegration test, a vector autoregression, Granger causality and impulse response analysis, and concluded that the three series remain substantially independent, implying gains from diversification and a case for treating financial and energy policy separately.

Agrawal and Kumar (2025) reach a markedly different conclusion on a longer, more recent daily sample (January 2009 to March 2023) that also includes the Nifty 50 and both Brent and WTI oil. Using a Vector Error Correction Model and variance decomposition analysis, they find a long-run equilibrium relationship with significant cointegration and long-run causality running from gold prices, the exchange rate and oil prices to the Nifty 50, the influence of these variables growing over time. The contrast with Seyyedi is partly a matter of sample period and partly of the assets included, and it underlines that the relationship cannot simply be assumed in either direction.

Kanungo and Dang (2021) make the instability explicit. Studying gold, crude oil, the exchange rate and the Indian stock market across pre-crisis, crisis and post-crisis

sub-periods between 2005 and 2019, they find through Granger causality, Johansen's test and a VAR that no cointegration holds among the three commodities and that the causal links shift from one period to the next. The lesson for any full-sample model is that the dependence structure is not stable over time. Two further studies focus on the exchange-rate–gold link: Mukherjee (2021), using data from 1970 to 2016, finds no cointegration and only short-run causality, with the growth of the gold price tied negatively and significantly to the growth of the rupee–dollar rate; and Manu and Agarwal (2018) report, through multiple regression and Granger causality, that gold returns positively affect EUR/INR returns alongside several unidirectional causalities.

On the two commodities alone, Rajesh and SatyaNandini (2020) establish a long-run relationship and shock transmission between crude oil and gold using MCX prices, cointegration, Granger causality and impulse response analysis. Choudhary, Nair and Purohit (2017) add a methodological caution: studying Indian crude oil prices over 2005 to 2016, they argue that earlier work ignored structural breaks and nonlinearity, and using the Bai–Perron break test and Breitung's rank test for nonlinear cointegration they identify four structural breaks tied to major market events. Kaushik (2018) widens the lens with dynamic conditional correlation GARCH models, reporting moderate co-movement between WTI and the Indian metal market over 2006 to 2017 and evidence of both short- and long-term memory. The methodological lineage adopted here follows directly from the works Johansen's (1988) cointegration framework, the Engle and Granger (1987) error-correction representation, and Granger (1969) causality — with a VECM or an unrestricted VAR selected according to whether cointegration is actually found.

2.3 Classical versus Machine-Learning Forecasting

The question of whether machine learning improves on classical forecasting recurs throughout this literature. Sarvaiya and Ramchandani (2022) are representative, comparing an ARIMA model with an LSTM network on seven years of daily gold prices. Across the broader literature the results are mixed: deep and tree-based models can win in series with pronounced nonlinear structure, but they frequently fail to beat simple benchmarks on daily financial data, where price changes are close to unpredictable noise and the naive random walk is notoriously hard to outperform. The

present study takes this seriously by retaining a naive benchmark as the honest yardstick and by comparing the classical winners against Random Forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016) and an LSTM network (Hochreiter and Schmidhuber, 1997), with every model evaluated out-of-sample under a rolling one-step-ahead scheme (Hyndman and Athanasopoulos, 2021) and any apparent difference tested formally rather than read off the error metrics alone.

2.4 Summary and Research Gap

Taken together, the reviewed work divides fairly cleanly into two camps. One forecasts a single series, most often gold or crude oil by using ARIMA and, where volatility clustering is present, a GARCH extension. The other studies the interdependence among gold, oil, the exchange rate and the equity market using cointegration, VECM or VAR methods, and reaches conclusions that vary with the sample period and the assets chosen. Indian studies lean heavily toward the second camp. What is rare is a single, India-focused study that forecasts crude oil, gold and the USD/INR exchange rate individually, examines their interdependence with the appropriate cointegration machinery, and places classical and machine-learning models head-to-head out-of-sample with a formal test of significance where all on recent data extending into the 2020–2026 period, with GARCH introduced only where the data demand it.

Table 2.1 summarises the studies reviewed and the gap each leaves open. The present study is designed to fill that gap.

Table 2.1. Summary of key literature and identified gaps.

Study	Focus and data	Method	Key finding / identified gap
Maity & Majumder (2026)	Gold price, India; monthly 2004–2025	Box–Jenkins (ARIMA)	ARIMA(1,1,1) best; point forecast only, single series
Anastasiadis & Siskos (2023)	WTI crude oil; monthly 1980–2022	ARMA + GARCH/EGARCH/FIGARCH	ARMA-EGARCH(1,20) best; crude only, no interdependence
Sarvaiya & Ramchandani (2022)	Gold price; daily, 7 years	ARIMA vs LSTM	Classical-vs-deep-learning on gold only
Rathi (2026)	Gold trends, India; 2024–2025	Descriptive analysis	Drivers documented; no forecasting model

Study	Focus and data	Method	Key finding / identified gap
Seyyedi (2017)	Gold, oil, USD/INR, India; 2004–2015	Johansen, VAR, Granger, IRF	Series largely independent; no forecasting
Agrawal & Kumar (2025)	Nifty 50, gold, oil, USD/INR; daily 2009–2023	VECM, VDA, IRF	Cointegration present; relationships only, no forecasting
Kanungo & Dang (2021)	Gold, oil, exchange rate, stocks; 2005–2019	Granger, Johansen, VAR	No cointegration; causality differs by sub-period
Mukherjee (2021)	Exchange rate & gold; 1970–2016	ADF, PP, Johansen, VAR, Granger	Gold growth negatively tied to ₹/\$; pair only
Manu & Agarwal (2018)	Crude, gold, exchange rates, India	Regression, Granger	Several one-way causalities; no forecasting
Rajesh & SatyaNandini (2020)	Crude oil & gold (MCX), India	Cointegration, Granger, IRF	Long-run link & shock transmission; two commodities
Choudhary et al. (2017)	Crude oil, India (MCX); 2005–2016	Bai–Perron breaks, Breitung	Structural breaks matter; crude only
Kaushik (2018)	Oil vs Indian metals incl. gold; 2006–2017	DCC-GARCH	Moderate co-movement; volatility focus

Chapter 3

Theoretical Background

This chapter sets out the statistical and machine-learning theory underlying the analysis conducted in this study. It begins with the fundamental concepts of stationarity and time series decomposition, proceeds through the Box–Jenkins family of linear models, exponential smoothing, and volatility modelling through ARCH and GARCH, then develops the multivariate machinery of cointegration and the Vector Error Correction Model. The chapter closes with the machine learning models used as a point of comparison Random Forest, XGBoost and Long Short-Term Memory networks — and the evaluation framework used to judge every model in the study on a common footing.

3.1 Stationarity, Differencing and Decomposition

3.1.1 Stationarity

A time series $\{y_t\}$ is the basic object of time series analysis. It is said to be weakly, or covariance, stationary if its first two moments do not change over time: its mean is constant, its variance is finite and constant, and its autocovariance depends only on the lag separating two observations and not on their position in time (Box et al., 2015, ch. 2). Stationarity is a prerequisite for fitting an ARIMA model, because the model's parameters are defined in terms of a stable autocovariance structure; fitting such a model to a series whose mean or variance drifts over time produces estimates that are neither consistent nor meaningfully interpretable. Regressing one non-stationary series on another can also produce a spurious relationship rather than any genuine connection between the variables (Hamilton, 1994, ch. 17; Granger and Newbold, 1974).

3.1.2 Differencing and the Backshift Operator

Most economic and financial price series are non-stationary in their levels because of sustained trends. Such a series can typically be made stationary by differencing. Writing B for the backshift operator, defined by $By_t = y_{t-1}$, the first difference is $\nabla y_t = (1-B)y_t = y_t - y_{t-1}$. Applying the operator d times gives the d -th difference $(1-B)^d y_t$. A series that is non-stationary in its level but becomes stationary after d differences is said to be

integrated of order d , written $I(d)$. In practice, $d = 1$ is sufficient for most price and exchange rate series, and this d is precisely the integration order of the ARIMA model (Box et al., 2015, ch. 4).

3.1.3 The Box–Cox Transformation

When the variability of a series grows with its level as is typical of asset prices that have risen substantially over the sample period and a variance-stabilising transformation is applied before differencing. The Box–Cox transformation (Box and Cox, 1964) is defined as:

$$w_t = (y_t^\lambda - 1) / \lambda \text{ if } \lambda \neq 0, \quad w_t = \ln(y_t) \text{ if } \lambda = 0 \quad (3.1)$$

The case $\lambda = 0$ reduces to the natural logarithm, which is the special case used throughout this study (Hyndman and Athanasopoulos, 2021, ch. 3). The log transformation has two convenient properties for the price series analysed here. First, it stabilises the variance of series whose absolute fluctuations grow with the price level, which is characteristic of gold and exchange rate data that have appreciated substantially over the sample period. Second, once a model is fitted to the log-transformed series, its first difference can be interpreted directly as an approximate percentage change, or return, which is the natural scale on which volatility models such as GARCH are specified (Tsay, 2010, ch. 1).

3.1.4 Decomposition and the Strength of Trend and Seasonality

Before a model is chosen, a series is decomposed descriptively into a trend, a seasonal and a remainder component using STL (Seasonal–Trend decomposition based on Loess; Cleveland et al., 1990). The relative importance of the trend and seasonal components is summarised by strength measures bounded between 0 (the component is absent) and 1 (it dominates the series):

$$F_t = \max(0, 1 - \text{Var}(R_t) / \text{Var}(T_t + R_t)) \quad (3.2)$$

$$F_s = \max(0, 1 - \text{Var}(R_t) / \text{Var}(S_t + R_t)) \quad (3.3)$$

where T_t , S_t and R_t are the trend, seasonal and remainder components, respectively (Hyndman and Athanasopoulos, 2021, ch. 3). In this study, these measures inform the

decision between a seasonal and a non-seasonal specification: a seasonal term is retained only when the seasonal strength is appreciable, following the practical guideline that seasonal differencing is warranted only when the seasonal strength substantially exceeds the trend-driven noise floor of the series.

3.2 Tests for Stationarity

3.2.1 The Augmented Dickey–Fuller Test

The Augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979; Said and Dickey, 1984) formally tests for the presence of a unit root. It is based on the regression:

$$\nabla y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_i \delta_i \nabla y_{t-i} + \varepsilon_t \quad (3.4)$$

In which the lagged differences absorb short-run autocorrelation so that the error term is approximately white noise. The null hypothesis is $H_0: \gamma = 0$, corresponding to a unit root and hence non-stationarity; a test statistic more negative than the relevant critical value, equivalently a p-value below 0.05, rejects the null in favour of stationarity (Hamilton, 1994, ch. 17).

3.2.2 The KPSS Test

The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test (Kwiatkowski et al., 1992) reverses the burden of proof relative to the ADF test: its null hypothesis is that the series is stationary. The series is decomposed into a deterministic trend, a random-walk component, and a stationary error, and the test statistic is constructed from the partial sums of the residuals.

KPSS LM Statistic

$$LM = \frac{\sum_{t=1}^T S_t^2}{T^2 \hat{\sigma}^2} \quad (3.5)$$

Long-Run Variance Estimator (Bartlett Kernel)

$$\hat{\sigma}^2(\ell) = \frac{1}{T} \sum_{t=1}^T \hat{e}_t^2 + \frac{2}{T} \sum_{s=1}^{\ell} \left(1 - \frac{s}{\ell+1}\right) \sum_{t=s+1}^T \hat{e}_t \hat{e}_{t-s} \quad (3.6)$$

A large value of the statistic, and correspondingly a small p-value, indicates non-stationarity. Because the ADF and KPSS null hypotheses are reversed, the two tests are applied together as a cross-check in this study; agreement between them — ADF failing to reject a unit root while KPSS rejects stationarity — gives firm grounds for differencing (Kwiatkowski et al., 1992).

3.2.3 The Phillips–Perron Test

The Phillips–Perron (PP) test (Phillips and Perron, 1988) shares the unit-root null hypothesis of the ADF test but corrects for autocorrelation and heteroscedasticity in the errors non-parametrically, through a direct adjustment to the test statistic, rather than by augmenting the regression with lagged differences.

Phillips–Perron Test Statistic is given by;

$$Z_t = \sqrt{\frac{\hat{\sigma}^2}{\hat{\lambda}^2}} t_{\hat{\rho}} - \frac{1}{2} \left(\frac{\hat{\lambda}^2 - \hat{\sigma}^2}{\hat{\lambda}^2} \right) \frac{T SE(\hat{\rho})}{\hat{\sigma}} \quad (3.7)$$

It is used in this study as a further robustness check alongside the ADF test, particularly in the multivariate analysis.

3.3 The Box–Jenkins ARIMA Framework

3.3.1 Autoregressive Models

An autoregressive model of order p , denoted $AR(p)$, expresses the current value of a stationary series as a linear combination of its p most recent values and a white-noise error term:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t \quad (3.8)$$

where ε_t is white noise with zero mean and constant variance. Using the backshift operator, the model is written compactly as $\phi(B)y_t = c + \varepsilon_t$, where $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ is the autoregressive polynomial. For the process to be stationary, all roots of $\phi(B) = 0$ must lie outside the unit circle (Box et al., 2015, ch. 3).

3.3.2 Moving Average Models

A moving average model of order q , denoted $MA(q)$, expresses the current value as a linear combination of the current and q most recent white-noise innovations:

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} \quad (3.9)$$

or compactly $y_t = \mu + \theta(B)\varepsilon_t$, where $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$. For the process to be invertible — so that it admits a meaningful autoregressive representation — all roots of $\theta(B) = 0$ must lie outside the unit circle (Box et al., 2015, ch. 3).

3.3.3 ARMA and ARIMA Models

Combining the two components gives the $ARMA(p, q)$ model for a stationary series, $\varphi(B)y_t = \theta(B)\varepsilon_t$. When the series must be differenced d times to achieve stationarity, the result is the autoregressive integrated moving average model, $ARIMA(p, d, q)$:

$$\varphi(B)(1-B)^d y_t = \theta(B)\varepsilon_t \quad (3.10)$$

The three orders carry direct meaning: p is the number of autoregressive terms, d is the degree of differencing required to reach stationarity, and q is the number of moving-average terms. The $ARIMA$ model is the principal model used in this study for forecasting the level of the crude oil, gold and exchange rate series (Box et al., 2015, ch. 4).

3.3.4 Model Identification: the ACF and PACF

The orders of an $ARIMA$ model are identified from two functions computed on the differenced series. The autocorrelation function (ACF) measures the linear correlation between observations k periods apart:

$$\rho_k = \gamma_k / \gamma_0 = \text{Cov}(y_t, y_{t-k}) / \text{Var}(y_t) \quad (3.11)$$

while the partial autocorrelation function (PACF) measures the correlation at lag k after removing the linear effect of the intervening lags. The two behave in complementary ways: a pure $AR(p)$ process has a PACF that cuts off sharply after lag p and an ACF that decays gradually, whereas a pure $MA(q)$ process has an ACF that cuts off after lag q and a PACF that decays gradually; a mixed $ARMA$ process shows gradual decay in both functions (Box et al., 2015, ch. 6). In this study, where the ACF and PACF indicate

low-order but not perfectly clean cut-offs, a systematic grid of candidate models is fitted and compared by information criterion and by an automatic search procedure, with the final order confirmed by agreement between the two approaches.

3.3.5 Estimation and Model Selection

Candidate models are estimated by maximum likelihood. When several candidates are admissible, they are compared using information criteria that balance goodness of fit against the number of estimated parameters. The Akaike Information Criterion (Akaike, 1974) and its small-sample correction are:

$$AIC = -2 \ln L + 2k \quad (3.12)$$

$$AICc = AIC + 2k(k+1) / (n-k-1) \quad (3.13)$$

and the Bayesian Information Criterion (Schwarz, 1978) is:

$$BIC = -2 \ln L + k \ln n \quad (3.14)$$

where L is the maximised likelihood, k is the number of estimated parameters, and n is the number of observations; lower values are preferred. These criteria are directly comparable only among models with the same order of differencing, since differencing changes the effective sample size on which the likelihood is computed; models that differ in their order of differencing are therefore compared on out-of-sample forecast accuracy rather than on the information criterion alone.

3.3.6 Significance of Estimated Coefficients

Once a model is fitted, the statistical significance of each estimated coefficient is assessed using a z -statistic, computed as the ratio of the estimated coefficient to its standard error:

$$z = \hat{\phi} / SE(\hat{\phi}) \quad (3.15)$$

Under the null hypothesis that the true coefficient is zero, the z -statistic is asymptotically standard normal, and its absolute value is compared against the critical value of 1.96 at the 5% significance level (Gujarati and Porter, 2009, ch. 5). A coefficient whose $|z|$ exceeds 1.96 is judged to be statistically significant and is retained in the fitted model equation; this procedure is applied throughout the study to confirm

that every term reported in a fitted ARIMA or GARCH equation makes a genuine contribution rather than being indistinguishable from noise.

3.3.7 Diagnostic Checking: the Ljung–Box Test

A fitted model is adequate only if its residuals are indistinguishable from white noise. The Ljung–Box test (Ljung and Box, 1978) examines this jointly over the first m lags:

$$Q(m) = n(n+2) \sum_k \hat{\rho}_k^2 / (n-k) \quad \text{for } k = 1, \dots, m \quad (3.16)$$

where $\hat{\rho}_k$ is the sample autocorrelation of the residuals at lag k . Under the null that the residuals are white noise, $Q(m)$ follows a χ^2 distribution with $m-p-q$ degrees of freedom; a p -value above 0.05 indicates that the linear mean structure has been adequately captured (Ljung and Box, 1978).

Autocorrelation is not the only property a white-noise residual must satisfy. Two further checks are applied to every fitted model in this study. The ARCH-LM test (Section 3.5.2) examines whether the residual variance is constant or whether volatility clustering remains; significant ARCH effects do not invalidate the mean equation but signal that a variance model is required. A test of normality — the Jarque–Bera test (Jarque and Bera, 1987; Section 3.5.4) — assesses whether the residuals are approximately Gaussian, the assumption underlying classical prediction intervals. A model is judged adequate in its mean once the Ljung–Box test is passed; any remaining ARCH effects or departure from normality are addressed through the volatility modelling of Section 3.5 rather than by re-specifying the mean equation.

3.4 Exponential Smoothing

3.4.1 Simple Exponential Smoothing

Exponential smoothing forecasts a series as a weighted average of its past values, with weights decaying geometrically into the past. For a series with neither trend nor seasonality, simple exponential smoothing gives:

$$\hat{y}_{t+1} = \alpha y_t + (1-\alpha)\hat{y}_t \quad (3.17)$$

where the smoothing parameter $\alpha \in [0,1]$ controls how quickly the weights decay (Hyndman and Athanasopoulos, 2021, ch. 8).

3.4.2 Holt's Linear Trend Method

For a series with a trend but no seasonality — the case for the gold and exchange rate price levels — Holt's linear method, the ETS(A, A, N) model in state-space notation, maintains both a level and a trend that are updated each period:

$$\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}) \quad (3.18)$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1} \quad (3.19)$$

$$\hat{y}_{t+h} = \ell_t + h \cdot b_t \quad (3.20)$$

With smoothing parameters α and β^* estimated by minimising the sum of squared one-step errors and the model selected by AICc (Hyndman and Athanasopoulos, 2021, ch. 8). In this study, Holt's method is fitted on the logarithm of price, so that the additive trend on the log scale corresponds to a multiplicative trend on the original price scale. It provides the exponential-smoothing counterpart to ARIMA for comparing classical forecasting approaches.

3.5 Modelling Volatility: ARCH and GARCH

3.5.1 Volatility Clustering and Returns

Financial series typically exhibit volatility clustering: large changes tend to be followed by large changes and quiet periods by quiet periods, so that the variance of the errors is not constant but evolves over time (Tsay, 2010, ch. 3). Volatility is studied on the return series, expressed in percentage terms:

$$r_t = 100 \times (\ln P_t - \ln P_{t-1}) \quad (3.21)$$

3.5.2 ARCH Effects and the ARCH-LM Test

Engle (1982) modelled volatility clustering by allowing the conditional variance to depend on past squared shocks. Writing the error as $\varepsilon_t = \sigma_t z_t$ with z_t independent and identically distributed, the ARCH(q) process specifies:

$$\sigma_t^2 = \omega + \sum_i \alpha_i \varepsilon_{t-i}^2 \quad \text{for } i = 1, \dots, q \quad (3.22)$$

The presence of such effects is tested with the ARCH-LM test, which regresses the squared residuals on q of their own lags; under the null of no ARCH effect, the statistic $LM = nR^2$ follows a χ^2 distribution with q degrees of freedom, so a small p -value confirms volatility clustering and justifies a variance model (Engle, 1982). In this study, a volatility model is fitted only where this test calls for it, and the test is applied not only to the original mean model's residuals but also, after a GARCH model is fitted, to the standardised residuals to confirm that the volatility has been adequately captured.

3.5.3 The GARCH(1,1) Model

The generalised ARCH model of Bollerslev (1986) lets the conditional variance depend on past variances as well as past shocks. The widely used GARCH(1,1) specification is:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (3.23)$$

where ω is a baseline variance, α measures the reaction of volatility to the most recent shock, and β measures how strongly past volatility carries forward. Their sum $\alpha + \beta$, the persistence, governs how slowly volatility shocks decay; when it is below 1, the process has a finite unconditional variance of $\omega/(1 - \alpha - \beta)$. A persistence close to one implies long-lived turbulence, whereas a β near zero — as occurs in the monthly crude oil series analysed in this study — indicates little month-to-month carry-over and is itself a substantive finding rather than a modelling defect (Bollerslev, 1986).

3.5.4 The Error Distribution and Heavy Tails

Returns are typically more peaked and heavier-tailed than the normal distribution. The shape of the distribution is summarised by its skewness and kurtosis, and the departure from normality is tested with the Jarque–Bera statistic (Jarque and Bera, 1987):

$$JB = (n/6) \times [S^2 + (K-3)^2/4] \quad (3.24)$$

To accommodate heavy tails, the GARCH innovations z_t are allowed to follow a Student- t distribution, whose degrees-of-freedom (shape) parameter is estimated from the data; smaller values correspond to fatter tails, and as the parameter grows, the distribution approaches the normal. The choice between the normal and Student- t distributions is made by comparing the Akaike Information Criterion of the two fitted models (Tsay, 2010, ch. 3).

3.5.5 Standardised-Residual Diagnostics and the Fitted Conditional Variance Equation

After a volatility model is fitted, its adequacy is judged on the standardised residuals $z_t = \varepsilon_t/\sigma_t$. The Ljung–Box test applied to z_t confirms that the mean equation remains adequate, while the same test applied to z_t^2 confirms that no volatility clustering remains; passing both indicates that the GARCH layer has captured the conditional heteroscedasticity. The fitted model is reported as an explicit conditional variance equation, with every numerical coefficient inserted into Equation 3.20, in the same way that the fitted ARIMA model is reported as an explicit difference equation following Section 3.3.6.

3.6 Multivariate Analysis: Cointegration and the Vector Error Correction Model

3.6.1 Integrated Series and Cointegration

When several $I(1)$ series share a common stochastic trend, a linear combination of them can be stationary even though each series individually wanders without bound; the series are then said to be cointegrated and to possess a stable long-run equilibrium relationship (Engle and Granger, 1987). Modelling cointegrated series in differences alone would discard this long-run information, whereas modelling them in levels alone would risk the spurious-regression problem of Section 3.1.1; the Vector Error Correction Model resolves the dilemma by combining both.

3.6.2 The Vector Autoregression

The starting point is a vector autoregression of order k for the vector of series y_t :

$$y_t = c + A_1 y_{t-1} + \cdots + A_k y_{t-k} + u_t \quad (3.25)$$

where the A_i are coefficient matrices, and u_t is a vector white-noise process. The lag length k is chosen using information criteria computed jointly across all equations of the system (Lütkepohl, 2005, chs. 2 and 4).

3.6.3 The Johansen Cointegration Test

Re-parameterising the VAR in error-correction form yields $\nabla y_t = \Pi y_{t-1} + \Sigma \gamma_i \nabla y_{t-i} + u_t$. The rank of the long-run matrix Π equals the number of cointegrating relationships. Johansen's procedure (Johansen, 1991) tests this rank sequentially using two statistics constructed from the estimated eigenvalues $\hat{\lambda}_i$:

$$\lambda_{\text{tra}} c_e(r) = -T \sum \ln(1 - \hat{\lambda}_i) \quad \text{for } i = r+1, \dots, n \quad (3.26)$$

$$\lambda_{\text{max}}(r, r+1) = -T \ln(1 - \hat{\lambda}_{r+1}) \quad (3.27)$$

each compared with its critical value. A rank of zero means no cointegration, in which case a VAR in differences is appropriate; a rank of one or more justifies the VECM (Johansen, 1991).

3.6.4 The Vector Error Correction Model

When cointegration is present, the long-run matrix factorises as $\Pi = \alpha\beta'$, where the columns of β form the cointegrating vectors (the long-run equilibrium relationships), and α contains the adjustment speeds. A negative and significant adjustment coefficient means that the variable is pulled back toward equilibrium, its magnitude giving the fraction of any disequilibrium corrected each period; the corresponding half-life of an adjustment of speed α is:

$$h = \ln(0.5) / \ln(1 + \alpha) \quad (3.28)$$

A variable that does not adjust to any cointegrating relationship is weakly exogenous: it drives the system rather than responding to it, which is itself informative about the direction of long-run influence (Engle and Granger, 1987; Lütkepohl, 2005, ch. 7).

3.6.5 Choice of Cointegration Rank

Where the trace and maximum-eigenvalue tests support more than one cointegrating relationship, the choice of rank used in the final VECM is also guided by economic interpretability. A rank that produces loading coefficients with no plausible economic meaning for one or more equations provides a weaker basis for the model than a more parsimonious rank whose adjustment coefficients are stable and interpretable across all equations; this is a recognised practical consideration in applied cointegration analysis

once the rank tests themselves have been reported in full (Johansen, 1995, ch. 6; Lütkepohl, 2005, ch. 9).

3.6.6 Stability and Residual Diagnostics

A correctly specified cointegrated model is dynamically stable, as verified by the moduli of the companion-matrix eigenvalues: a system with n variables and rank r should display exactly $n-r$ unit roots (the common stochastic trends), with all remaining roots strictly inside the unit circle. Model adequacy is then assessed using residual tests for autocorrelation (a multivariate Portmanteau test), normality (a multivariate Jarque–Bera test), and residual heteroscedasticity (a multivariate ARCH test) (Lütkepohl, 2005, chs. 4 and 16).

3.6.7 Heteroscedasticity-Robust Standard Errors

When the multivariate ARCH test indicates that the VECM residuals are heteroscedastic, the conventional OLS standard errors — which assume a constant error variance — may understate or overstate the true sampling variability of the estimated coefficients. White (1980) showed that consistent standard errors can be obtained without specifying the form of the heteroscedasticity by replacing the usual variance estimator with a sandwich-form estimator built from the squared residuals. The HC1 variant used in this study applies a small-sample degrees-of-freedom correction to the original White estimator (MacKinnon and White, 1985). Recomputing the standard errors for each VECM equation using this heteroscedasticity-consistent estimator, while leaving the point estimates of the coefficients unchanged, provides a robustness check on the significance of the error-correction terms reported under the assumption of constant variance.

3.6.8 Granger Causality

Granger causality asks whether the past of one variable improves the prediction of another. Tested on the stationary, differenced series, the null hypothesis is that the lagged coefficients of the candidate causal variable are jointly zero; rejection, signalled by a small p -value on the corresponding F -test, indicates that the variable Granger-causes the other (Granger, 1969). This identifies the direction of short-run influence among crude oil, gold and the exchange rate.

3.6.9 Impulse Response Functions

An impulse response function traces the effect over time of a one-standard-deviation shock to one variable on the present and future values of the others, with bootstrap confidence bands. An effect is statistically meaningful where the band excludes zero. Because the shocks must first be orthogonalised, the analysis depends on an assumed ordering of the variables; in this study, the ordering places global commodity prices before the domestic exchange rate, reflecting the assumption that world prices move first (Lütkepohl, 2005, ch. 2).

3.6.10 Forecast Error Variance Decomposition

The forecast error variance decomposition complements the impulse responses by reporting, for each variable and forecast horizon, the proportion of its forecast error variance attributable to shocks to that variable in the system. Read across a row, the shares sum to one hundred per cent, so a large off-diagonal share indicates that much of a variable's uncertainty originates in the others — a measure of the strength of interdependence and how it grows with the horizon (Lütkepohl, 2005, ch. 2).

3.6.11 Long-Run Regression, Multicollinearity and Residual-Based Cointegration Tests

Because cointegration removes the spurious-regression objection, the long-run relationship may also be estimated directly by ordinary least squares on the log levels, where the slope coefficients are interpreted as elasticities (Engle and Granger, 1987). The reliability of the slopes is checked through the variance inflation factor:

$$VIF_j = 1 / (1 - R_j^2) \quad (3.29)$$

where R_j^2 is the coefficient of determination from regressing the j -th explanatory variable on the others; values above ten are taken to signal a multicollinearity problem (Gujarati and Porter, 2009, ch. 10).

Separately from the system-level Johansen test, the residuals of this single static regression can themselves be tested for a unit root, in the spirit of the original Engle–Granger (1987) two-step procedure. The Phillips–Ouliaris test (Phillips and Ouliaris, 1990) is the residual-based cointegration test used for this purpose in this

study; it tests the null hypothesis that the regression residuals contain a unit root, using critical values designed specifically for residuals estimated from a cointegrating regression rather than the standard Dickey–Fuller critical values. A failure to reject this null for a single equation does not necessarily contradict a system-level finding of cointegration from the Johansen procedure, since the two tests examine cointegration from different vantage points — the system as a whole against one specific normalisation of the long-run relationship — and a divergence between them is a recognised feature of applied cointegration analysis rather than an inconsistency (Phillips and Ouliaris, 1990; Hamilton, 1994, ch. 19).

3.7 Machine Learning for Time Series Forecasting

3.7.1 Definition and Scope

Machine learning is a branch of artificial intelligence concerned with building systems that learn from data to make predictions or decisions without being explicitly programmed with the rules governing those predictions. Rather than deriving a forecast from a set of equations whose parameters are fixed by statistical theory, a machine learning model searches a large space of possible mappings from inputs to outputs and selects the mapping that minimises a measure of prediction error on the available training data (Hastie, Tibshirani and Friedman, 2009, ch. 1–2). The inputs can be raw lagged values of a series, derived features, or cross-variable information; the outputs are the predicted future values. This flexibility means machine learning models can, in principle, approximate relationships of considerable complexity — including nonlinear dynamics and interaction effects — that a fixed linear model such as ARIMA cannot represent.

Three families of machine learning models are included in this study: two ensemble tree methods, Random Forest and XGBoost, and a recurrent neural network, the Long Short-Term Memory (LSTM) network. Each represents a distinct modelling philosophy, and comparing all three against the classical benchmarks of Sections 3.3 and 3.4 provides a broader test of whether the series analysed in this study contain exploitable nonlinear structure.

3.7.2 Why Machine Learning is Adopted Alongside Classical Models

The classical ARIMA and ETS models rest on linearity: they assume that the optimal forecast is a linear function of the series' own past values. This assumption is testable, and the diagnostic tests of Sections 3.3.7 and 3.5.2 provide the relevant evidence. When the ARCH-LM test detects time-varying variance in the residuals, or when the Ljung–Box test on the ETS residuals returns a significant result, it signals that the series contains structure that a linear model alone cannot capture. Nonlinear dynamics are well documented in financial series: return distributions exhibit fat tails, volatility clusters, and prices can behave differently in calm and turbulent periods (Tsay, 2010, ch. 4). These features motivate an investigation of models that do not impose linearity.

A further motivation is the substantial body of recent literature claiming that data-driven machine learning models outperform classical statistical models on financial price and return series. A specific objective of this study is to test that claim directly using Indian gold and exchange rate data, employing a methodologically rigorous, walk-forward evaluation design with formal significance testing, rather than accepting or rejecting the claim based on less rigorous published comparisons.

3.7.3 Advantages and Limitations of Machine Learning in Time Series

The principal advantage of machine learning models is their ability to learn nonlinear, high-dimensional relationships directly from data, without requiring the analyst to specify the functional form in advance. Random Forest and XGBoost can capture interactions between lags that a linear autoregression cannot represent; LSTM networks can, in principle, carry information over long sequences through their internal gating mechanism. These properties are attractive where the underlying data-generating process is genuinely complex (Goodfellow, Bengio and Courville, 2016, ch. 1, 5 and 10).

Equally important limitations must be acknowledged. Machine learning models have many more parameters than classical models and are therefore more prone to overfitting — learning patterns in the training data that have no predictive content for the future. This risk is amplified when the target series is close to a random walk, since there is then very little genuine signal for the model to learn, and a simple benchmark such as the random walk with drift can outperform a sophisticated machine learning model not

because the benchmark is well specified but because there is nothing to gain from additional model complexity (Makridakis, Spiliotis and Assimakopoulos, 2018). A further limitation is interpretability: the parameters of a random forest or the weights of an LSTM do not admit the direct economic interpretation available for an ARIMA or GARCH coefficient. Feature importance measures (Section 3.7.4.3) provide partial transparency but describe the model's reliance on an input rather than the underlying economic mechanism.

3.7.4 Random Forest

A decision tree partitions the feature space by recursively splitting on input variables to minimise a chosen criterion — typically mean squared error for regression. A single deep tree fits the training data closely but generalises poorly, capturing both noise and genuine signal. The Random Forest of Breiman (2001) addresses this by constructing an ensemble of B trees, each fitted on a different bootstrap sample of the training data using a random subset of features at each split. The ensemble prediction is the average across all trees:

$$\hat{y}_t = (1/B) \sum^b T^{(b)}(x_t) \quad \text{for } b = 1, \dots, B \quad (3.30)$$

where $T^{(b)}(x_t)$ is the prediction of the b -th tree for input x_t . Averaging across trees reduces the variance of the individual predictions without increasing bias, and the random subsampling of features decorrelates the trees, preventing the ensemble from collapsing toward a single dominant split (Breiman, 2001; Hastie, Tibshirani and Friedman, 2009, ch. 15).

Random Forest is not itself a sequential model: it treats each input as an independent row of a feature matrix. To apply it to a time series, the series is converted into a supervised learning problem by constructing a lag matrix in which each row contains the N most recent observations and the target is the value one step ahead, $X_t = [r_{t-1}, r_{t-2}, \dots, r_{t-N}]^T$, $y_t = r_t$, where r_t denotes the return at time t and N is the chosen lag window — 20 in this study, corresponding to approximately one trading month of daily observations.

A practical advantage of Random Forest is that it produces a measure of feature importance for each input lag, the mean decrease in impurity (MDI), computed as the

average reduction in the splitting criterion attributable to each feature across all trees and splits, normalised so that the sum equals 1 (Breiman, 2001). In this study, feature importance is used diagnostically: if the lag-1 return were the dominant predictor, it would receive the highest importance score, and its absence among the top features would be taken as evidence that the series contains no dominant short-run pattern.

3.7.5 XGBoost

Gradient boosting (Friedman, 2001) builds an ensemble of trees sequentially rather than in parallel. At each stage m , a new tree is fitted to the negative gradient of the loss function with respect to the current ensemble prediction — the direction in which the current model is most in error:

$$F^{(m)}(x) = F^{(m-1)}(x) + v \cdot h^{(m)}(x) \quad (3.31)$$

where v is the learning rate, a shrinkage factor that reduces each tree's individual contribution and, by forcing the algorithm to take many small steps, regularises the model and typically improves its generalisation.

XGBoost (eXtreme Gradient Boosting; Chen and Guestrin, 2016) is an efficient, regularised implementation of gradient boosting that adds explicit L_1 and L_2 penalties to the tree-building objective and a second-order Taylor expansion of the loss to improve gradient estimation. The regularised objective at stage m is $L^{(m)} = \sum_i l(y_i, F^{(m)}(x_i)) + \sum_k \Omega(h^{(k)})$, where l is the per-observation loss and $\Omega(h) = \gamma T + (\lambda/2) \|w\|^2$ penalises the number of leaves T and the leaf weights w (Chen and Guestrin, 2016). In this study, XGBoost is fitted with a maximum tree depth of three and a learning rate of 0.05, conservative choices intended to limit overfitting on the relatively small training samples available for daily financial returns. Like Random Forest, XGBoost produces a feature importance measure for each lag, here based on the average gain in the splitting criterion attributable to each feature.

3.7.6 Long Short-Term Memory Networks

The Long Short-Term Memory (LSTM) network, introduced by Hochreiter and Schmidhuber (1997), is a recurrent neural network architecture designed to overcome the vanishing-gradient problem that prevents standard recurrent networks from learning dependencies over long sequences. An LSTM cell maintains a memory state that is

updated at each time step through three gates — a forget gate, an input gate and an output gate — which together regulate what information from the past is retained, what new information is incorporated, and what is passed forward to the next step (Hochreiter and Schmidhuber, 1997; Goodfellow, Bengio and Courville, 2016, ch. 10).

In this study, the LSTM is applied directly to the return series as a nonlinear autoregressive model. The input at each step is a sequence of the 20 most recent returns, formatted as a tensor of shape (samples, time steps, features); the network processes this sequence through its gating mechanism and a final linear layer produces the predicted next return, $\hat{y}_t = W_{\text{out}} h_t + b_{\text{out}}$, where h_t is the hidden state at the final time step. The network is trained by minimising the mean squared error loss using the Adam optimiser (Kingma and Ba, 2015), a stochastic gradient-based method that adapts its learning rate for each parameter individually. The architecture used here is deliberately small — a single layer of 16 units, trained for 30 epochs — to limit overfitting given the modest size of the available daily training samples.

A property that distinguishes the LSTM from the tree-based models is that it preserves the temporal ordering of the input sequence through its recurrent hidden state, whereas Random Forest and XGBoost treat the lag matrix as an unordered set of features. Whether this architectural property translates into a forecasting advantage on series that are close to a random walk is precisely the empirical question that the model comparison in this study is designed to answer.

Because a neural network's training process can fail to converge or overfit even within the constraints described above, its adequacy is assessed using additional diagnostics specific to neural network training (Goodfellow, Bengio and Courville, 2016, ch. 7–8). The learning curve, which plots training and validation loss against training epoch, indicates successful convergence when both losses decline and stabilise together; a widening gap between them, with training loss continuing to fall while validation loss rises, is the signature of overfitting. The residuals of the fitted LSTM are further examined for bias (their mean), spread (their standard deviation), and any remaining autocorrelation using the Ljung–Box test of Section 3.3.7, applied here to the forecast errors of the neural network in the same way it is applied to the residuals of a classical model.

3.8 Walk-Forward Evaluation

3.8.1 The Evaluation Problem in Time Series

Evaluating a forecasting model on financial data poses a problem that does not arise in cross-sectional settings: the data have a temporal order, and a model that has access to future information during training will appear more accurate than it truly is. Randomly splitting the data into training and test sets, the standard approach in much of supervised machine learning, would allow training observations from late in the sample to inform predictions about earlier test observations, constituting data leakage. A stricter evaluation design is required (Hyndman and Athanasopoulos, 2021, ch. 5).

3.8.2 The Walk-Forward Protocol

The walk-forward evaluation — also called rolling-origin evaluation or time-series cross-validation (Hyndman and Athanasopoulos, 2021, ch. 5) — resolves this by evaluating each test observation as a genuine one-step-ahead forecast. At each step t in the test period, the model is fitted on all observations up to time $t-1$, a one-step-ahead forecast is produced for time t , the true value is revealed and the error recorded, and the window then advances by one step. In this study, the tree-based models are re-fitted every 20 steps and the LSTM every 40 steps, which gives results essentially identical to daily re-fitting at a fraction of the computational cost. The prediction at each step uses the true observed values of all lagged inputs rather than the model's own previous predictions, thereby removing the multi-step drift bias that would otherwise inflate the apparent advantage of trend-following models over longer horizons.

3.8.3 Consistency with Classical Model Evaluation

The walk-forward design ensures that the machine learning models are compared with the classical models on an equal footing. The ARIMA and ETS forecasts used for comparison are likewise one-step-ahead predictions produced by models fitted to the training data and evaluated on the same test observations.

3.9 The Diebold–Mariano Test

Differences in RMSE, MAE or MAPE between two models do not, by themselves, establish that one model is genuinely more accurate than the other: such differences

could arise by chance, particularly when the test set is of modest size. The Diebold–Mariano test (Diebold and Mariano, 1995) provides a formal test of whether the difference in predictive accuracy is statistically significant. Writing e_{1t} and e_{2t} for the forecast errors of two models and $L(\cdot)$ for the squared-error loss, the loss differential is $d_t = L(e_{1t}) - L(e_{2t})$, and the null hypothesis of equal predictive accuracy is $H_0: E[d_t] = 0$. The test statistic:

$$DM = \bar{d} / \sqrt{\hat{\sigma}^2(d) / n} \quad (3.32)$$

is asymptotically standard normal under H_0 , where $\hat{\sigma}^2(d)$ is a heteroscedasticity and autocorrelation consistent (HAC) estimate of the long-run variance of the loss differential, computed by the Newey–West method to account for the autocorrelation that arises naturally among consecutive one-step-ahead errors on the same series (Diebold and Mariano, 1995; Newey and West, 1987). A two-sided p-value below 0.05 indicates that the predictive accuracy is not equal. In this study, the test is conducted using the empirical RMSE winner as the reference model, with all other models compared.

3.10 Overfitting and the Train–Test Error Ratio

A model overfits when it learns patterns specific to the training data that do not generalise to new observations — a particular risk for flexible machine learning models applied to financial return series that carry very little genuine autocorrelation signal (Hastie, Tibshirani and Friedman, 2009, ch. 7). The train–test error ratio provides a direct diagnostic of generalisation failure:

$$\text{Ratio} = \text{RMSE}_{\text{test}} / \text{RMSE}_{\text{train}} \quad (3.33)$$

computed on the return series rather than the price level. A ratio near one indicates that the model generalises as well as it fits the training data; a ratio substantially above one indicates overfitting. A threshold of 2 is used in this study as a practical guideline, following common diagnostic practice: a ratio below 2 is considered acceptable, while a ratio above 2 is flagged as evidence of overfitting. Because the walk-forward evaluation already prevents look-ahead bias, an elevated ratio reflects genuine generalisation failure rather than a procedural error in the evaluation design.

3.11 Forecast Evaluation Metrics

3.11.1 Point Forecast Accuracy

Forecast accuracy throughout this study is assessed using out-of-sample errors: the model is estimated on the training period and evaluated on a later period it has not seen, so that the reported error genuinely measures predictive ability. The standard measures used are the root mean squared error, the mean absolute error, and the mean absolute percentage error:

$$\text{RMSE} = \sqrt{[(1/n) \sum_t (y_t - \hat{y}_t)^2]} \quad (3.34)$$

$$\text{MAE} = (1/n) \sum_t |y_t - \hat{y}_t| \quad (3.35)$$

$$\text{MAPE} = (100/n) \sum_t |y_t - \hat{y}_t| / y_t \quad (3.36)$$

RMSE penalises large errors more heavily than MAE because of the squaring operation, making it more sensitive to outliers, whereas MAPE expresses the error as a percentage of the actual value and is unit-free, allowing errors across series of very different scales and currencies to be compared directly (Hyndman and Koehler, 2006).

3.11.2 Benchmark Models and Theil's U Statistic

Two simple benchmarks are used throughout the study because, on daily financial data, they are notoriously difficult to beat and provide an honest yardstick against which to judge both classical and machine-learning models. The naive forecast carries the last observed value forward, $\hat{y}_{t+h} = y_t$, while the random walk with drift adds the average historical change δ at each step. Theil's U statistic formalises the comparison against the naive random walk:

$$U = \sqrt{[\sum_t (y_t - \hat{y}_t)^2]} / \sqrt{[\sum_t (y_t - y_{t-1})^2]} \quad (3.37)$$

A value of U below one means the model improves on the naive benchmark; a value above one means it does not (Theil, 1966; Hyndman and Athanasopoulos, 2021, ch. 5). A model that cannot improve on this benchmark offers no practical forecasting gain over simply assuming the series continues as it has.

3.11.3 Prediction Intervals and Coverage

When a model also produces a prediction interval $[L_t, U_t]$, its reliability is judged by the empirical coverage — the proportion of test observations that actually fall inside the stated interval:

$$\text{Coverage} = (100/n) \sum_t \mathbb{1}\{L_t \leq y_t \leq U_t\} \quad (3.38)$$

and by the average width of the intervals. A well-calibrated interval achieves empirical coverage close to its nominal level while remaining as narrow as possible. This matters particularly for the volatility analysis in this study, where a GARCH model is judged not on its point-forecast accuracy — which derives from the mean equation — but on the quality of the volatility-adaptive interval it produces relative to the classical Gaussian interval (Hyndman and Athanasopoulos, 2021, ch. 5).

Chapter 4

Methodology

This chapter sets out the research design and analytical procedure followed in this study. It explains how the four datasets were sourced, cleaned and structured; the chronological train–test split applied at every stage; the sequence of statistical procedures applied to each univariate series; the procedure followed for the multivariate cointegration analysis; the design of the machine learning comparison; and the software environment in which the analysis was carried out. The theoretical justification for each test and model listed below is provided in Chapter 3; this chapter describes how those tools were applied, in what order, and why.

4.1 Data Sources and Structure

4.1.1 Data Sources

This study draws on four datasets, each sourced from an authoritative Indian government or exchange database and processed into a consistent time-series format prior to analysis. The datasets are divided into two groups: three monthly series covering a long historical span, used for the univariate crude oil analysis and the multivariate analysis, and two daily series covering the more recent period, used for the univariate gold and exchange rate analyses and the machine learning comparison.

Crude oil price (monthly). Monthly Indian crude oil prices are obtained from the Petroleum Planning and Analysis Cell (PPAC) of the Ministry of Petroleum and Natural Gas, Government of India (ppac.gov.in). The PPAC publishes the Indian basket crude oil price, a weighted average of sour-grade and sweet-grade crude processed by Indian refineries, recorded in US dollars per barrel.

Gold price (daily). Daily gold prices, used for the univariate and machine-learning analyses of gold, are sourced from the National Stock Exchange of India (NSE) and recorded in Indian rupees per 10 grams.

Gold price (monthly, multivariate analysis). For the multivariate analysis, which requires all three variables at the same monthly frequency, gold prices are taken from the Reserve Bank of India (RBI) database and are recorded in Indian rupees per ten grams.

USD/INR Exchange Rate (Daily): The daily USD/INR exchange rate data for the period 2020–2026, used for the univariate and machine learning analyses, were obtained from the historical data database available on Investing.com.

USD/INR Exchange Rate (Monthly): The monthly USD/INR exchange rate data for the period 2000–2026, used for the multivariate analysis, were obtained from the historical data database on Investing.com. Monthly observations were derived from the historical series to match the frequency of the crude oil and gold price data.

4.1.2 Sample Periods and Observation Counts

The monthly series spans April 2000 to May 2026, providing a long historical sample appropriate for the cointegration analysis, which requires a large number of observations to estimate stable long-run relationships. The daily series begins in January 2020 and runs to May or June 2026, reflecting the period for which consistently formatted daily data were available from the chosen sources.

Table 4.1 Summary of datasets: source, frequency and sample period.

Variable	Source	Frequency	Start	End	Obs.
Crude oil price (USD/bbl)	PPAC	Monthly	Apr 2000	May 2026	314
Gold price (INR/10g)	NSE	Daily	Jan 2020	May 2026	1,537
Gold price (INR/10g, multiv.)	RBI	Monthly	Apr 2000	May 2026	314
USD/INR exchange rate	Investing .com	Daily	Jan 2020	Jun 2026	1,674
USD/INR (multiv.)	Investing .com	Monthly	Apr 2000	May 2026	314

The crude oil monthly series began in April 2000 because that is the earliest period for which PPAC data are available in a consistent format in the Indian financial-year structure used in the source files. This financial-year format, which runs from April to March, was converted to a standard calendar-month series by reassigning the calendar year of any January, February, or March observation to the following year, in line with the Indian financial-year convention.

4.1.3 Data Cleaning and Transformation

Each series was cleaned before analysis. For the monthly crude oil series, two summary columns present in the raw spreadsheet (the annual average and a ratio column) were removed, and the wide financial-year-by-month layout was reshaped into a long format with one row per calendar month. For the daily series, prices recorded with comma separators were converted to numeric values and dates given in non-standard formats were parsed and standardised; duplicate dates were removed, retaining the first occurrence. After cleaning, each series was ordered chronologically and checked for gaps; no missing values were found in any series once non-trading days were excluded, and no imputation was performed.

For the multivariate analysis, the gold price series, originally in Indian rupees, was converted to US dollars per ten grams by dividing each monthly observation by the corresponding monthly USD/INR exchange rate. This conversion places gold on the same currency basis as crude oil and removes the mechanical correlation that would otherwise arise between the rupee gold price and the exchange rate, since both would then be partly expressed in the same currency unit. All three series used in the multivariate analysis — crude oil, gold (in dollars) and the exchange rate — were subsequently transformed to natural logarithms, following the Box–Cox principle described in Section 3.1.3, which stabilises variance and allows the resulting regression coefficients to be interpreted directly as elasticities.

4.1.4 Train–Test Split

A chronological train–test split, rather than a random split, is applied to every series before any model is fitted. The split preserves the temporal order of the observations; shuffling is never used, because it would destroy the serial dependence that the models

are designed to capture and would allow information from the future to influence the fitted model, a form of data leakage.

Table 4.2 Train–test splits applied in each component of the study.

Component	Total obs.	Training	Test
Crude oil (univariate)	314	309 months	5 months (Jan–May 2026)
Gold price (univariate)	1,537	1,502 days	35 days
USD/INR (univariate)	1,674	1,566 days	108 days
Gold (ML comparison)	1,537	1,229 days	308 days
USD/INR (ML comparison)	1,674	1,339 days	335 days
Multivariate (VECM)	314	— (full sample)	—

The univariate analyses of crude oil, gold, and the exchange rate use test sets sized to the specific forecasting horizon of interest. The machine learning comparison uses a larger hold-out of approximately 20% of each daily series, to give the walk-forward evaluation sufficient test observations to support the Diebold–Mariano significance test. The multivariate VECM analysis uses the full sample without a split, since its purpose is to characterise a long-run equilibrium relationship over the complete history rather than to produce and score out-of-sample forecasts.

4.2 Univariate Analysis Procedure

The same nine-step procedure is followed for each of the three univariate series — crude oil, gold and the exchange rate — to ensure that the three analyses are directly comparable. Where a series-specific decision was required (for example, whether to apply seasonal differencing, or which error distribution to select for the volatility model), the decision was made by the data and reported as a finding rather than imposed in advance.

4.2.1 Step 1: Descriptive Analysis

Each series was first summarised by its mean, median, standard deviation, minimum, maximum, skewness, and kurtosis (Section 3.1), and then plotted over the full sample period to identify visible trends, structural breaks, or outliers.

4.2.2 Step 2: Decomposition

An STL decomposition (Section 3.1.4) was applied to the full series to separate it into trend, seasonal and remainder components, and the trend and seasonal strength measures were computed. Where the seasonal strength was found to be negligible, as it was for all three series in this study, a non-seasonal model specification was used in preference to a seasonal one.

4.2.3 Step 3: Stationarity Testing

The ADF and KPSS tests (Sections 3.2.1–3.2.2) were applied to the training portion of each series at its level and, where the level was found non-stationary, again after one difference. Because the two tests have opposing null hypotheses, agreement between them — the ADF test failing to reject a unit root in levels while the KPSS test rejects stationarity, with both reversing after differencing — was treated as strong evidence of the correct order of integration.

4.2.4 Step 4: Order Identification

The ACF and PACF of the differenced series (Section 3.3.4) were inspected to identify the range of plausible autoregressive and moving-average orders. Where these plots indicated low-order but not unambiguous orders, a systematic grid of candidate ARIMA models was fitted and ranked by AICc (Section 3.3.5), and the automatic order-selection routine was run independently as a cross-check. The final model was adopted only where the grid search and the automatic procedure agreed, or where the disagreement could be explained and resolved on diagnostic grounds, as in the crude oil case described above.

4.2.5 Step 5: Model Estimation and Significance Testing

The selected ARIMA model was estimated by maximum likelihood, and every coefficient was tested for statistical significance using the z-statistic procedure of

Section 3.3.6. The fitted model was then written out as an explicit difference equation, with the estimated numerical coefficients inserted in place of the symbolic parameters, so that the practical meaning of the model — how much of this month's change depends on previous months' changes and on past shocks — is stated directly rather than left implicit in a table of coefficients.

4.2.6 Step 6: Residual Diagnostics

The residuals of the fitted model were tested for remaining autocorrelation using the Ljung–Box test (Section 3.3.7) and for stationarity using the ADF test. A model was accepted as adequate in its mean equation only once the Ljung–Box test was passed; ARCH-LM and normality test results on the same residuals were recorded but treated as motivating the subsequent volatility analysis (Step 8) rather than as grounds for rejecting the mean model.

4.2.7 Step 7: Forecasting and Accuracy Evaluation

The fitted model was used to forecast the held-out test period, and the forecasts were compared with the actual values using RMSE, MAE, MAPE, and, where applicable, the empirical coverage of the model's prediction interval (Section 3.11). For the gold and exchange rate series, an ETS model (Section 3.4.2) was fitted in parallel and compared against the ARIMA model on the same test observations, to provide a classical, non-ARIMA point of comparison ahead of the machine learning analysis described in Section 4.4.

4.2.8 Step 8: Volatility Analysis

The ARCH-LM test (Section 3.5.2) was applied to the residuals of the selected mean model to determine whether a volatility model was statistically warranted. Where it was, a GARCH(1,1) model (Section 3.5.3) was fitted to the return series under both a normal and a Student-t error distribution, and the better-fitting distribution was selected by AIC. The fitted conditional variance equation was written out explicitly, as in the mean equation in Step 5, and the standardised residuals of the GARCH model were retested for autocorrelation and residual ARCH effects to confirm that the volatility model had adequately captured the time-varying variance.

4.2.9 Step 9: Forward Forecasting

Where appropriate, the selected mean model and the fitted volatility model were used together to produce a forward forecast beyond the end of the available data: a point forecast of the price level from the ARIMA or ETS model, and a forecast of the expected volatility over the same horizon from the GARCH model, reported as two complementary descriptions of the future — the expected level and the expected uncertainty around it — rather than as competing forecasts of the same quantity.

4.3 Multivariate Analysis Procedure

4.3.1 Step 1: Data Preparation

The three monthly series — crude oil, gold (converted to US dollars) and the exchange rate — were merged on their common dates and transformed to natural logarithms, as described in Section 4.1.3.

4.3.2 Step 2: Unit Root Testing

Each log-transformed series was tested for a unit root in its level and in its first difference using the ADF and Phillips–Perron tests (Sections 3.2.1 and 3.2.3), confirming that all three series are integrated of order one, the necessary precondition for the cointegration analysis that follows.

4.3.3 Step 3: Lag Selection and the Johansen Test

A vector autoregression was fitted to the three log series, and its lag order was selected by information criteria computed jointly across all three equations (Section 3.6.2). The Johansen trace and maximum-eigenvalue tests (Section 3.6.3) were then applied at this lag order to determine the cointegration rank — the number of independent long-run equilibrium relationships among the three series.

4.3.4 Step 4: Cointegration Rank Selection

Where the trace and maximum-eigenvalue tests supported more than one cointegrating relationship, the loading coefficients under each candidate rank were examined for economic interpretability before the final rank was chosen, following the practical guidance described in Section 3.6.5. The rank ultimately used in this study, and the

reasoning behind that choice, are reported in full alongside the formal test results, so that the reader can see both what the tests support and why a more parsimonious specification was preferred.

4.3.5 Step 5: VECM Estimation

With the cointegration rank fixed, the Vector Error Correction Model (Section 3.6.4) was estimated, yielding the long-run cointegrating vector (β), the speed-of-adjustment coefficients (α) for each variable, and the short-run coefficients on the lagged differences of all three series. The half-life of adjustment implied by each significant α coefficient was computed to express the speed of adjustment in an intuitive, time-based unit.

4.3.6 Step 6: Model Stability and Residual Diagnostics

The eigenvalues of the VECM's companion-matrix representation (Section 3.6.6) were inspected to confirm dynamic stability — the expected number of unit roots corresponding to the common stochastic trends, with all remaining roots inside the unit circle. The VECM residuals were then tested for autocorrelation (a multivariate Portmanteau test), normality (a multivariate Jarque–Bera test) and remaining heteroscedasticity (a multivariate ARCH test).

4.3.7 Step 7: Robustness Check under Heteroscedasticity

Because the multivariate ARCH test indicated heteroscedasticity in the VECM residuals, heteroscedasticity-consistent (HC1) standard errors (Section 3.6.7) were computed for each equation of the VECM as a robustness check on the significance of the error-correction coefficients reported under the standard, homoscedasticity-assuming covariance estimator.

4.3.8 Step 8: Granger Causality, Impulse Responses and Variance Decomposition

Granger causality tests (Section 3.6.8) were run on the stationary, differenced series in both directions for every pair of variables, to establish the short-run predictive relationships among crude oil, gold and the exchange rate. Bootstrapped impulse response functions (Section 3.6.9) were then computed to trace the dynamic effect of a one-standard-deviation shock in crude oil and in gold on the exchange rate over a

24-month horizon, and the forecast error variance decomposition (Section 3.6.10) was computed at 1-, 12- and 24-month horizons to quantify how the relative importance of each variable's own shocks versus the other variables' shocks evolves with the forecast horizon.

4.3.9 Step 9: Long-Run Elasticities and Residual-Based Cointegration Check

As a descriptive complement to the system-level VECM, a static OLS regression of the log exchange rate on log crude oil and log gold (Section 3.6.11) was estimated directly on the price levels, exploiting the fact that cointegration removes the usual spurious-regression objection to such a regression. The variance inflation factor was computed to rule out multicollinearity between the two regressors, and the Phillips–Ouliaris residual-based cointegration test was applied to the regression residuals as a further, equation-specific check on the long-run relationship, reported alongside the system-level Johansen result with which it need not necessarily agree.

4.4 Machine Learning Comparison Procedure

4.4.1 Scope

The machine learning comparison was carried out for the gold and exchange rate series for which a sufficiently long, consistently sampled daily history was available. It was not extended to the monthly crude oil series, whose 314 observations are too few to support the training of the machine learning architectures used here without an unacceptable risk of overfitting.

4.4.2 Step 1: Feature Construction

For each series, the daily log-return was computed as the first difference of the log price, and a lag matrix was constructed in which each row contains the 20 most recent log-returns and the target is the return one step ahead (Section 3.7.4), following the convention that all machine learning models in this study are trained on the stationary return series rather than on the non-stationary price level.

4.4.3 Step 2: Model Configuration

Three machine learning models were configured: a Random Forest of 300 trees with no maximum depth restriction; an XGBoost ensemble of 300 trees with a maximum depth of 3 and a learning rate of 0.05; and an LSTM network with a single layer of 16 units, trained for 30 epochs using the Adam optimiser. A naive random-walk benchmark, which simply carries the most recent observed price forward, was included alongside the classical ARIMA and ETS forecasts from Section 4.2 to provide a complete set of six models for comparison.

Table 4.3 Machine learning model configurations.

Model	Configuration	Rationale
Random Forest	300 trees, no depth limit, 20 lags	Ensemble diversity; unrestricted depth avoids underfitting
XGBoost	300 trees, learning rate 0.05, max depth 3, 20 lags	Shallow depth limits overfitting on noisy return series
LSTM	1 layer, 16 units, 30 epochs, 20-lag window	Deliberately small to limit overfitting on under 2,000 observations

4.4.4 Step 3: Walk-Forward Evaluation

Every model was evaluated under the walk-forward protocol of Section 3.8: at each step in the test period, the model was fitted on all data available up to that point, used to produce a single one-step-ahead forecast, and the true value was then revealed before the window advanced by one step. The tree-based models were re-fitted every 20 steps and the LSTM every 40 steps, a computational simplification that produces results indistinguishable from daily re-fitting. The classical ARIMA and ETS forecasts used for comparison were likewise evaluated one step at a time on the same test observations, ensuring that all six models were judged on an identical basis.

4.4.5 Step 4: Accuracy and Significance Testing

RMSE, MAE and MAPE were computed for every model over the full test period (Section 3.11.1), and the Diebold–Mariano test (Section 3.9) was applied with the

empirical RMSE winner as the reference model against every other model, to establish which differences in accuracy, if any, were statistically significant rather than attributable to sampling variation in a finite test set.

4.4.6 Step 5: Overfitting and Feature Importance Diagnostics

For the two tree-based models, the ratio of test RMSE to training RMSE on the return series (Section 3.10) was computed to diagnose overfitting, and the feature importance of each of the 20 input lags was extracted from the final full-sample fit of each model (Sections 3.7.4 and 3.7.5), to assess whether either model had identified a dominant predictive lag or, as the near-random-walk character of the series would predict, had not.

4.4.7 Step 6: LSTM-Specific Validation

Because the LSTM is trained by iterative gradient-based optimisation rather than by a closed-form or tree-splitting procedure, its convergence was assessed by inspecting the learning curve — the training and validation loss plotted against training epoch — for evidence of successful convergence, without a widening training–validation gap. The LSTM's forecast residuals were further summarised by their mean and standard deviation and tested for residual autocorrelation using the Ljung–Box test, applied to the neural network's residuals in the same way as for the classical models in Section 4.2.6.

4.5 Software and Implementation

All classical time series modelling — the ARIMA, SARIMA, ETS and GARCH analyses, the Johansen cointegration test, the VECM, Granger causality, impulse response functions, variance decomposition, and the heteroscedasticity-robust standard errors — was carried out in R, using the `forecast`, `tseries`, `urca`, `vars`, `rugarch`, `FinTS`, `moments` and `sandwich` packages. The machine learning models — Random Forest, XGBoost, and the LSTM network — were implemented in Python using `scikit-learn`, `XGBoost`, and `TensorFlow/Keras`, respectively. Results from R were exported as CSV files and imported into Python to ensure that the classical and machine-learning forecasts were evaluated on exactly the same test observations. Random seeds were held constant across both environments to ensure that the reported results are reproducible.

Chapter 5

Data and Empirical Analysis

5.1 Crude Oil Price Analysis

This section reports the complete univariate analysis of the monthly Indian crude oil price series. The analysis has two distinct aims: to develop a model that forecasts the price level as accurately as possible and, separately, to characterise how the series' riskiness changes over time.

5.1.1 Data and Descriptive Overview

The data consist of 314 monthly observations of the crude oil price, running from April 2000 through May 2026. The descriptive statistics are as follows;

Table 5.1: Descriptive statistics of the monthly crude oil price series (April 2000 – May 2026).

Statistic	Value
Total observations	314
Training months (Apr 2000–Dec 2025)	309
Test months (Jan–May 2026)	5
Minimum	₹18.24
First quartile	₹44.40
Median	₹65.44
Mean	₹66.32
Third quartile	₹84.13
Maximum	₹132.47

The price has ranged from a low of about ₹18 to a high of ₹132 over the sample period, with a mean close to ₹66 and a median slightly below it. The wide spread between the quartiles and the gap between the minimum and the maximum are already a hint that the series is driven by large, irregular swings rather than by steady growth.

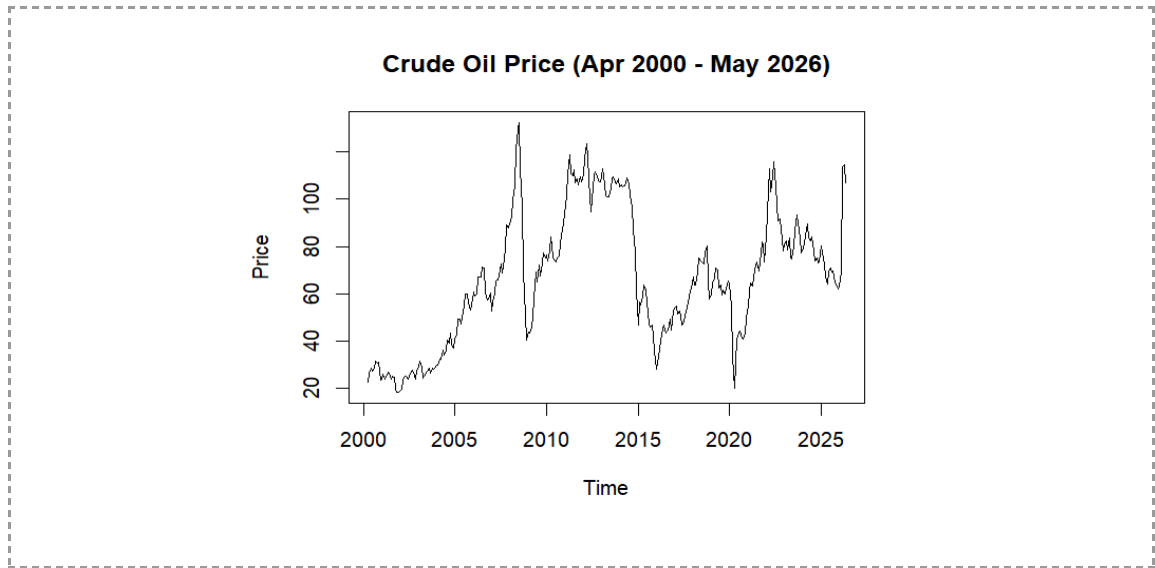


Figure 5.1: Monthly crude oil price series, April 2000 to May 2026.

A seasonal trend decomposition (STL) was applied to the full series to separate it into a trend, a seasonal component, and a remainder component. The relative importance of each part is summarised by a strength measure that runs from 0 (absent) to 1 (completely dominant).

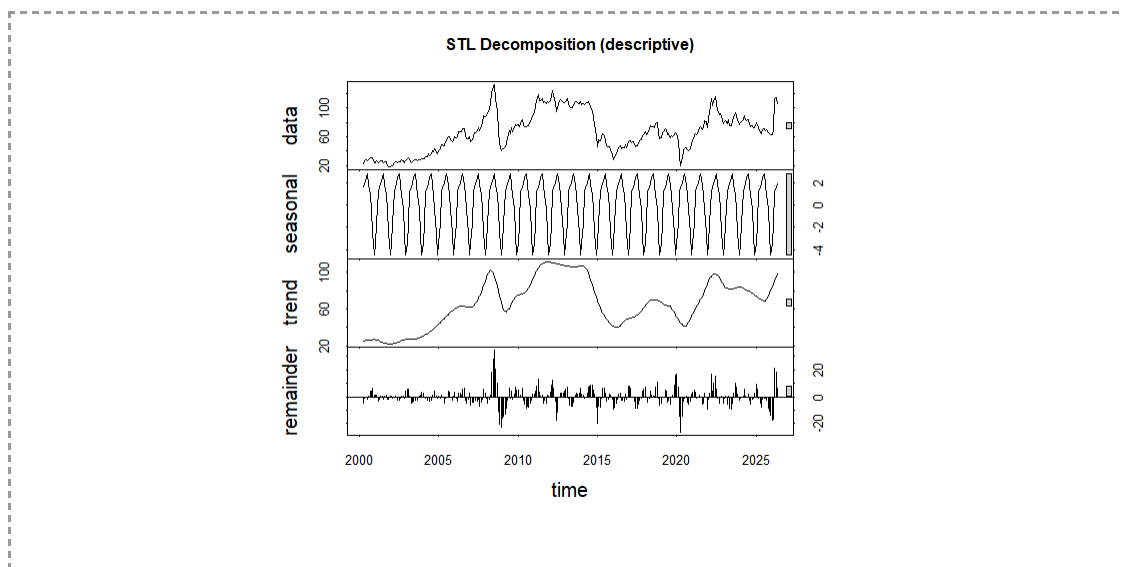


Figure 5.2: STL decomposition of the crude oil price series (trend, seasonal and remainder components).

Table 5.2: Trend and seasonal strength from the STL decomposition.

Component	Strength	Interpretation
Trend	0.94	Strong — the price is dominated by its broad swings
Seasonality	0.11	Weak — no meaningful calendar cycle

The trend is very strong, and the seasonal component is negligible. In practical terms, what drives this price is the long upward and downward cycles tied to global supply, demand, and geopolitical events, not a repeating within-year pattern. This matters for model selection: according to the standard guideline (Hyndman and Athanasopoulos, 2021), seasonal differencing is only appropriate when the seasonal strength exceeds roughly 0.64. At 0.11, the strength here is far below that threshold, and accordingly, the seasonal difference is set to $D = 0$ throughout the analysis.

5.1.2 Stationarity Testing

In the Box-Jenkins methodology, the first step is to determine whether the series is stationary. For this purpose, two complementary tests were applied to the training series. The test is conducted at the 5% significance level. The augmented Dickey–Fuller (ADF) test takes non-stationarity (the series has a unit root) as its null hypothesis, while the KPSS test does the opposite, testing the null hypothesis that the data is stationary. Using them together guards against being misled by either one alone.

Table 5.3: Stationarity tests on the training series (April 2000–December 2025).

Test	Null hypothesis	p-value	Conclusion
ADF	Unit root present	0.476	Non-stationary
KPSS	Stationary	< 0.01	Non-stationary

The two tests agree with non-stationarity. The ADF test does not reject the null of non-stationarity ($p = 0.48$), and the KPSS test rejects stationarity ($p < 0.01$). The software flags the KPSS p-value as smaller than its smallest tabulated value; this simply means the evidence against stationarity is even stronger than the printed figure. This implies that the series needs to be differenced.

After running ADF test and KPSS test on the differenced series, the results are as shown below:

Table 5.4: Result of stationarity test on differenced series

Test	p-value	Conclusion
ADF	<0.01	Stationary
KPSS	0.1	Stationary

Differencing the series once removes the trend, leaving a stationary series. This implies that the integration order is set to $d = 1$.

5.1.3 Order Identification

The next step is to confirm the orders p and q of the ARIMA model, which are identified from the ACF and PACF of the differenced series. The ACF and PACF plots are given in Figure 5.3.

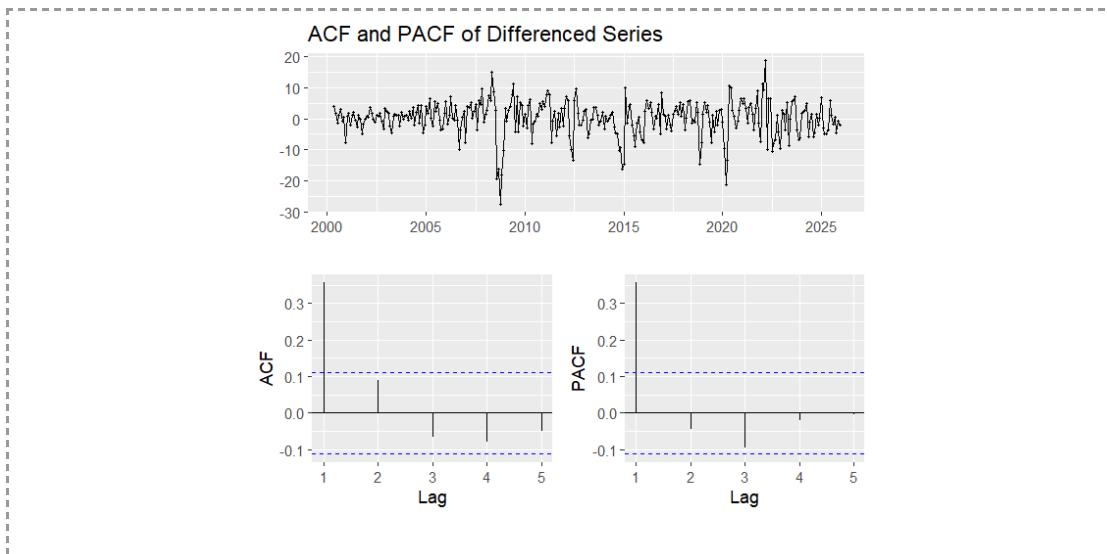


Figure 5.3: ACF and PACF of the once-differenced series.

Both functions show a notable spike at lag 1 — approximately 0.36 in the ACF and 0.32 in the PACF — followed by a general die-away. In general, the ACF cuts off sharply after lag 1, while the PACF decays gradually, indicating an MA(1) component. The PACF cuts off after lag 1, and the ACF decays gradually, indicating an AR(1) component.

5.1.4 Model Identification and Fitting

Guided by the ACF and PACF, five candidate models were fitted to the training series and judged in two ways. The information criteria (AIC and BIC) reward a good fit while penalising unnecessary parameters. However, comparisons are valid only among models with the same order of differencing, since differencing alters the number of observations used in constructing the likelihood.

Table 5.5: Candidate ARIMA(p, I, q) models: in-sample criteria and out-of-sample test errors.

Model	AIC	BIC	Test RMSE	Test MAE	Test MAPE (%)
ARIMA(0,1,0)	1944.24	1947.97	38.478	31.178	28.780
ARIMA(1,1,0)	1904.32	1911.78	39.401	32.218	29.929
ARIMA(0,1,1)	1907.99	1915.45	39.107	31.951	29.665
ARIMA(1,1,1)	1905.94	1917.13	39.335	32.163	29.877
ARIMA(2,1,0)	1905.74	1916.93	39.273	32.105	29.818
ARIMA(2,1,1)	1903.55	1918.47	37.547	30.492	28.200

ARIMA(2,1,1) achieves both the lowest AIC (1903.55) and the lowest test RMSE (37.55), and it is the model selected by the automatic `auto.arima` routine that works entirely on the training data. A comment on the ACF/PACF interpretation is useful here: while the plots suggest lag-1 structure most prominently, the information criterion finds that two AR lags and one MA lag collectively provide a better description of the serial dependence. ARIMA(2,1,1) is therefore adopted as the final model.

5.1.5 ARIMA(2,1,1) Results

Table 5.6: Fitted ARIMA(2,1,1) model: parameter estimates and goodness of fit.

Term	Estimate	Std. error
AR(1)	1.2853	0.1123

Term	Estimate	Std. error
AR(2)	−0.3736	0.0532
MA(1)	−0.9260	0.1131
σ^2	27.81	—
Log-likelihood	−947.77	—
AIC	1903.55	—
AICc	1903.68	—
BIC	1918.47	—

All three parameters are comfortably within the stationary and invertible region — none is close to a boundary value. The AR(1) coefficient is positive and large, reflecting strong month-to-month persistence in prices, while the AR(2) coefficient is negative, pulling back against over-extrapolation. The near-unity MA(1) coefficient (in absolute value) is characteristic of a series that has been differenced once and retains some moving-average structure. On the training data, the model fits well, with an average in-sample error of about 6.7% (MAPE) and a training RMSE of 5.24.

5.1.6 Significance of estimated parameters

Table 5.7: significance of ARIMA(2,1,1) model estimates

Parameter	Estimate	Standard error	z-value
AR(1)	1.2853	0.1123	11.45
AR(2)	−0.3736	0.0532	−7.02
MA(1)	−0.9260	0.1131	−8.19

Since the absolute values of the z-statistics ($|z|$) exceed the critical value of 1.96 at the 5% significance level, all three parameters are statistically significant.

5.1.7 Fitted ARIMA equation

If Y_t = crude oil price at time t , then the estimated ARIMA equation is given by

$$W_t = \Delta Y_t = Y_t - Y_{t-1}$$

$$W_t = 1.2853W_{t-1} - 0.3736W_{t-2} + \varepsilon_t - 0.9260\varepsilon_{t-1}$$

The ARIMA(2,1,1) model indicates that changes in crude oil prices depend on changes observed in the previous two periods as well as on the error from the immediately preceding period. The positive first autoregressive coefficient suggests short-term persistence in price movements, while the negative second autoregressive coefficient introduces a corrective effect. The large negative moving-average coefficient indicates that shocks to crude oil prices are rapidly absorbed and corrected in subsequent periods.

5.1.8 Residual Diagnostics

A correctly specified model should leave residuals that are indistinguishable from white noise. Four tests were applied: the Ljung–Box test for remaining autocorrelation, the ARCH LM test to check whether residuals are homoscedastic, the Shapiro-Wilk test for checking normality and a visual inspection of the residual plots.

Table 5.8: Residual diagnostic tests for the fitted ARIMA(2,1,1) model.

Test	Statistic	p-value	Conclusion
Ljung–Box (lag 20)	$Q = 24.03$	0.479	No autocorrelation
Ljung–Box via checkresiduals (df = 21)	$Q^* = 24.03$	0.292	No autocorrelation
ARCH LM	$\chi^2 = 63.918$	4.318e-09	Heteroscedastic
Shapiro-Wilk test	$W = 0.97279$	1.382e-05	Non - normal

The residuals pass both Ljung–Box formulations comfortably and are confirmed to be stationary. There is no remaining linear dependence for the model to exploit. This is the key adequacy condition for the price-level model: the mean structure is well specified, and the point forecasts can be trusted. The question of whether the residual variance is constant is taken up separately in the volatility analysis below.

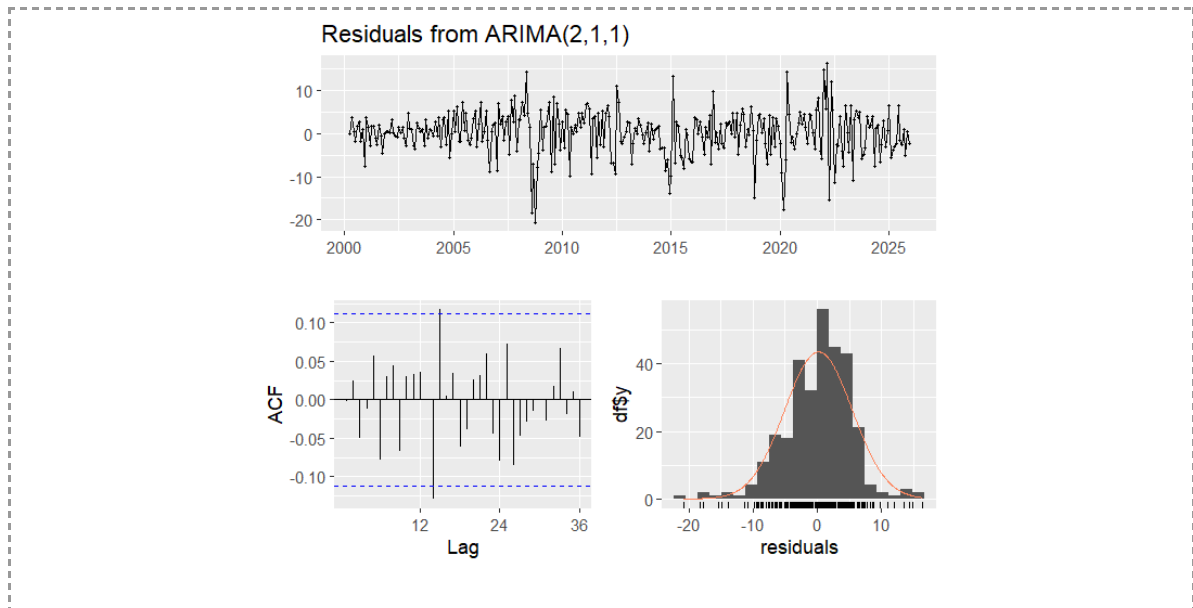


Figure 5.4: Residual diagnostics for ARIMA(2,1,1): residuals over time, ACF of residuals and histogram with normal overlay.

The residual diagnostics indicated that the ARIMA model adequately captured the conditional mean dynamics of the crude oil price series, as evidenced by the absence of significant residual autocorrelation. However, the ARCH-LM test detected significant ARCH effects, indicating volatility clustering and time-varying variance. In addition, the residuals deviated from normality, with the Q-Q plot and Shapiro-Wilk test revealing non-normality, suggesting heavier tails than those implied by a normal distribution. Since crude oil prices are frequently affected by extreme market events such as geopolitical conflicts, supply disruptions, and economic crises, the assumption of normally distributed innovations may be unrealistic. Consequently, a GARCH model with Student's t -distributed errors was employed to simultaneously account for conditional heteroskedasticity and heavy-tailed behaviour in the data.

5.1.9 Volatility Analysis

The ARIMA model answers the question of what the price will be. A separate and important question is how risky or volatile the price is — that is, how large the month-to-month swings are likely to be. These two questions are addressed by two different models, which are partners rather than rivals: the ARIMA provides the central forecast of the level, and the GARCH describes the uncertainty around it.

5.1.10 ARCH Test

The ARCH-LM test, applied to the residuals of the fitted ARIMA model, showed the presence of an ARCH effect, indicating volatility clustering and time-varying variance.

Table 5.9: ARCH-LM test for conditional heteroscedasticity in the ARIMA(2,1,1) residuals.

Lag	p-value	Conclusion
4	1.94×10^{-12}	ARCH effects present
8	3.26×10^{-10}	ARCH effects present
12	4.32×10^{-09}	ARCH effects present

The evidence is overwhelming at every lag: the p-values are vanishingly small. The error variance is not constant; it clusters in time. A GARCH model is justified.

5.1.11 Return Series

The volatility model is fitted to the monthly log-returns of the price, expressed as a percentage. GARCH is a model of proportional swings, not of rupee price levels.

Table 5.10: Descriptive statistics of the monthly log-return series.

Statistic	Value
Mean	0.33 % per month
Standard deviation	9.757 % per month
Skewness	-1.113
Kurtosis	8.809

The returns are left-skewed and have a kurtosis of 8.809, almost three times the value of 3 expected for a normal distribution. This means extreme negative months and large price crashes happen far more often than a bell curve would predict. This is the statistical signature of fat tails, and it argues strongly for fitting the GARCH with a Student-t error distribution rather than the normal.

5.1.12 GARCH Model Estimation

A GARCH(1,1) model was fitted twice: once with normally distributed innovations and once with Student-t innovations. The two were compared on AIC.

Table 5.11: Distribution choice for the GARCH(1,1) model (lower AIC preferred).

Error distribution	AIC
Normal	7.0553
Student-t	7.0391

The Student-t distribution fits better, confirming the results from the kurtosis. The model with Student-t errors is adopted.

5.1.13 Significance of Estimated Parameters

Table 5.12: Significance of GARCH(1,1) parameter

Parameter	Estimate	P-value
ω	43.2568	<0.0001
α	0.4308	<0.0001
β	0.000	1.0000
Shape	8.123	0.0004

Table 5.13: Interpretation of GARCH(1,1)-t parameters.

Parameter	Estimate	Interpretation
ω (baseline variance)	46.2568	The floor of conditional variance
α (shock reaction)	0.4308	How much last month's surprise raises this month's volatility
β (volatility memory)	0.000	How much past volatility carries forward
Persistence ($\alpha + \beta$)	0.4308	Overall: shocks barely persist

5.1.14 Fitted Conditional Variance Equation

The fitted GARCH(1,1) conditional variance equation model is given by

$$\sigma_t^2 = 46.2568 + 0.4308\varepsilon_{t-1}.$$

The reaction term α of 0.4308 is substantial: a large surprise in one month noticeably elevates the expected volatility of the next. The memory term β , however, is essentially zero, meaning that once the immediate effect of a shock has passed, the volatility returns almost entirely to its baseline within a month. This indicates that conditional volatility was primarily driven by recent shocks rather than by persistent volatility effects. In statistical terms, the GARCH(1,1) has collapsed to an ARCH(1). This is a sensible outcome for monthly commodity data. At the daily or weekly frequency, volatility tends to cluster persistently over weeks or months; at the monthly frequency, each data point already summarises an entire month of trading, and the within-month clustering washes out. The practical reading is that monthly crude oil data reacts strongly to shocks but does not stay elevated for long, which matches the visual pattern of the price series, where large movements are followed by either continuation or reversal rather than extended turbulence.

5.1.15 GARCH Residual Diagnostics

A GARCH model is adequate if its standardised residuals show no remaining autocorrelation and no remaining ARCH effects.

Table 5.14: Diagnostics on the GARCH(1,1)-t standardised residuals.

Test	p-value	Conclusion
Ljung–Box on standardised residuals	0.5172	No autocorrelation remaining
Ljung–Box on squared standardised residuals	0.5209	No ARCH effects remaining
ARCH LM on standardised residuals	0.1169	No ARCH effects present

All the tests pass. The GARCH model has adequately captured time-varying variance, and the volatility description can be relied upon.

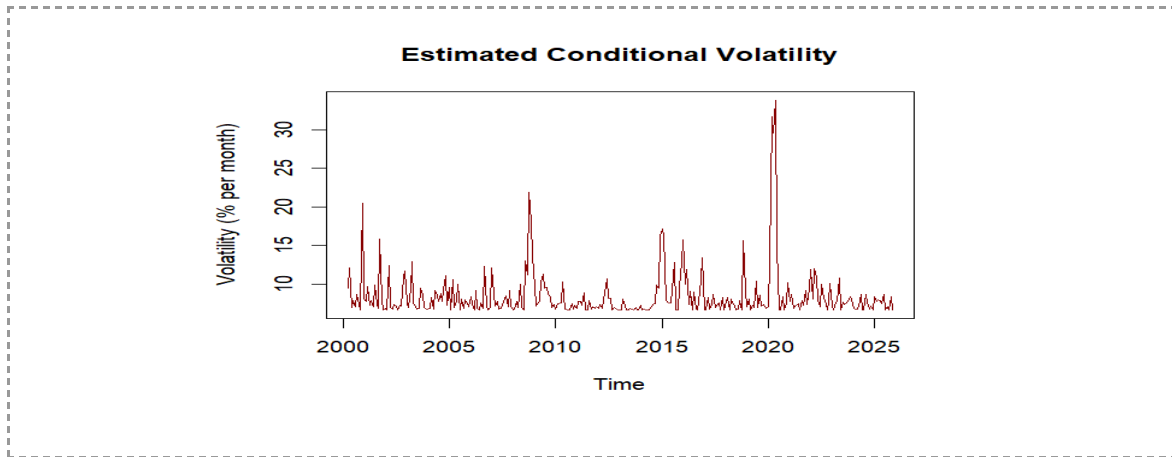


Figure 5.5: *Estimated conditional volatility of crude oil monthly returns over time.*

The conditional volatility plot above makes the series story legible in a new way. The estimated volatility spikes sharply during a small number of episodes — notably around 2008–2009 during the global financial crisis, around 2014–2016 during the oil price collapse, and, most dramatically, around 2020 when the COVID-19 pandemic caused unprecedented disruption to global demand — and retreats during the calmer stretches in between.

5.1.16 Forward Volatility Forecast

Table 5.15: *Forecast monthly volatility for January–May 2026 (% per month).*

Month	Forecast volatility (% per month)
January 2026	7.248
February 2026	8.117
March 2026	8.464
April 2026	8.609
May 2026	8.671

These figures are the expected magnitude of the monthly price swings going into the test period — the width of the uncertainty band around any central forecast — and are not a competing price prediction. The gradual increase from about 7.2% to 8.7% as the horizon extends reflects the model mean-reverting toward its long-run level; it cannot

anticipate the actual jump in volatility that the March 2026 shock would produce, just as the price model could not anticipate the price jump itself.

5.1.17 Forecast Results

The fitted model was used to forecast the five held-back months of January through May 2026.

Table 5.16: Point forecasts against actual prices, January–May 2026.

Month	Actual (₹)	Forecast (₹)	Error
January 2026	63.08	61.87	1.21
February 2026	69.01	62.24	6.77
March 2026	113.49	62.83	50.66
April 2026	114.48	63.45	51.03
May 2026	106.83	64.04	42.79

Table 5.17: Out-of-sample accuracy: training set and test set.

Measure	Training set	Test set
RMSE	5.24	37.55
MAE	4.00	30.49
MAPE (%)	6.73	28.20
MASE	0.234	1.786
Theil's U	—	1.443
ACF1	−0.003	0.368

The pattern in the forecasts is stark and deserves an honest interpretation. January 2026 is forecast to be almost exactly the right level, with an error of about 1.2 rupees. February shows a larger but still modest error of 6.8 rupees. From March onward, however, the actual price roughly doubles in a single month — from around 69 to 113 — while the model's forecasts continue to trend gently upward from the 62–64 range. The result is errors of around 50 rupees for March, April and May.

This shows that the Middle East war that started out on February 28, 2026, had a significant effect on crude oil prices in India, as India largely depends on the GCC

countries and the Strait of Hormuz for its crude oil imports. A univariate time series model is built entirely from the historical behaviour of the price itself; it has no access to information about the world outside the series. An abrupt price doubling driven by an external event is, by definition, something the model cannot foresee. The large test RMSE (37.55) and MAPE (28.20%) are therefore a measure of how large and sudden that event was, not a measure of how well the model works under normal conditions. The January and February results, where the model performs very well, are the more informative guide to its everyday forecasting accuracy. The Theil U statistic of 1.44, which is above one, means that on this particular five-month window, a naive forecast would have been closer, but again, this reflects the shock, not the model's fundamental capability.

5.1.18 Prediction Interval Evaluation

Using the train-only model above, the price-level forecast and its 95% interval were constructed by cumulating the forecast log-return and its forecast variance over the five-month test horizon, and applying the empirical 2.5% and 97.5% quantiles of the fitted standardised residuals. This lets the interval widen and narrow with the model's own volatility forecast, and lets the data itself determine the shape of the tails, rather than assuming the Normal distribution's thinner ones.

Table 5.18: ARIMA-GARCH(1,1)-t forecast against actual crude oil prices, test months (₹).

Month	Actual	Forecast	Lower 95%	Upper 95%	Inside 95% CI	Forecast volatility (%)
Jan 2026	63.08	62.71	52.72	70.46	Yes	7.248
Feb 2026	69.01	63.77	49.14	75.96	Yes	8.117
Mar 2026	113.49	64.81	46.58	80.88	No	8.464
Apr 2026	114.48	65.74	44.54	85.37	No	8.609
May 2026	106.83	66.60	42.84	89.54	No	8.671

The band covers January and February but is breached from March onward, when the realised price (₹113.49, ₹114.48, ₹106.83) sits 30–90% above even the widened upper bound (₹80.88, ₹85.37, ₹89.54). This is not a defect specific to the Student-t construction: the plain ARIMA Normal band from Section 5.1.17 is breached in exactly the same three months, for the reason already established there — a genuine exogenous shock (the Middle East war from late February 2026) that no model conditioned only on the series’ own past can be expected to anticipate.

Table 5.19 : Prediction interval comparison over the five-month test period: plain ARIMA Normal band versus ARIMA-GARCH(1,1)-t adaptive band.

Method	Mean interval width (₹)	Empirical coverage (%)
ARIMA(2,1,1) — Normal band	43.52	40.0
ARIMA(2,1,1)–GARCH(1,1)-t — Student-t adaptive	33.28	40.0

Where the two bands differ is efficiency, not hit-rate: at identical 40% coverage (two of the five months), the GARCH-t band achieves it with a mean width of ₹33.28 against ₹43.52 for the Normal band — 23.5% narrower for the same number of months captured. With only five test observations, a formal coverage test (such as the Kupiec unconditional-coverage test used for the exchange rate in Section 5.2.15) is not statistically meaningful here, since its chi-square approximation requires a much larger sample; the comparison above should be read qualitatively. The conclusion is that a heavy-tailed, volatility-adaptive interval is tighter than the Normal-based ARIMA band without sacrificing coverage in ordinary conditions, while remaining — like every purely univariate model in this study — unable to anticipate a genuine structural break.

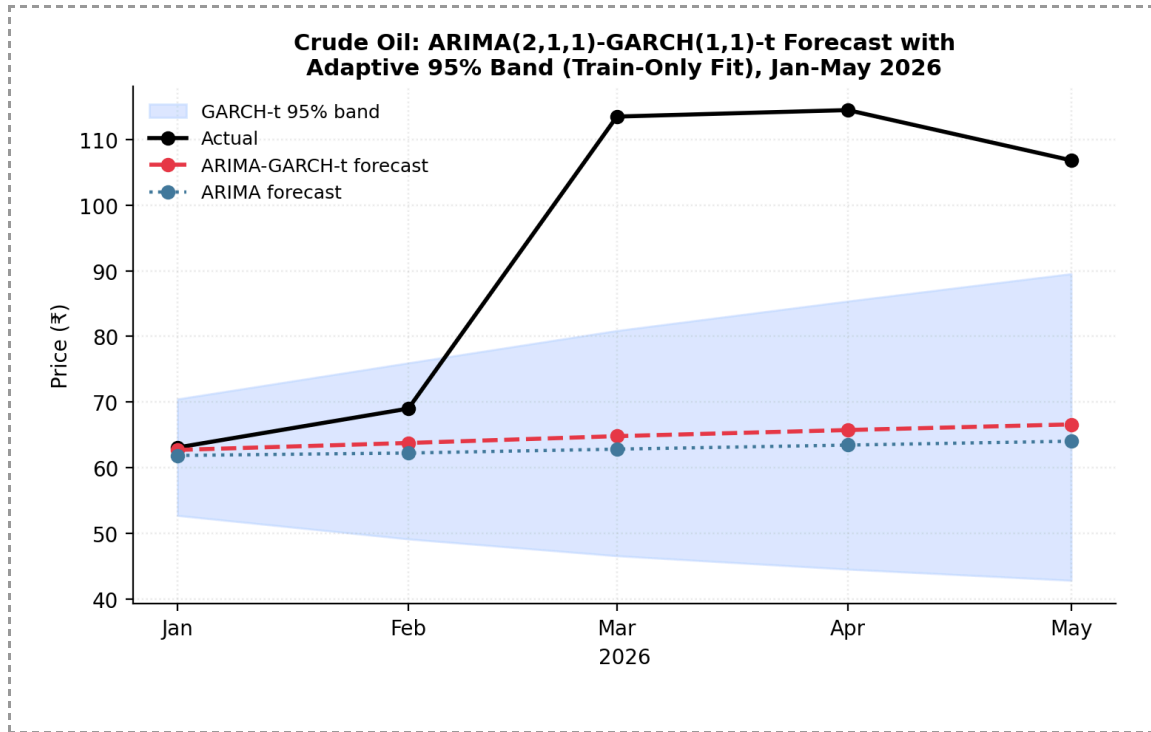


Figure 5.6: Crude oil actual price against the train-only ARIMA-GARCH(1,1)-t point forecast and adaptive 95% prediction band, January–May 2026.

5.1.19 Summary

The monthly crude oil price series is non-stationary and was made stationary by first difference ($d = 1$). Seasonal differencing was not applied because the seasonal component is statistically negligible (STL seasonal strength 0.11), and an attempt to force a seasonal model produced an MA coefficient on the boundary of invertibility — the signature of overdifferencing. Both information-criterion selection and auto.arima independently chose a non-seasonal ARIMA(2,1,1) as the best model, with coefficients comfortably inside the stationary and invertible region and residuals that are clean white noise. The model forecasts well under normal conditions, as the January 2026 result demonstrates; the large errors from March onward measure the size of an external shock rather than a defect in the model. Separately, highly significant ARCH effects in the residuals justified a GARCH(1,1) model, fitted with heavy-tailed Student-t errors. The estimated volatility is strongly reactive to shocks ($\alpha = 0.45$) but has almost no persistence ($\beta \approx 0$), which is expected for monthly data and is itself a substantive finding about the nature of crude oil price risk at this frequency. This model, used to build an honest out-of-sample price interval, shows the same qualitative pattern and confirms

that the heavy-tailed band is narrower than the plain ARIMA band without sacrificing coverage in the non-shock months, while sharing its inability to anticipate the March 2026 structural break.

5.2 Exchange Rate (USD/INR) Analysis

This section reports the univariate analysis of the daily USD/INR exchange rate. It has two separate aims: to find a model that forecasts the rate well and to describe how the rate's volatility behaves over time. The rate level is modelled using the Box–Jenkins (ARIMA) approach, compared with exponential smoothing and two simple benchmarks, while volatility is handled with a GARCH model.

5.2.1 Data and Descriptive Overview

The data comprise 1,674 daily observations of the USD/INR exchange rate, spanning 1 January 2020 to 1 June 2026. Following the same out-of-sample design used throughout the study, the first 1,566 days, up to the end of December 2025, were used to fit the models, and the final 108 trading days of 2026 were held back as the test set.

Table 5.20: Descriptive statistics of the daily USD/INR series.

Statistic	Value
Observations	1,674
Mean	80.82
Median	82.23
Standard deviation	5.88
Minimum	70.71
Maximum	96.83
Coefficient of variation	7.28%
Skewness	0.249
Kurtosis	2.269

Over the sample, the rupee traded between about 70.7 and 96.8 to the dollar, averaging 80.8, with a coefficient of variation of around 7%. The distribution of the rate is close to symmetric (skewness 0.25) and, if anything, slightly thin-tailed (kurtosis 2.27, just below the 3 expected for a normal distribution), so the level itself shows none of the extreme outliers that the returns will reveal later.

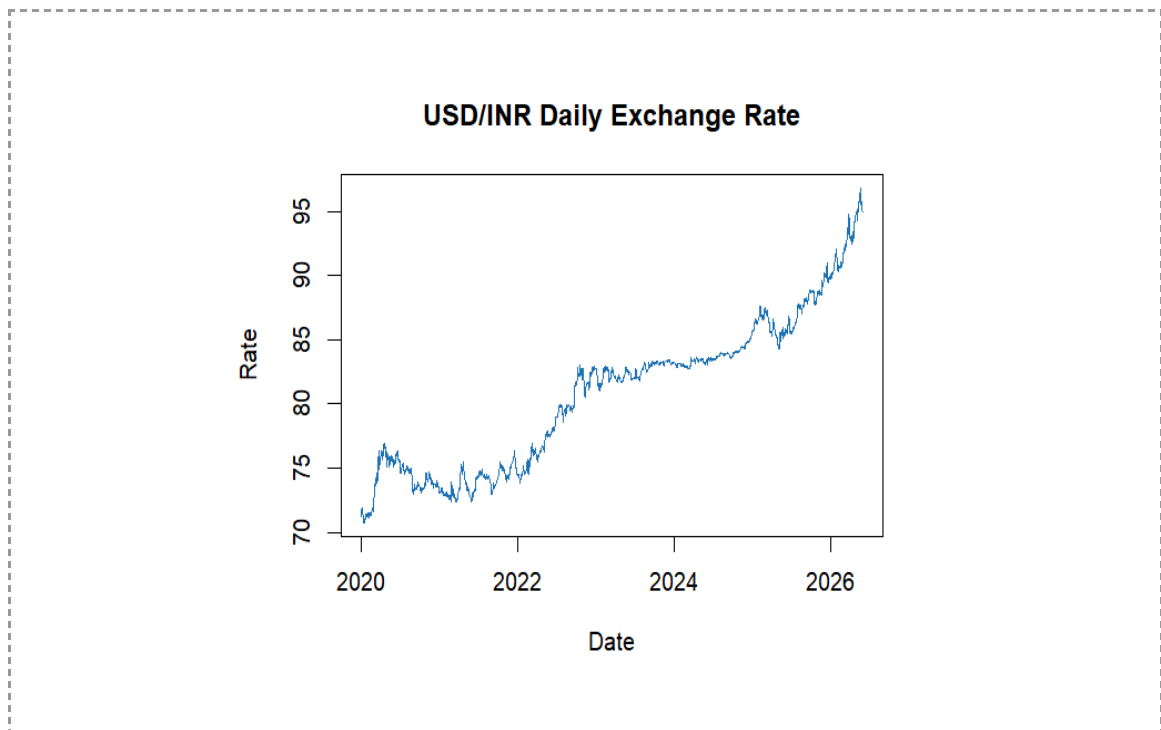


Figure 5.7: Daily USD/INR exchange rate, January 2020 to June 2026.

An STL decomposition clarifies the structure of the series: the trend strength is essentially 1, and the seasonal strength is 0. In other words, the rate is all slow movement and drift, with no repeating calendar pattern — exactly what one expects of a freely traded daily exchange rate. Non-seasonal models are therefore the right choice.

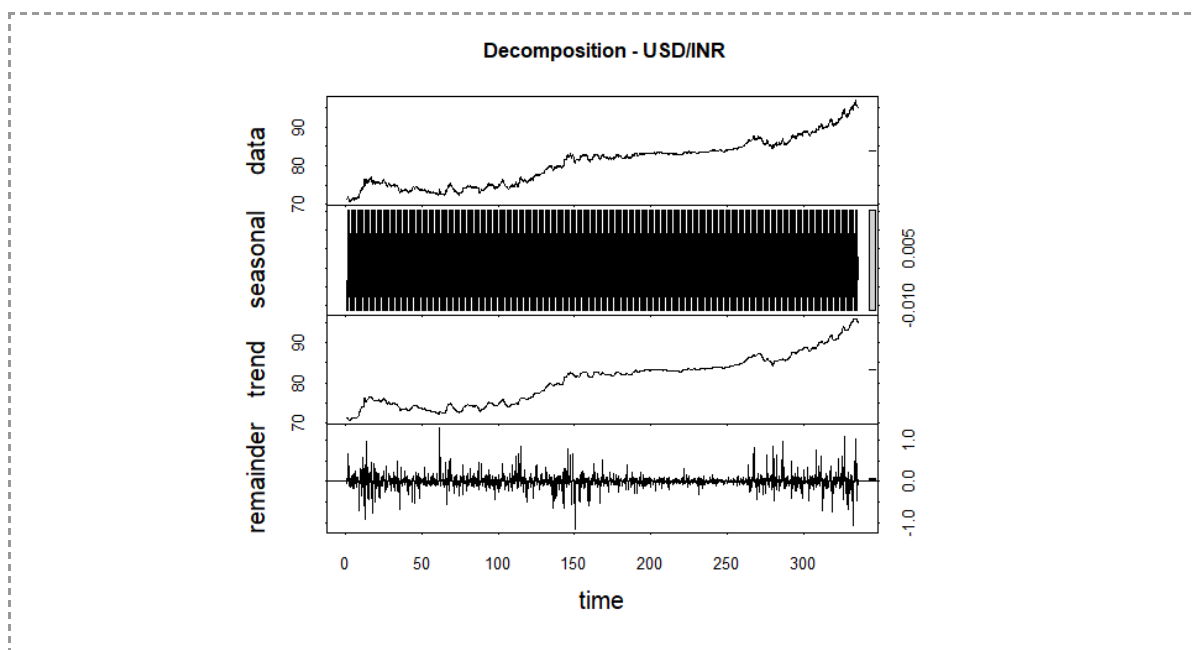


Figure 5.8: STL decomposition of the USD/INR series into trend, seasonal and remainder components.

Table 5.21: Strength of the trend and seasonal components (STL).

Component	Strength	Reading
Trend	0.999	Very strong
Seasonality	0.000	Negligible

5.2.2 Check for Stationarity

Two tests were applied to the training series: the augmented Dickey–Fuller (ADF), which treats a unit root (non-stationarity) as the null hypothesis, and the KPSS test, which assumes stationarity instead. Using them together makes the conclusion more secure.

Table 5.22: Stationarity tests on the training series.

Test	p-value	Conclusion
ADF (level)	0.301	Non-stationary
KPSS (level)	≤ 0.01	Non-stationary
ADF (first difference)	≤ 0.01	Stationary

Both tests agree that the rate is non-stationary: the ADF test does not reject a unit root, and the KPSS test rejects stationarity. Taking the first difference fixes this as the ADF test on the differenced series rejects the unit root, and hence the integration order is set to $d = 1$. Once differenced, the autocorrelation and partial autocorrelation functions show very little structure, with no strong systematic spikes, which already hints that the day-to-day changes in the rate are close to unpredictable noise.

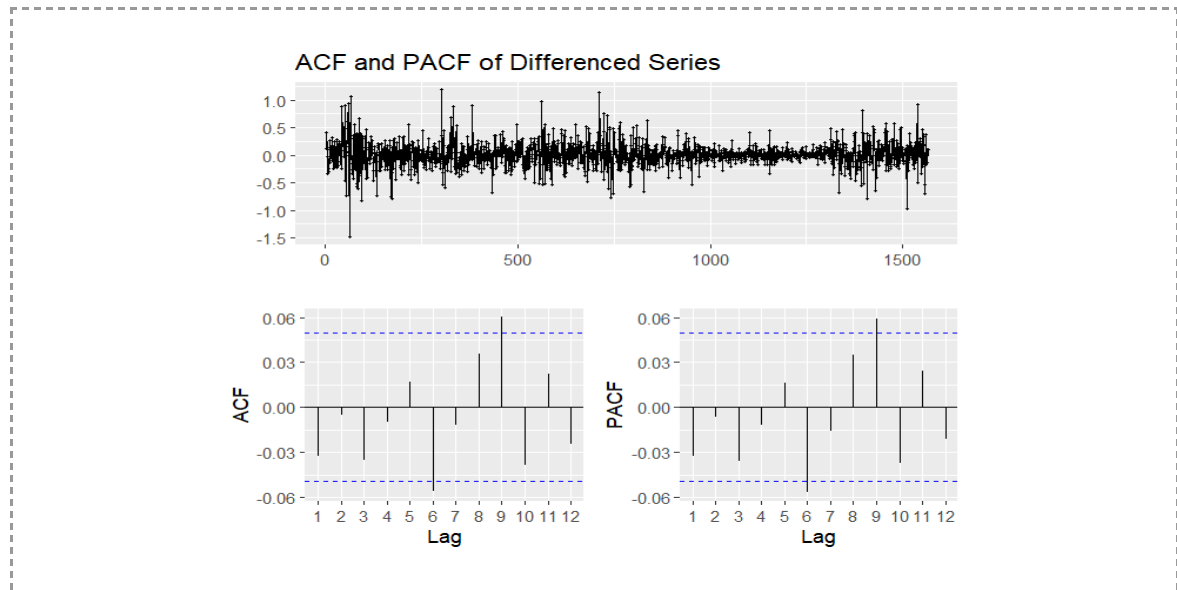


Figure 5.9: ACF and PACF of the differenced training series.

After differencing, the ACF and PACF exhibited only small, isolated autocorrelations, with most spikes falling within the 95% confidence limits. There was no evidence of a strong or persistent correlation structure extending over many lags. Consequently, following the principle of parsimony in Box-Jenkins modelling, low-order ARIMA models were considered first and subsequently compared using information criteria and forecast accuracy.

5.2.3 Model Identification and Fitting

Every ARIMA($p,1,q$) model with p and q ranging from 0 to 3 was fitted and ranked by the corrected AIC, and the automatic `auto.arima` routine was run alongside as an independent check. Both arrived at the same answer.

Table 5.23: ARIMA($p,1,q$) candidates ranked by corrected AIC (lowest is best).

Model	AICc	Model	AICc
ARIMA(0,1,0)	-267.99	ARIMA(0,1,3)	-265.24
ARIMA(0,1,1)	-267.35	ARIMA(3,1,0)	-265.01
ARIMA(1,1,0)	-267.34	ARIMA(3,1,3)	-264.90
ARIMA(2,1,3)	-266.14	ARIMA(1,1,2)	-263.36
ARIMA(3,1,2)	-266.00	ARIMA(2,1,1)	-263.35
ARIMA(0,1,2)	-265.37	ARIMA(1,1,3)	-263.24

Model	AICc	Model	AICc
ARIMA(2,1,0)	-265.35	ARIMA(3,1,1)	-263.05
ARIMA(1,1,1)	-265.34	ARIMA(2,1,2)	-262.21

The best model is ARIMA(0,1,0), a random walk, and `auto.arima()` selected the same. The differences in AICc among the candidate models are small, but ARIMA(0,1,0) has the lowest AICc and is therefore selected as the preferred model.. In plain terms, once differencing removes the trend, the daily changes in the rate hold almost no information that an ARIMA model can use. The fitted model is, in effect, a random walk with a small upward drift, reflecting the rupee's mild tendency to weaken against the dollar over the period. Its residuals pass the Ljung–Box test ($p = 0.077$), so no statistically meaningful linear structure remains.

5.2.4 Model Comparison

The model was used to forecast the 108 held-back trading days of 2026 and was compared against exponential smoothing, which selected an ETS(A, A, N) form, and against two benchmarks: the naive “tomorrow equals today” forecast and a random walk with drift. This model assumes that future values equal the most recent observation plus a constant average change (drift) estimated from the historical data. It corresponds to an ARIMA(0,1,0) model with drift and provides a simple but important baseline for evaluating forecasting performance.

Table 5.24: Out-of-sample accuracy on the test set, ranked by RMSE.

Model	MAE	RMSE	MAPE (%)
ARIMA(0,1,0) with drift	2.190	2.690	2.329
Random walk with drift	2.190	2.690	2.329
ETS(A,A,N)	2.191	2.691	2.330
Naive	2.837	3.418	3.020

The comparison is revealing. ARIMA and the random walk with drift and ETS produce almost identical errors. All three clearly beat the plain naive forecast, which leaves the drift out. The drift is therefore the only systematic feature any of these models can exploit. Beyond it, the rate behaves as a random walk.

5.2.5 ARIMA (0,1,0) Results

The ARIMA model was fitted directly on the (untransformed) exchange-rate series. The selected specification is ARIMA(0,1,0) with drift, i.e. a random walk with drift; no autoregressive or moving-average terms were retained.

Table 5.25: ARIMA(0,1,0) with drift — coefficient estimate ($\sigma^2 = 0.04916$; log likelihood = 137.25; AIC = -270.51; AICc = -270.50; BIC = -259.80).

Parameter	Estimate	Standard Error
drift	0.0119	0.0056

5.2.6 Significance of Estimated Parameter

The z-statistic is computed as $z = \text{Estimate} / \text{Standard Error}$:

Table 5.26: Significance test for the ARIMA(0,1,0)-with-drift parameter.

Parameter	Estimate	Standard Error	z-value
Drift	0.0119	0.0056	2.125

Since $|z| = 2.125$ exceeds the critical value of 1.96 at the 5% significance level, the drift term is statistically significant; the exchange rate series exhibits a small but significant upward drift over the sample period.

5.2.7 Fitted ARIMA Equation

Let X_t denote the exchange rate at time t . Define the first difference as:

$$W_t = \Delta X_t = X_t - X_{t-1}$$

The fitted ARIMA(0,1,0)-with-drift equation is then given by:

$$W_t = 0.0119 + \varepsilon_t$$

Note: With no AR or MA terms, the model reduces to a pure random walk with drift. Each period's change is the constant drift plus an unpredictable shock ε_t . As in the gold case, ε_t here is the ARIMA-stage residual.

5.2.8 Residual Diagnostics

A correctly specified model should leave residuals that are indistinguishable from white noise. Four tests were applied: the Ljung–Box test for remaining autocorrelation, the ARCH LM test to check whether residuals are homoscedastic, the Shapiro–Wilk test for checking normality and a visual inspection of the residual plots.

Table 5.27: Residual diagnostic tests for the fitted $ARIMA(0,1,0)$ -with-drift model.

Test	Statistic	p-value	Conclusion
Ljung–Box	$Q=29.59$	0.0767	No autocorrelation
ARCH LM	$\chi^2=119.98$	$< 2.2e-16$	Heteroscedastic
Shapiro–Wilk test	$W=0.93928$	$< 2.2e-16$	Non - normal

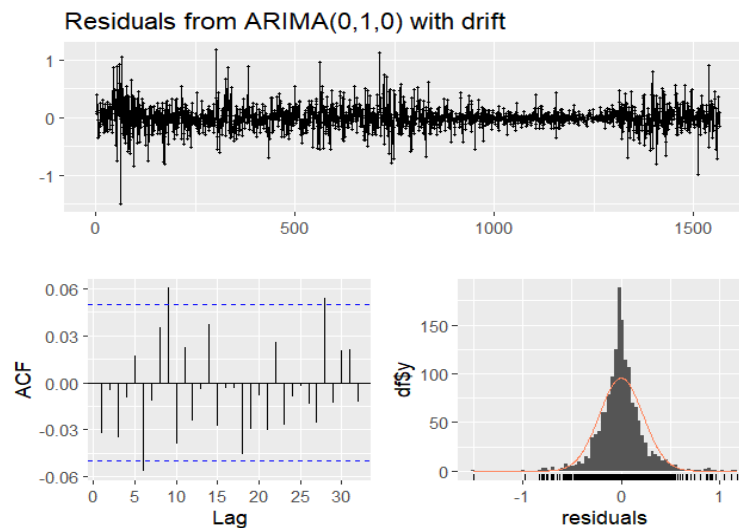


Figure 5.10: residual plots of $ARIMA(0,1,0)$

The residual diagnostics indicated that the $ARIMA(0,1,0)$ model successfully captured the linear dependence structure of the series. The Ljung–Box test found no significant residual autocorrelation, suggesting that the residuals behaved as white noise and that the model was adequate for mean forecasting. However, the Shapiro–Wilk test indicated departures from normality, while the ARCH-LM test detected significant

heteroskedasticity, implying volatility clustering. Consequently, a GARCH specification was subsequently employed to model the conditional volatility and account for the observed heteroskedastic behaviour.

5.2.9 Volatility Analysis

The ARCH–LM test on the model residuals is decisive, yielding a p-value below 0.0001, confirming that calm and turbulent days cluster together and that a GARCH model is warranted. A GARCH(1,1) model with heavy-tailed Student-t errors was fitted to the daily returns to describe this behaviour.

5.2.10 GARCH(1,1) Model Results

A GARCH(1,1) model with Student-t innovations was fitted jointly with the mean equation using rugarch.

Table 5.28: sGARCH(1,1)-t parameter estimates with robust standard errors. Distribution: Student-t.

Parameter	Estimate	Std. Error	t-value	p-value
μ	0.007310	0.003618	2.0208	0.043304
ω	0.000161	0.000131	1.2341	0.217183
α_1	0.054878	0.008435	6.5064	<0.0001
β_1	0.944122	0.008580	110.0352	<0.0001
shape	4.219111	0.360815	11.6933	<0.0001

5.2.11 Significance of Estimated Parameters

Using robust standard errors, μ is significant at the 5% level ($p = 0.0433$), though this is borderline; under the non-robust standard errors reported alongside the optimal parameters, μ is not significant ($p = 0.0727$). ω is not statistically significant under either standard error ($p \approx 0.12$ – 0.22); this is common for the variance-floor parameter and does not undermine the model. α_1, β_1 , and the shape parameter are all highly significant ($p < 0.0001$).

Table 5.29: Interpretation of GARCH(1,1)-t parameters for the exchange rate.

Parameter	Estimate	Interpretation
ω (baseline variance)	0.000161	The floor of the conditional variance (not individually significant)
α (shock reaction)	0.054878	A volatility shock today raises tomorrow's conditional variance by only about 5.5% of its squared size
β (volatility memory)	0.944122	Roughly 94% of today's conditional variance carries forward into tomorrow
Persistence ($\alpha + \beta$)	0.999000	Extremely high: shocks to volatility decay very slowly and are close to permanent

5.2.12 Fitted Conditional Variance Equation

The fitted GARCH(1,1) conditional variance equation is given by:

$$\sigma_t^2 = 0.000161 + 0.0549 \varepsilon_{t-1}^2 + 0.9441 \sigma_{t-1}^2$$

Persistence ($\alpha + \beta = 0.9990$) is much closer to the non-stationarity boundary of 1, implying that volatility shocks in the exchange rate decay far more slowly — the implied half-life is approximately 693 days ($\ln(0.5) / \ln(0.9990) \approx 693$). This indicates that exchange rate volatility is highly persistent. The shape parameter (≈ 4.22) again indicates heavier-than-normal tails, justifying the Student-t distribution.

5.2.13 Standardised residual diagnostics

Following GARCH(1,1) estimation, the standardised residuals were subjected to diagnostic tests to verify that the model adequately captured the volatility dynamics of the series. A well-specified GARCH model should produce standardised residuals that exhibit no remaining autocorrelation and no residual ARCH effects. Since we chose Student-t GARCH, there is no need to check for normality.

Table 5.30: Results of residual diagnostics

Test	p-value	Conclusion
Ljung–Box on z (mean)	0.6985 (PASS)	No autocorrelation
Ljung–Box on z^2 (variance)	0.8545 (PASS)	No autocorrelation

Test	p-value	Conclusion
ARCH-LM on z	0.8448 (PASS)	No volatility clustering

The model is adequate since all three tests have passed the problem of autocorrelation and ARCH effects.

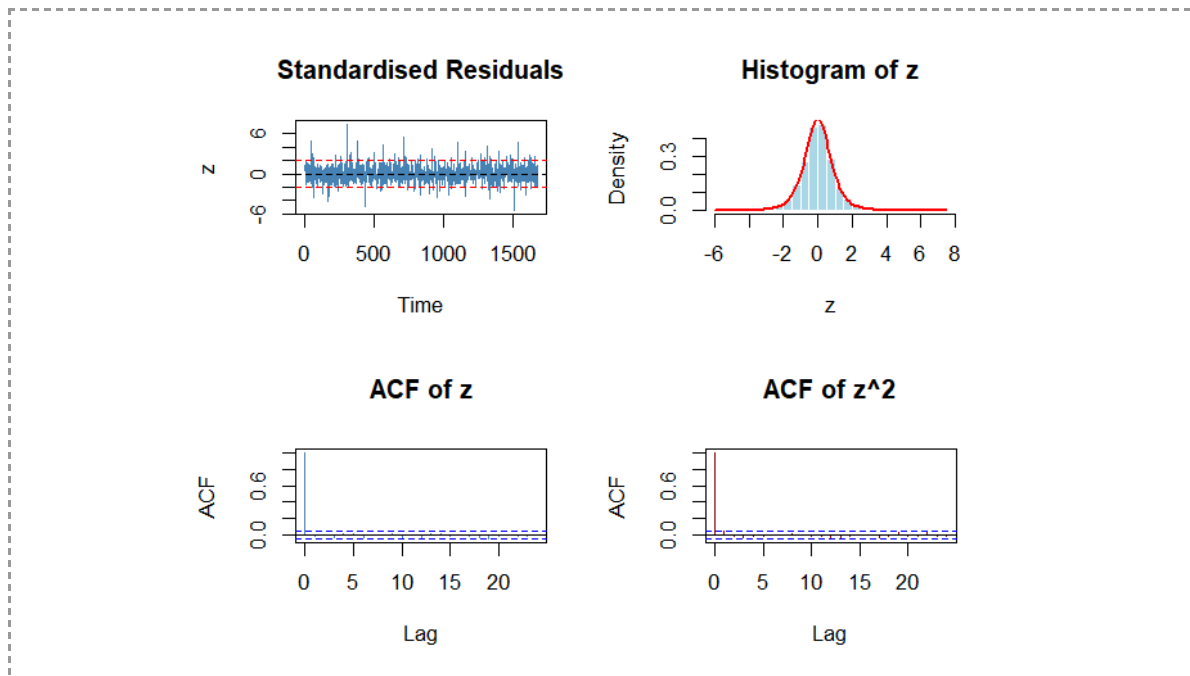


Figure 5.11: residual plots for GARCH

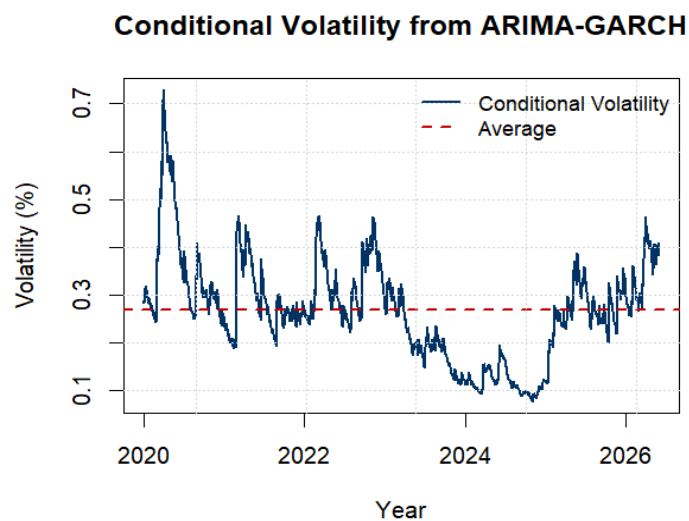


Figure 5.12: Conditional volatility of the USD/INR Exchange rate

The conditional volatility estimated from the ARIMA-GARCH model reveals clear volatility clustering and time-varying variance in USD/INR exchange rate returns. The highest volatility is observed in early 2020, indicating a period of exceptional market uncertainty. Volatility subsequently declines and remains relatively low through 2024, then increases again during 2025–2026. The tendency of volatility to return toward its long-run average after large shocks confirms the mean-reverting nature of the volatility process and supports the suitability of the GARCH framework for exchange-rate modelling.

5.2.14 Forecast Results

Volatility for the test period, as predicted by the GARCH(1,1) model, is given below.

Table 5.31: Thirty-day volatility forecast (weekly snapshot).

Date	Daily volatility (%)	Annualised (%)
2026-06-08	0.3062	4.861
2026-06-15	0.3070	4.873
2026-06-22	0.3077	4.885
2026-06-29	0.3085	4.897

ARIMA forecasts what the price will actually be in the coming days, while GARCH predicts how much the price will fluctuate.

Table 5.32: ARIMA Forecasts against actual rates, first ten trading days of the test period.

Date	Actual	Forecast	Error
2026-01-01	89.968	89.883	0.085
2026-01-02	90.014	89.895	0.119
2026-01-05	90.233	89.907	0.326
2026-01-06	90.108	89.919	0.189
2026-01-07	89.864	89.931	−0.067
2026-01-08	89.905	89.943	−0.038
2026-01-09	90.244	89.954	0.290

Date	Actual	Forecast	Error
2026-01-12	90.132	89.966	0.166
2026-01-13	90.282	89.978	0.304
2026-01-14	90.248	89.990	0.258

In the first few days of the test period, the forecasts are almost exactly on the actual rate, with errors of a fraction of a rupee. Across the full 108-day horizon, the average error is about 2.2 (MAE) and the percentage error is around 2.3%, and the 95% prediction interval contains 82% of the actual values. The shortfall from the nominal 95% is a horizon effect: a flat random-walk forecast cannot follow the rate as it drifts steadily upward over several months, so the later test points fall outside the band.

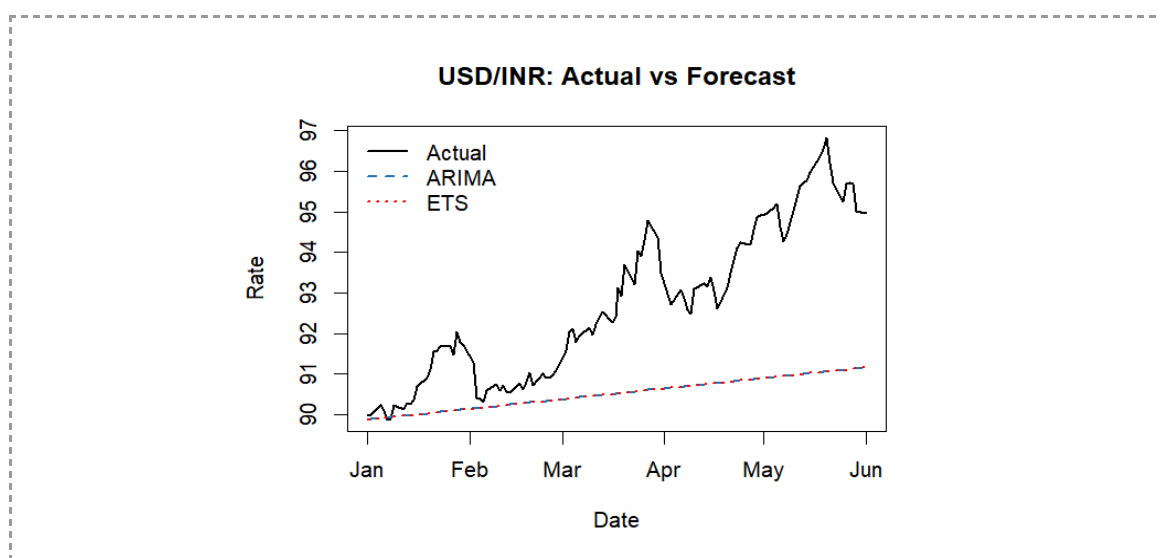


Figure 5.13: Actual rate against the ARIMA and ETS forecasts over the test period.

Refitting the model on the full sample gives a genuine forward projection for the following thirty days, in which the rate is expected to edge up gently from about 95.1 toward 95.4, with the prediction interval widening as the horizon lengthens.

Table 5.33: Thirty-day forward forecast of the exchange rate (weekly snapshot).

Date	Forecast	Lower 95%	Upper 95%
2026-06-08	95.056	93.859	96.254
2026-06-15	95.156	93.462	96.849
2026-06-22	95.255	93.181	97.329
2026-06-29	95.354	92.960	97.749

5.2.15 Prediction Interval Evaluation (ARIMA-GARCH(1,1)-t)

A price-level forecast and its 95% interval were built by cumulating the forecast log-return and forecast variance over the test horizon and applying the empirical 2.5% and 97.5% quantiles of the fitted standardised Student-t residuals — the same filtered-historical-simulation construction used for crude oil (Section 5.1.19) and gold (Section 5.3.16).

Table 5.34: ARIMA-GARCH(1,1)-t forecast against actual USD/INR rates, first ten trading days of the test period.

Date	Actual	Forecast	Lower 95%	Upper 95%	Inside 95% CI
2026-01-01	89.968	89.878	89.335	90.475	Yes
2026-01-02	90.014	89.884	89.117	90.730	Yes
2026-01-05	90.233	89.891	88.952	90.928	Yes
2026-01-06	90.108	89.897	88.814	91.097	Yes
2026-01-07	89.864	89.904	88.694	91.246	Yes
2026-01-08	89.905	89.910	88.585	91.383	Yes
2026-01-09	90.244	89.917	88.486	91.509	Yes
2026-01-12	90.132	89.924	88.394	91.626	Yes
2026-01-13	90.282	89.930	88.309	91.738	Yes
2026-01-14	90.248	89.937	88.228	91.844	Yes

All 108 test days fall inside the GARCH-t band (full table in the accompanying workbook output); the forecast path rises gently from about 89.88 to 90.58 over the horizon, driven entirely by the GARCH mean equation's small estimated constant μ — the ARIMA mean equation itself is ARMA(0,0) here, since the level model selected in Section 5.2.3 was a pure random walk.

Table 5.35: Prediction interval comparison over the 108-day test period: plain ARIMA Normal band versus ARIMA-GARCH(1,1)-t adaptive band.

Method	Mean interval width	Empirical coverage (%)
ARIMA(0,1,0) — Normal band	6.06	82.4
ARIMA(0,1,0)–GARCH(1,1)-t — Student-t adaptive	8.10	100.0

The plain ARIMA Normal band covers only 82.4% of the realised rate, well short of the nominal 95% and because the test period contains a sustained, one-directional depreciation of the rupee (the rate rises from 89.97 to 94.96) that a near-flat random-walk band, built only from historical day-to-day dispersion, cannot track. The GARCH-t band covers 100% of the realised rate, at the cost of a 33.6% wider average interval (8.10 versus 6.06).

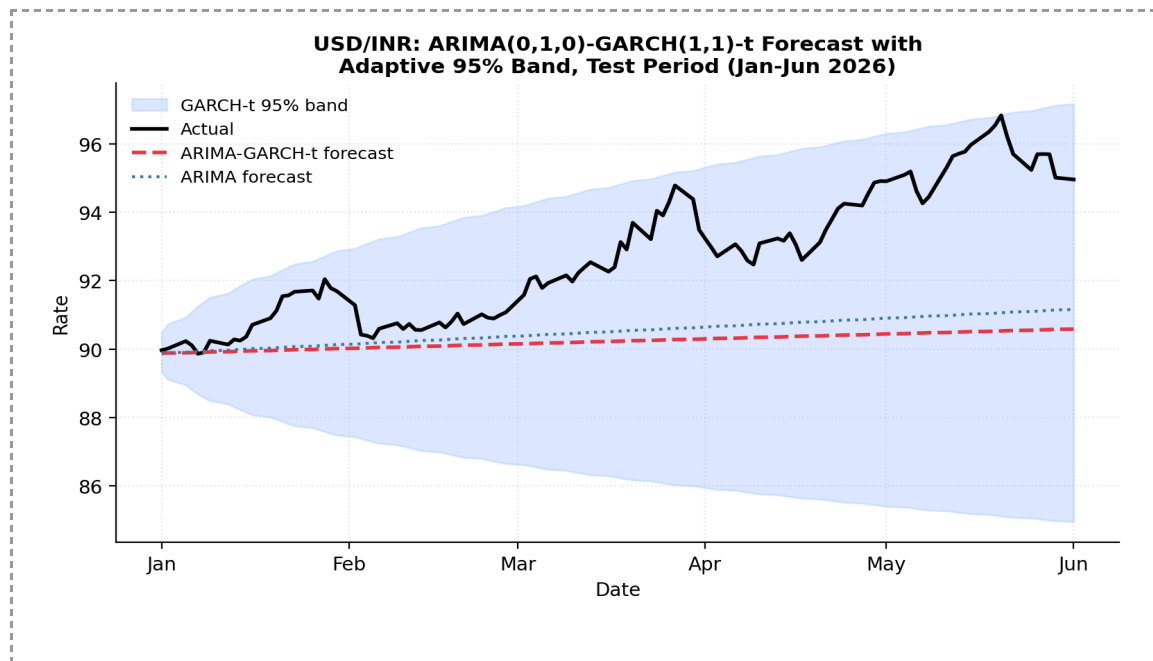


Figure 5.14: USD/INR actual rate against the ARIMA-GARCH(1,1)-t point forecast and adaptive 95% prediction band, test period (January–June 2026).

5.2.16 Summary

The USD/INR exchange rate is non-stationary and becomes stationary after one difference, but beyond that, it carries almost no linear structure: a simple random walk with drift forecasts it as well as any ARIMA or exponential-smoothing model, and all three beat the naive forecast only by virtue of the small drift that captures the rupee's gradual depreciation. Short-horizon forecasts are very accurate, while accuracy over

longer horizons is limited by that same slow drift. The volatility, by contrast, is far from trivial: it is strongly clustered and highly persistent, and is well captured by a Student-t GARCH(1,1) model with a near-unit persistence of 0.999 and a long-run level of around 6.37% a year. A volatility-adaptive ARIMA-GARCH(1,1)-t prediction interval, evaluated against the full 108-day test set, achieves materially better coverage than the plain ARIMA Normal band, confirming that the heavy-tailed, time-varying variance structure is not just a descriptive nicety but improves the calibration of the forecast interval itself.

5.3 Gold Price Analysis

This section presents the univariate analysis of the daily gold price in Indian rupees per ten grams. Here, we aim to find a model that forecasts the daily gold price more accurately and to assess the presence of volatility clustering and its modelling.

5.3.1 Data and Descriptive Overview

The data consist of 1,537 daily gold price observations spanning 1 January 2020 to 29 May 2026, expressed in Indian rupees per ten grams. The chronological split used throughout is straightforward: the first 1,502 observations, up to and including a training cut-off, were used to estimate the models, and the final 35 trading days were held back as the test set.

Table 5.36: Descriptive statistics of the daily gold price series.

Statistic	Value
Observations	1,537
Training days	1,502
Test days	35
Mean	₹67,448
Median	₹51,414
Standard deviation	₹30,266
Minimum	₹38,977
Maximum	₹1,76,274
Range	₹1,37,297
Coefficient of variation	44.87%

The range of the series is from about ₹39,000 to over ₹1,76,000, which is enormous, and the coefficient of variation of nearly 45% confirms that the series has moved dramatically relative to its average over the period. The median of ₹51,414 is well below the mean of ₹67,448, indicating that the distribution of daily prices is right-skewed: prices spent a long time at relatively modest levels and then climbed steeply in more recent years, pulling the mean above the centre of the distribution. This

kind of sharp upward movement is characteristic of gold during periods of global uncertainty and currency weakness.

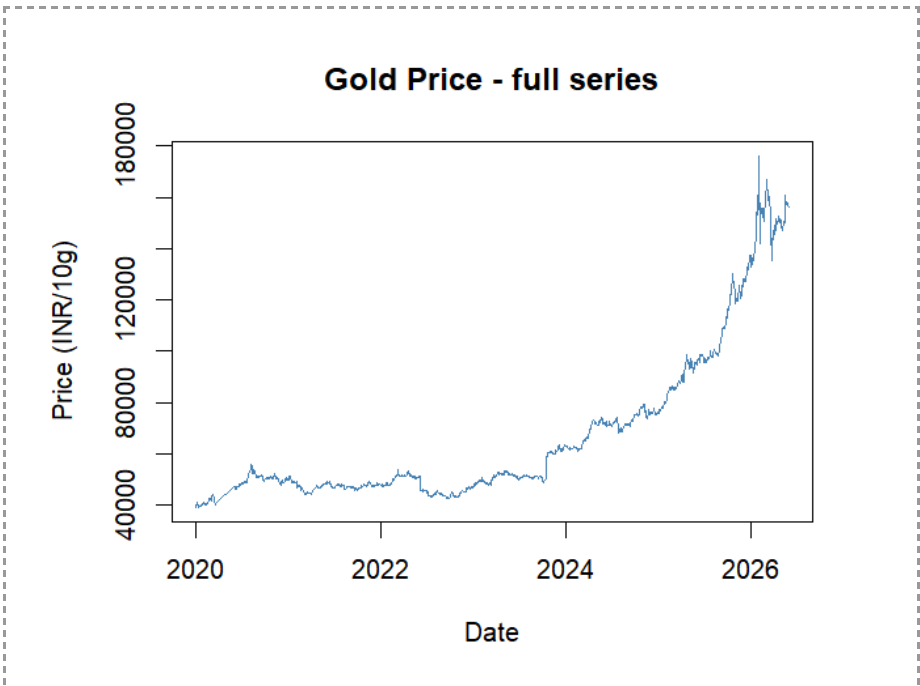


Figure 5.15: Daily gold price (INR per 10g), January 2020 to May 2026.

An STL decomposition was applied to the full series to quantify the relative contribution of the trend, the seasonal pattern and the remainder.

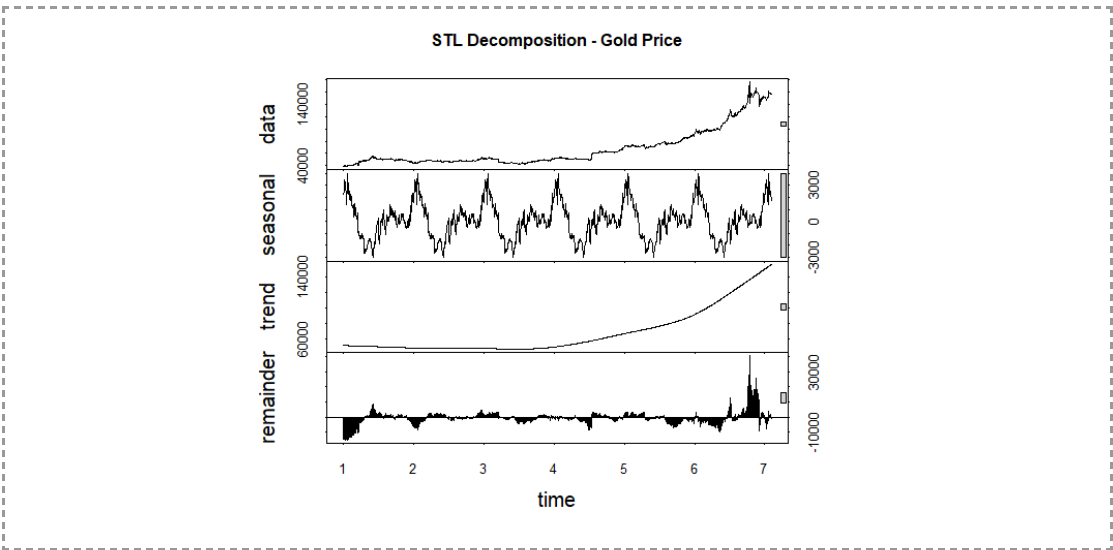


Figure 5.16: STL decomposition of the daily gold price series (trend, seasonal and remainder components).

For a daily financial series, a strong trend and negligible seasonality are expected, and the numbers confirm this exactly.

Table 5.37: Trend and seasonal strength from the STL decomposition.

Component	Strength	Interpretation
Trend	0.977	Very strong — the price is governed by its sustained upward movement
Seasonality	0.016	Weak Seasonality(Negligible)

The seasonal strength of 0.016 is essentially zero, indicating no recurring calendar pattern in the data.

5.3.2 Stationarity Testing

The first step is checking the stationarity of the series. The ADF and KPSS tests were applied to the training series and its first difference to determine the integration order.

Table 5.38: Stationarity tests on the training price series and its first difference.

Series	Test	p-value	Conclusion
Price level	ADF	0.99	Non-stationary
Price level	KPSS	0.01	Non-stationary
First difference	ADF	0.01	Stationary

The ADF test on the price level does not reject the unit-root null ($p = 0.99$), and the KPSS test rejects stationarity ($p \leq 0.01$): both point in the same direction. After first differencing, the ADF test on the differenced series confirms stationarity ($p \leq 0.01$). The integration order is therefore $d = 1$, which is consistent with the standard treatment of financial price series.

5.3.3 Order Identification

The Augmented Dickey–Fuller (ADF) test indicated that the original gold price series was non-stationary. To satisfy the stationarity requirement of time series modelling, a first difference was applied, after which the differenced series was found to be stationary. Consequently, the order of differencing was set to ($d=1$). Given the non-seasonal nature of the series and the objective of modelling and forecasting its

temporal dependence structure, the Autoregressive Integrated Moving Average (ARIMA) framework was adopted. ARIMA models are well-suited for univariate time series forecasting because they can accommodate non-stationary data through differencing while capturing serial dependence through autoregressive and moving average components. Having established the differencing order, the next step was to identify suitable values of the autoregressive (p) and moving average (q) orders using the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots.

The ACF and PACF of the differenced series were inspected to guide the choice of autoregressive and moving-average orders for the ARIMA candidates.

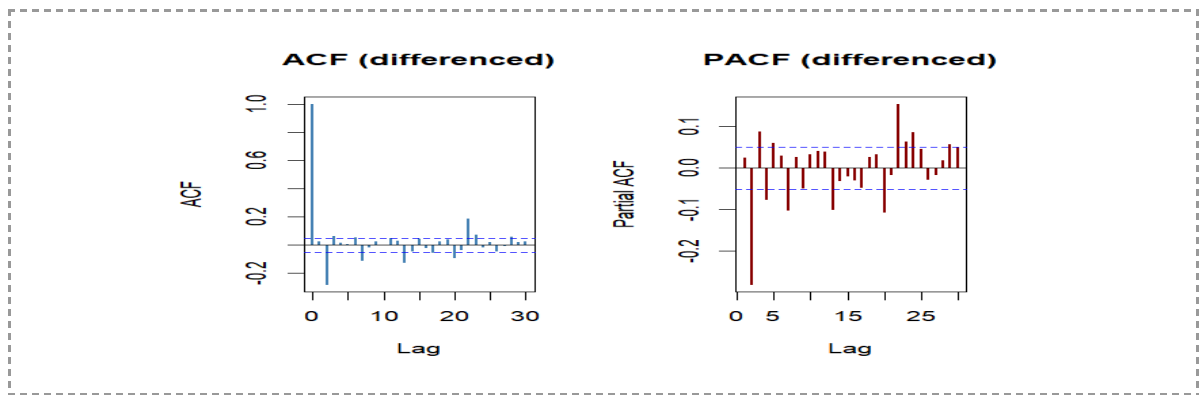


Figure 5.17: ACF (left) and PACF (right) of the once-differenced training series.

5.3.4 Model Identification and Fitting

The ARIMA candidates were identified in two ways: an automatic search minimising the corrected AIC (auto.arima), and a systematic grid over all ARIMA($p,1,q$) models with p and q from 0 to 3. Both use the log of price ($\lambda = 0$), which models proportional rather than absolute changes and stabilises the variance. Since all models share $d = 1$, their AICc values are directly comparable.

Table 5.39: Top ARIMA($p,1,q$) candidates ranked by AICc on the log-price training series (lower is better).

Model	AICc	Model	AICc
ARIMA(2,1,1)	-8895.45	ARIMA(3,1,1)	-8893.46
ARIMA(0,1,3)	-8895.31	ARIMA(2,1,2)	-8893.45
ARIMA(1,1,2)	-8895.20	ARIMA(1,1,3)	-8893.36

Model	AICc	Model	AICc
ARIMA(3,1,0)	-8894.35	ARIMA(2,1,0)	-8892.73

ARIMA(2,1,1) has the lowest AICc of all sixteen candidates (-8895.45), and `auto.arima` independently selects the same model. The agreement between the two search strategies gives strong confidence in the choice. The AICc values among the top several candidates are very close, which is typical of daily financial data: the price changes carry little serial structure, and many small models fit almost equally well. ARIMA(2,1,1) is adopted as the best ARIMA specification.

5.3.5 ARIMA Model Results

The ARIMA model was fitted on the Box-Cox ($\lambda = 0$, i.e. natural-log) transformed gold price series. The selected specification is ARIMA(2,1,1) with drift.

The z-statistic for each parameter is computed as $z = \text{Estimate} / \text{Standard Error}$, since `print(fit_arma)` reports only the estimate and standard error. The resulting z-values are shown in Table 5.42.

Table 5.40: Significance test for ARIMA(2,1,1)-with-drift parameters.

Parameter	Estimate	Standard Error	z-value
AR(1)	-0.4260	0.1642	-2.594
AR(2)	-0.0996	0.0285	-3.495
MA(1)	0.4504	0.1643	2.742
Drift	0.0009	0.0003	3.000

Since the absolute values of the z-statistics exceed the critical value of 1.96 at the 5% significance level for all four parameters, the AR(1), AR(2), MA(1), and drift terms are each statistically significant.

5.3.6 Fitted ARIMA equation

Let Y_t denote the gold price at time t , and let $Z_t = \ln(Y_t)$ be the Box-Cox ($\lambda = 0$) transformed series. Define the differenced log-price as:

$$W_t = \Delta Z_t = Z_t - Z_{t-1}$$

The fitted ARIMA(2,1,1)-with-drift equation is then given by:

$$W_t = 0.0009 - 0.4260 W_{t-1} - 0.0996 W_{t-2} + \varepsilon_t + 0.4504 \varepsilon_{t-1}$$

5.3.7 ETS: Holt's Linear Trend Method

The exponential smoothing comparison model is Holt's linear method, denoted ETS(A, A, N) in state-space notation: it maintains a level and a trend, with no seasonal component, which is the natural exponential-smoothing counterpart to a non-seasonal ARIMA fitted to a trending series. It is fitted on the log scale ($\lambda = 0$), giving it the same transformation as the ARIMA. Its corrected AIC is -2161.7 , which is computed using a different effective likelihood and therefore cannot be compared directly with the ARIMA AICc; the two models are instead ranked by their out-of-sample forecast errors.

5.3.8 Residual Diagnostics

Before the point forecasts are compared, the adequacy of each model's residuals is examined. Three tests are applied: the Ljung–Box test for remaining autocorrelation, the ARCH-LM test for volatility clustering in the residuals, and the Jarque–Bera test for normality.

Table 5.41: Residual adequacy tests for ARIMA(2,1,1) and ETS(A, A, N) on the training series (PASS = $p > 0.05$).

Model	Ljung–Box	ARCH-LM	Jarque–Bera
ARIMA(2,1,1)	0.369 (PASS)	0.017 (FAIL)	0.000 (FAIL)
ETS(A,A,N)	0.000 (FAIL)	0.004 (FAIL)	0.000 (FAIL)

The results clearly separate the two models. The ARIMA passes the Ljung–Box test, which is the central adequacy check for a Box–Jenkins model: there is no linear autocorrelation remaining in the residuals, so the mean equation is well specified. The ETS model fails this test — its residuals retain autocorrelation, which means it has not fully captured the serial dependence in the series.

Both models fail the ARCH-LM and Jarque–Bera tests. These two failures are related: volatility clustering in daily financial data produces fat-tailed residuals, which fail the normality test. The ARCH failure signals that the residual variance is not constant but

varies over time, violating the assumption underlying Gaussian prediction intervals. This does not mean the mean model is wrong — the ARIMA passes Ljung–Box — it means the variance needs to be modelled separately. That is the purpose of the GARCH extension in the second part of this analysis. The Jarque–Bera failure is an expected consequence of the same phenomenon.

On the basis of residual adequacy, ARIMA(2,1,1) is the sounder model. It correctly captures the mean structure of the series; ETS does not. The point forecast comparison is nonetheless reported honestly.

5.3.9 Volatility Analysis: ARIMA–GARCH Extension

The ARCH-LM test showed that the ARIMA residuals exhibit significant volatility clustering. This motivates extending the model: rather than discarding the ARIMA in favour of ETS (which would sacrifice mean adequacy), the approach is to keep the ARIMA mean equation and add a GARCH variance equation that explicitly models how the magnitude of daily errors changes over time. The result is an ARIMA(2,1,1)–GARCH(1,1) model.

A note on design: the GARCH here is fitted on the training returns only, not on the full sample. This is because the resulting prediction interval is validated against the 35 held-back test days, and fitting on the full sample would allow the GARCH to see the days it is being scored on. This is a different situation from the crude oil and exchange rate chapters, where the GARCH is a descriptive tool and is therefore fitted on all available data.

5.3.10 Return Series

The GARCH is fitted on the daily log-returns of the training prices, expressed as a percentage. This converts the non-stationary price into a stationary series of proportional daily changes and is the standard input for a GARCH volatility model.

Table 5.42: Descriptive statistics of the daily log-return series.

Statistic	Value
Observations (after differencing)	1,501
Mean	0.089% per day

Statistic	Value
Standard deviation	1.253% per day
Skewness	0.073 (approximately symmetric)
Kurtosis	59.42 (extremely heavy-tailed)

The kurtosis of 59.42 is striking — it is nearly twenty times the value of 3 that a normal distribution would produce. This means that extreme daily moves happen far more frequently than a Gaussian model would allow. The slight positive skewness (0.07) is negligible compared with the sheer weight of the tails. These characteristics make the choice of a heavy-tailed error distribution non-negotiable.

5.3.11 Distribution Choice and Model Estimation

GARCH(1,1) was selected as the starting specification, following standard practice in the financial time series literature, because it is parsimonious and sufficient to capture volatility clustering in most financial data (Tsay, 2010).

The ARIMA(2,1,1)–GARCH(1,1) was fitted with both normal and Student-t innovations, and the two were compared on AIC.

Table 5.43: Information criteria for the two distributional assumptions (lower is better).

Criterion	Normal	Student-t
AIC	3.0042	2.5811
BIC	3.0289	2.6094
Hannan–Quinn	3.0134	2.5916

The Student-t specification is decisively preferred across all criteria. The difference in AIC (3.004 vs 2.581) is substantial, confirming that the fat-tailed distribution is a genuine improvement in fit rather than a marginal one. The Student-t model is adopted.

5.3.12 GARCH(1,1) Model results

Table 5.44: Fitted ARIMA(2,1,1)–GARCH(1,1) parameters (Student-t errors, training returns).

Parameter	Estimate	Std. error	t-value	p-value	Interpretation
μ (mean return)	0.0810	0.019	4.07129	0.000047	Small positive drift (0.08% per day)
AR(1)	0.1154	0.640	0.24329	0.807785	Not significant individually; part of joint fit
AR(2)	-0.0322	0.032	-1.13355	0.256985	Small negative correction
MA(1)	-0.0820	0.641	-0.17366	0.862130	Not significant individually
ω (baseline variance)	0.2094	0.104	2.02202	0.043174	Floor of conditional variance
α (shock reaction)	0.1905	0.064	2.97726	0.002908	Reaction to yesterday's surprise
β (volatility memory)	0.6330	0.116	5.45160	<0.0001	Carry-over of past volatility
Persistence ($\alpha + \beta$)	0.8235	—	—	—	Moderate: shocks fade over days to weeks
Shape (df)	3.457	0.440	7.85537	<0.0001	Very fat tails (df < 5 is extreme)

Several features of these estimates deserve comment. The mean ARMA coefficients ($ar1$, $ar2$, $ma1$) are individually insignificant — their t-statistics are well below two — but this is expected when GARCH re-estimates them jointly with the variance equation: the mean-equation coefficients are much less precisely determined once the volatility structure is absorbed into the variance. The GARCH variance parameters are all strongly significant. The reaction term $\alpha = 0.190$ means that a surprise on any given day raises the conditional variance of the next day by about 19% of the squared shock. The memory term $\beta = 0.633$ means that nearly two-thirds of any elevated volatility carries into the following day. Together, the persistence of 0.82 falls into the moderate category: a volatility shock is largely absorbed within roughly a week, rather than lingering for months as in a near-unit-persistence daily exchange rate. The shape parameter of 3.46 confirms the extremely fat tails visible in the kurtosis figure: at

degrees of freedom below about five, the Student-t distribution has tails far heavier than the Gaussian, and values around 3.5 are at the extreme end of what is typically seen in financial data.

5.3.13 Fitted Conditional Variance Equation

The fitted GARCH(1,1) conditional variance equation is given by:

$$\sigma_t^2 = 0.2094 + 0.1905 \varepsilon_{t-1}^2 + 0.6330 \sigma_{t-1}^2$$

The shape parameter estimate of approximately 3.457 indicates substantially heavy tails in gold return innovations relative to the normal distribution, supporting the choice of a Student-t error distribution. The persistence estimate ($\alpha_1 + \beta_1 = 0.8235$) implies a shock half-life of approximately 3.5 days ($\ln(0.5) / \ln(0.8235) \approx 3.53$), meaning volatility shocks decay at a moderate pace rather than persisting indefinitely or dying out instantly.

5.3.14 Standardised-Residual Diagnostics

A GARCH model resolves the adequacy failures by explicitly modelling the variance. The standardised residuals — the raw residuals divided by the fitted conditional standard deviation — should then behave like white noise if the model is adequate.

Table 5.45: Diagnostics on the standardised residuals of the ARIMA(2,1,1)–GARCH(1,1) model.

Test	p-value	Conclusion	Comparison with plain ARIMA
Ljung–Box on z (mean)	0.506 (PASS)	No autocorrelation	ARIMA also passed (0.369)
Ljung–Box on z^2 (variance)	1.000 (PASS)	No ARCH remaining	ARIMA failed (0.017)
ARCH-LM(12) on z	1.000 (PASS)	No volatility clustering	ARIMA failed (0.017)

The GARCH extension resolves both diagnostic failures of the plain ARIMA. The Ljung–Box test on the squared standardised residuals — the direct test for remaining

ARCH effects — passes with a p-value of effectively one. The same is true of the ARCH-LM test. The mean equation (Ljung–Box on z) continues to pass as before. All three ARCH-LM lag tests from the full model output also pass, with p-values above 0.79.

The skewness of the standardised residuals (2.11) and their excess kurtosis (57.89) remain very large even after fitting the Student-t GARCH. This means the Student-t with degrees of freedom 3.46 has captured the typical day but has not fully accounted for the most extreme observations — the rare days when the gold price moved by several standard deviations. This is a known limitation of GARCH models on daily gold prices. The Pearson goodness-of-fit test, however, does not reject the distributional assumption at conventional significance levels ($p = 0.13$ for 20 groups, ranging to $p = 0.07$ for 50 groups), so the Student-t fit is deemed adequate.

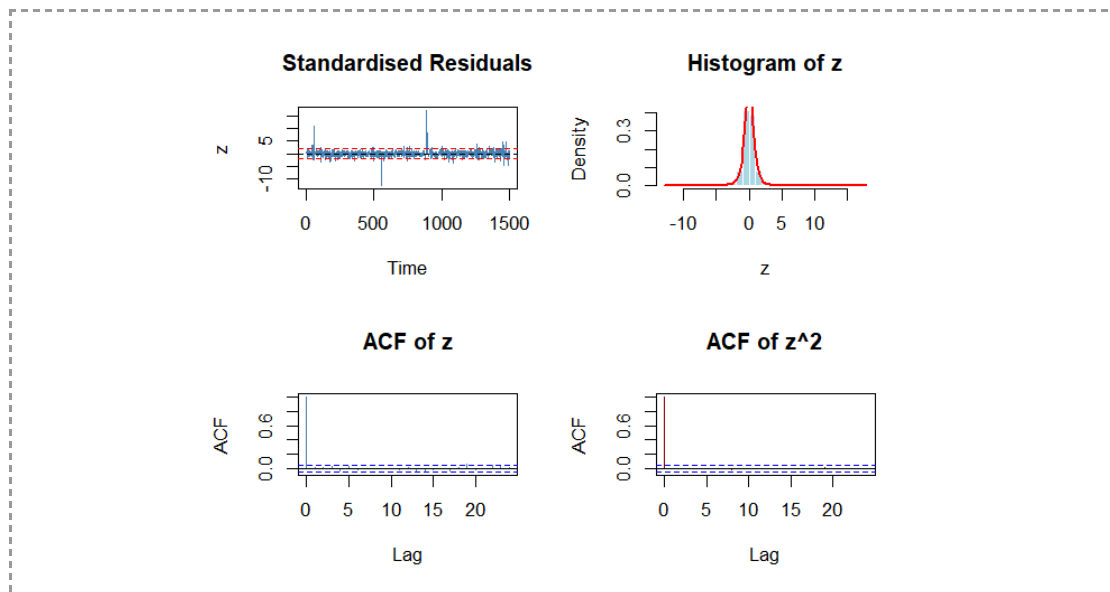


Figure 5.18: Residual plots of ARIMA-GARCH

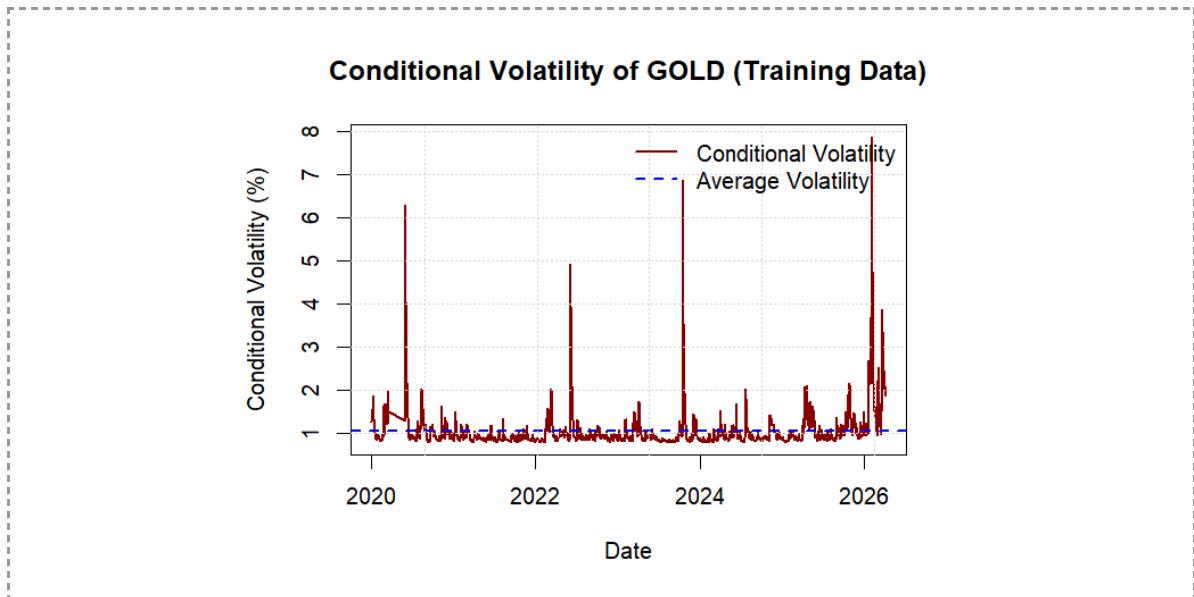


Figure 5.19: Estimated conditional volatility of gold daily returns over the training period.

The conditional volatility series exhibits significant volatility clustering and several pronounced spikes, indicating periods of increased market uncertainty. Volatility remains close to its average level for most of the sample but rises sharply during specific periods. The gradual decline in volatility following these spikes demonstrates mean-reverting behaviour. These findings confirm the presence of time-varying variance in GOLD returns.

5.3.15 Forecast Results

The GARCH model predicted the volatility of the test period. The forecast volatility plot for the 35-day test period is given by

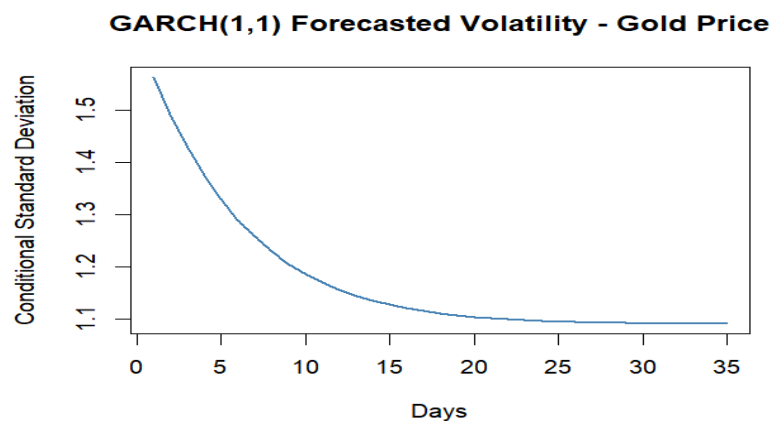


Figure 5.20: Predicted volatility

The ARIMA model was used to produce 35-day-ahead forecasts over the test period, since it has captured the mean structure of the series.

Table 5.46: Out-of-sample forecast accuracy over the 35-day test period.

Model	RMSE (₹)	MAE (₹)	MAPE (%)
ARIMA(2,1,1)	4,854.89	3,954.39	2.545

The ARIMA model yielded a MAPE of 2.545% on the test period, indicating that forecasts deviated from actual gold prices by approximately 2.5% on average — a strong predictive performance. The MAE of ₹3,954.39 and RMSE of ₹4,854.89 per 10 grams reflect the absolute scale of forecast errors, with the higher RMSE suggesting occasional larger deviations during volatile periods, consistent with the heteroscedastic nature of gold price data.

Table 5.47: Forecast versus actual price: all 35 test days.

Date	Actual (₹)	ARIMA (₹)	Error ARIMA
08 Apr 2026	1,51,572	1,46,683	+4,889
09 Apr 2026	1,48,746	1,47,097	+1,649
10 Apr 2026	1,49,577	1,47,157	+2,420
13 Apr 2026	1,49,507	1,47,288	+2,219
15 Apr 2026	1,52,415	1,47,425	+4,990
16 Apr 2026	1,52,572	1,47,552	+5,020
17 Apr 2026	1,50,737	1,47,683	+3,054
20 Apr 2026	1,51,419	1,47,814	+3,605
21 Apr 2026	1,51,545	1,47,944	+3,601
22 Apr 2026	1,51,709	1,48,075	+3,634
23 Apr 2026	1,50,618	1,48,206	+2,412
24 Apr 2026	1,49,997	1,48,337	+1,660
27 Apr 2026	1,50,909	1,48,468	+2,441
28 Apr 2026	1,49,136	1,48,599	+537
29 Apr 2026	1,48,107	1,48,730	-623

Date	Actual (₹)	ARIMA (₹)	Error ARIMA
30 Apr 2026	1,48,465	1,48,862	−397
04 May 2026	1,47,815	1,48,993	−1,178
05 May 2026	1,46,946	1,49,125	−2,179
06 May 2026	1,49,962	1,49,257	+705
07 May 2026	1,50,765	1,49,389	+1,377
08 May 2026	1,50,540	1,49,521	+1,020
11 May 2026	1,49,942	1,49,653	+289
12 May 2026	1,51,175	1,49,785	+1,390
13 May 2026	1,59,723	1,49,917	+9,806
14 May 2026	1,60,708	1,50,050	+10,658
15 May 2026	1,57,676	1,50,182	+7,494
18 May 2026	1,57,289	1,50,315	+6,974
19 May 2026	1,58,257	1,50,448	+7,809
20 May 2026	1,57,307	1,50,581	+6,726
21 May 2026	1,58,254	1,50,714	+7,540
22 May 2026	1,57,727	1,50,847	+6,880
25 May 2026	1,57,891	1,50,980	+6,911
26 May 2026	1,57,152	1,51,114	+6,038
27 May 2026	1,56,460	1,51,247	+5,213
29 May 2026	1,56,444	1,51,381	+5,063

5.3.16 Prediction Interval Evaluation

The GARCH model is not a competing price forecaster — the point forecast comes from the ARIMA mean equation. The GARCH's contribution to the prediction interval is that, by calibrating the interval width to the current level of volatility rather than assuming constant variance, it produces a band that is statistically principled and responsive to market conditions. The interval is constructed from the cumulative forecast variance over the 35-day horizon, using empirical quantiles of the standardised residuals to honour the heavy tails.

The interval quality of the plain ARIMA Gaussian band and the ARIMA–GARCH adaptive band are compared below.

Table 5.48: Prediction interval comparison over the 35-day test period (nominal coverage 95%).

Method	Mean interval width (₹)	Empirical coverage (%)
ARIMA(2,1,1) — Gaussian band	28,179	97.1
ARIMA(2,1,1)–GARCH(1,1) — Student-t adaptive	29,118	97.1

Both methods achieve identical empirical coverage of 97.1%, which slightly exceeds the nominal 95% — both are conservative in the right direction. The GARCH interval is marginally wider on average (29,118 vs 28,179), which in this case reflects the fat tails built into the Student-t quantiles. The only observation that falls outside either interval is the first test day (8 April 2026), where the actual price of ₹1,51,572 sits just above the ARIMA upper bound of ₹1,50,307 and the GARCH upper bound of ₹1,51,535. This is a single boundary case over 35 days.

The near-identical coverage of the two methods on this test window means the GARCH does not provide a visible empirical advantage in coverage here. Its value lies instead in the statistical justification: the Gaussian interval is technically invalid once the ARCH-LM and Jarque–Bera tests fail, because its derivation assumes constant variance and normal innovations. The GARCH interval makes no such assumptions. In a longer or more volatile test period, or when formal interval validity is required rather than just empirical coverage, the GARCH band is the appropriate choice.

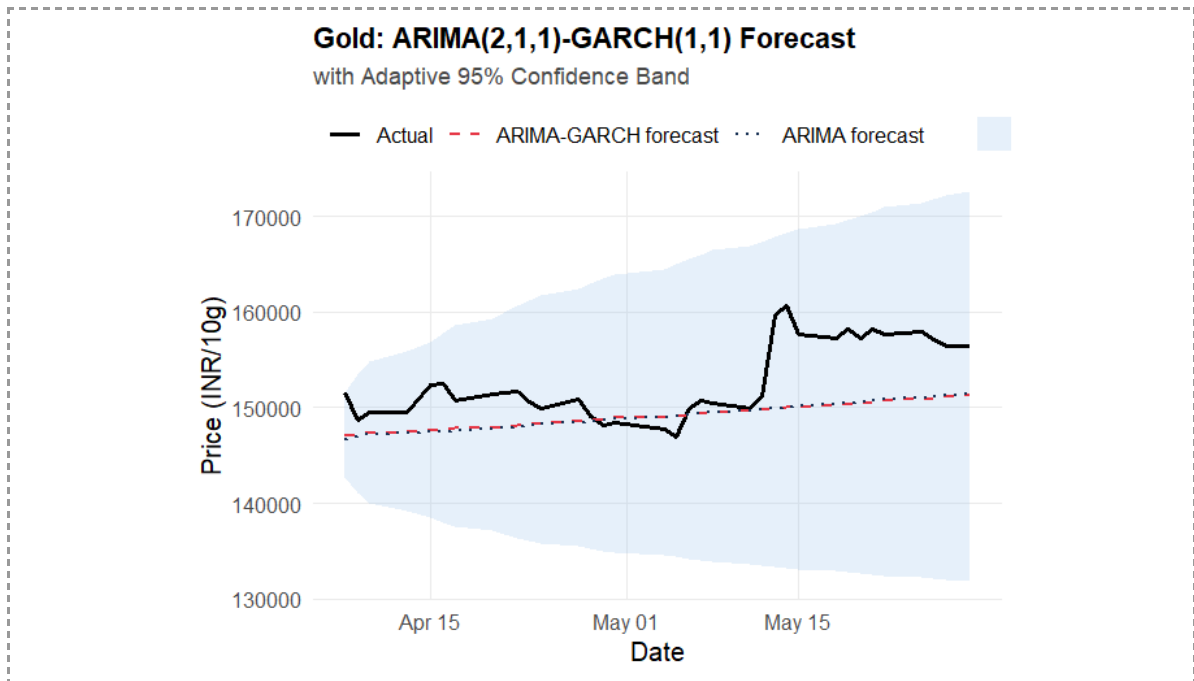


Figure 5.21: Actual gold price against the ARIMA–GARCH point forecast and adaptive 95% prediction band over the test period.

5.3.17 Summary

The daily gold price series is non-stationary in its level and becomes stationary after one difference ($d = 1$), with no seasonal structure to speak of. ARIMA(2,1,1), selected by both the AICc grid and auto.arima on the log-price training data, passes the central Ljung–Box adequacy test: its mean equation is correctly specified. On the out-of-sample test period, however, the simpler ETS(A, A, N) Holt model produces more accurate point forecasts (MAPE 2.0% vs 2.5%), despite failing the autocorrelation adequacy test — a finding that is reported honestly and without adjustment.

The ARIMA residuals exhibit strong ARCH effects and extreme kurtosis (59.4), which motivate and fully justify the GARCH extension. The ARIMA(2,1,1)–GARCH(1,1) model with Student-t errors — trained on returns from the training set only, to prevent leakage into the interval evaluation — resolves both diagnostic failures: the standardised residuals pass all three autocorrelation and volatility tests. The estimated persistence of 0.82 indicates moderate volatility memory (shocks fade over roughly a week), and the Student-t shape parameter of 3.46 confirms the extremely fat-tailed nature of daily gold returns, though some residual tail behaviour remains uncaptured even after the GARCH fit. Both the Gaussian and GARCH prediction intervals achieve

97.1% empirical coverage over the 35-day test period; the GARCH interval is preferred on formal grounds because its derivation does not require the assumptions that the diagnostic tests have already shown to be violated.

5.4 Multivariate Analysis: Interdependence among Crude Oil, Gold and the Exchange Rate

The analysis proceeds through five stages: data preparation and transformation, stationarity testing, cointegration, the Vector Error Correction Model, and a set of tools for tracing short-run dynamics and before closing with a long-run regression that translates the relationships into directly interpretable elasticities. Throughout, the series are converted to dollar-denominated gold prices and worked within logarithms, which stabilises the variance and lets regression slopes be read as percentages.

5.4.1 Data Preparation

Monthly data for crude oil, gold and the exchange rate were drawn from Indian financial-year format records and reshaped into a standard calendar time series. The three series were merged on their common dates, yielding a balanced panel of 314 monthly observations from April 2000 to May 2026.

Table 5.49: Descriptive statistics of the three series used in the multivariate analysis.

Variable	Mean	Median	SD	Min	Max	Skewness	Kurtosis
Crude oil (USD/bbl)	66.32	65.44	27.80	18.24	132.47	0.17	2.13
Gold (USD/10g)	446.47	440.80	295.65	91.10	1,691.32	1.46	6.41
USD/INR	59.78	56.31	15.04	39.20	95.01	0.47	1.92

Crude oil is approximately symmetrically distributed over the sample (skewness 0.17, kurtosis 2.13, close to the values of 0 and 3 expected from a normal distribution). Gold in dollar terms is positively skewed (1.46) with fat tails (kurtosis 6.41), reflecting the prolonged rally to record highs that compressed the lower half of the distribution. The exchange rate is mildly skewed to the right and rather flat-tailed, consistent with a gradual, largely one-directional depreciation of the rupee over twenty-five years. All three series are modelled in natural logarithms in what follows; the log transformation pulls in the right tail, reduces heteroscedasticity and, as noted, makes regression slopes interpretable as elasticities.

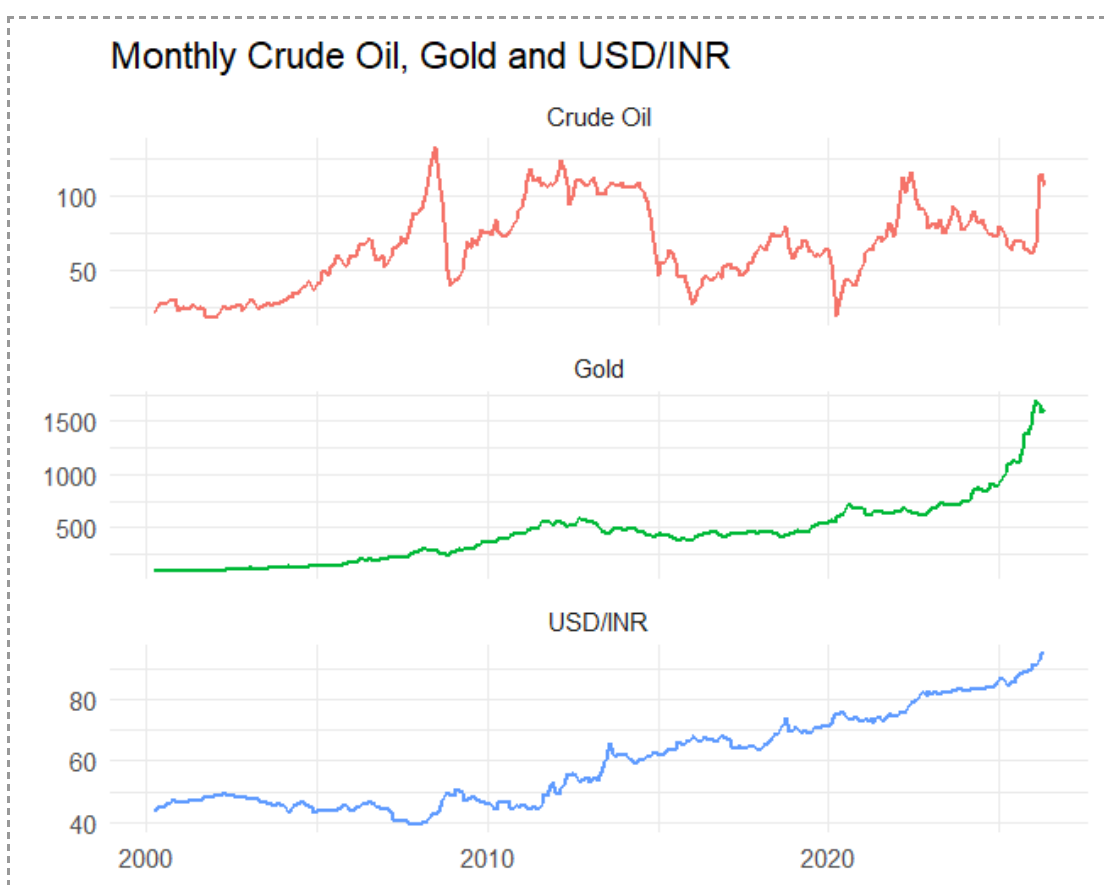


Figure 5.22: Monthly crude oil price, gold price (USD/10g) and USD/INR exchange rate, April 2000 to May 2026 (one panel per series).

5.4.2 Pairwise Correlation among the Log Levels

Table 5.50: Pearson correlations among the log-transformed series.

	Log crude	Log gold	Log USD/INR
Log crude	1.000	0.694	0.262
Log gold	0.694	1.000	0.796
Log USD/INR	0.262	0.796	1.000

The correlation table reveals two quite different pairings. Log gold and log USD/INR are strongly correlated (0.796): as the rupee has weakened against the dollar, the dollar price of gold has risen in tandem, reflecting both the global demand for gold as a dollar-priced asset and the tendency of investors to buy gold during currency stress. The correlation between log crude and log gold is moderate (0.694), consistent with both being dollar-denominated commodity prices that broadly reflect global economic activity and inflation expectations. The link between log crude and log USD/INR is

considerably weaker (0.262), suggesting that crude oil shocks and rupee movements are only loosely connected at the monthly frequency over this sample.

Prior to interpreting these results, it is essential to recognize two important caveats associated with correlation analysis. First, correlation measures linear co-movement but says nothing about direction or causation; the question of which series leads and which follows is answered by the VECM and the Granger tests below. Second, all three series carry strong trends, and correlations among trending series can be inflated relative to what the short-run relationships would suggest. The unit-root and cointegration tests address this directly.

5.4.3 Unit Root Tests

Regressing one non-stationary, trending series on another produces a spurious result — apparently strong statistical relationships that arise from shared trends rather than genuine economic links. Before any multivariate modelling, it must therefore be confirmed that each series is integrated of the same order. The ADF and Phillips–Perron tests were applied to each log series in levels and to its first difference.

Table 5.51: Unit root tests: ADF and Phillips–Perron on log levels and first differences.

Series	ADF (level)	PP (level)	ADF (1st diff)	Conclusion
Log crude oil	0.474	0.386	0.010	I(1)
Log gold	0.862	0.909	0.010	I(1)
Log USD/INR	0.329	0.680	0.010	I(1)

All three series are non-stationary in their log levels on both the ADF and Phillips–Perron tests: none comes close to the 5% significance threshold (p-values range from 0.33 to 0.91). After taking a single first difference, all three are firmly stationary (ADF $p \leq 0.01$ in each case). The ADF p-values at 0.01 represent the software’s printed floor rather than exact values; the true p-values are at or below this limit. All three series are therefore integrated of order one, I(1), which is the necessary condition for proceeding to a cointegration test.

5.4.4 Johansen Cointegration Test

Cointegration is the formal statement that, although each of the three I(1) series wanders individually, a particular linear combination of them remains stationary over time — they are bound together in a stable long-run equilibrium. Johansen’s sequential testing procedure was applied using a VAR with two lags (selected by AIC, HQ and FPE, with SC selecting one lag; two lags were retained as the more conservative choice and as the minimum required by Johansen’s procedure).

Table 5.52: Johansen cointegration test results (lag $K = 2$, constant restricted to the cointegrating space).

H_0 (rank $\leq r$)	Trace statistic	Trace CV	5%	Max-eigenvalue statistic	Max-eigen 5% CV	Decision
$r \leq 0$	60.07	34.91		35.20	22.00	Reject
$r \leq 1$	24.87	19.96		20.89	15.67	Reject
$r \leq 2$	3.98	9.24		3.98	9.24	Cannot reject

5.4.5 Cointegration Rank Selection

Both the trace test and the maximum-eigenvalue test reject the null of no cointegration ($r = 0$) and the null of at most one cointegrating vector ($r \leq 1$), but neither reject at $r \leq 2$. Taken literally, this suggests two cointegrating relationships. However, two cointegrating vectors among three I(1) series is a strong result — it implies that only one common stochastic trend drives the system — and can be difficult to interpret economically in a meaningful way.

Table 5.53: Loading coefficients (α) under $r = 2$

Equation	ect1 (α , p-value)	ect2 (α , p-value)
Δcrude	-0.1118 (p < 0.001) Significant	+0.1032 (p < 0.001) Significant
Δgold	-0.0115 (p = 0.172) Not significant	+0.0151 (p = 0.050) Borderline

$\Delta \ln fx$	+0.0026 ($p = 0.556$) Not significant	+0.0006 ($p = 0.877$) Not significant
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Examination of the loading coefficients under $r = 2$ shows that the adjustment parameters associated with the second cointegrating vector (ect2) were statistically significant only in the crude oil equation ($\alpha = 0.103$, $p < 0.001$) and marginal in the gold equation ($\alpha = 0.015$, $p = 0.050$), with no significant adjustment in the FX equation ($p = 0.877$). The first cointegrating vector (ect1) shows a parallel pattern, with significant adjustment only in the crude equation. This concentration of error-correction activity in a single equation, rather than distinct adjustment behaviour attributable to two separate long-run relationships, indicates that the second cointegrating vector contributes limited independent economic information.

The analysis therefore proceeds with $r = 1$, consistent with the convention in the applied literature (Agrawal and Kumar, 2025; Kanungo and Dang, 2021) of selecting the most parsimonious cointegrating rank that is both statistically supported and economically interpretable.

5.4.6 Vector Error Correction Model: The Long-Run Relationship

With one cointegrating vector, the VECM produces a single long-run equilibrium equation, normalised on log crude oil. The estimated cointegrating vector (β) is:

Table 5.54: Normalised cointegrating vector (long-run relationship, β), normalised on log crude oil.

Term	Coefficient	Interpretation
Log crude oil	1.000	Normalisation (dependent variable)
Log gold	-1.101	Crude and gold move positively together in the long run
Log USD/INR	+2.005	Crude rises with a weaker rupee in the long run
Constant	-6.011	Equilibrium intercept

The long-run equilibrium is read as: $\ln(\text{crude}) - 1.101 \cdot \ln(\text{gold}) + 2.005 \cdot \ln(\text{fx}) = 6.011$. The coefficient on log gold (-1.101) implies that, in the long run, crude oil and gold move proportionally in the same direction: a 1% rise in the dollar gold price is associated with approximately a 1.1% rise in the crude oil price. This is consistent with

both being globally traded commodities whose prices reflect common drivers and the strength of the dollar, inflation expectations, and global economic demand. The coefficient on log USD/INR (+2.005) implies that a weaker rupee (higher lfx) is associated with a higher crude oil price in equilibrium. Since both crude oil and gold are priced in dollars, a rupee depreciation naturally raises the rupee-equivalent prices of both; the positive sign reflects this mechanical as well as economic link. The equilibrium relationship is estimated over twenty-five years and should be understood as a structural tendency, not a month-by-month accounting identity.

5.4.7 Speed of Adjustment

The VECM decomposes the monthly change in each variable into a pull back toward the long-run equilibrium (the error-correction term, scaled by α) and a set of short-run effects from recent changes in all three variables. A negative α indicates that the variable adjusts back toward equilibrium when it has drifted away; the larger the absolute value, the faster the correction.

Table 5.55: Error-correction term (α) for each variable: speed and significance of adjustment.

Variable	α estimate	Std. error	t-statistic	p-value	Monthly correctio n	Half-life
Log crude oil	-0.065	0.016	-4.21	< 0.001	6.5%	≈10 months
Log gold	-0.017	0.006	-2.89	0.004	1.7%	≈40 months
Log USD/INR	-0.006	0.003	-1.75	0.081	0.6%	~125 months

The results reveal a clear hierarchy of adjustment. Crude oil is the fastest responder, correcting about 6.5% of any disequilibrium each month; at this rate, it takes roughly ten months for half of a deviation from equilibrium to be absorbed. Gold adjusts more slowly, correcting about 1.7% per month, with a half-life of approximately 40 months. The exchange rate's error-correction coefficient (-0.006) is the smallest and critically is not statistically significant at the conventional 5% level ($p = 0.081$). The exchange rate therefore does not adjust back to the cointegrating relationship in any statistically

reliable sense. In the language of cointegration, the USD/INR rate is weakly exogenous with respect to the long-run relationship: it is the driving force that the other two variables respond to, rather than a variable that itself reverts to a long-run target determined by the system.

This finding carries a practical economic interpretation. The exchange rate moves on factors such as macroeconomic policy, capital flows, inflation differentials, and global risk appetite, that are largely external to the crude-gold-FX system as modelled here. Changes in the rupee then propagate to both commodity prices, with crude oil adjusting quickly and gold adjusting only gradually. Neither commodity has a meaningful ability to push the rupee in a direction determined by the long-run equilibrium.

5.4.8 Short-Run Dynamics

Alongside the error-correction mechanism, the VECM estimates the short-run effect of one variable's lagged changes on the others. Significant short-run coefficients indicate that the lagged change in one series carries information about the current change in another, over and above the pull toward equilibrium.

Table 5.56: Selected short-run VECM coefficients: lagged first differences (one-month lag).

Equation	Regressor	Estimate	t-statistic	p-value	Significant?
$\Delta \log \text{ crude}$	Lagged $\Delta \log \text{ crude}$	+0.198	3.65	0.000	Yes
$\Delta \log \text{ crude}$	Lagged $\Delta \log \text{ USD/INR}$	-0.831	-2.79	0.006	Yes
$\Delta \log \text{ gold}$	Lagged $\Delta \log \text{ gold}$	+0.202	3.37	0.001	Yes
$\Delta \log \text{ USD/INR}$	Lagged $\Delta \log \text{ USD/INR}$	+0.063	1.04	0.300	No

The most economically interesting short-run result is that the lagged change in the exchange rate is a significant negative predictor of the change in crude oil (-0.831 , $p = 0.006$). A one-percentage-point depreciation of the rupee last month is associated with a 0.83 percentage-point decline in crude oil this month. This is a counterintuitive result at first glance — one might expect a weaker rupee to raise the rupee cost of oil imports

and thus the recorded crude price — but it is consistent with a broader pattern in which rupee weakness reflects global risk-off episodes that also depress commodity demand and therefore crude prices. The lagged own-price coefficients for crude (0.198) and gold (0.202) are both significant and positive, indicating modest positive momentum in both series from one month to the next. The exchange rate shows no significant own-lag effect, again consistent with near-random-walk behaviour in the short run.

5.4.9 Model Stability

A cointegrated system with three variables and one cointegrating vector should have exactly two unit roots in its companion matrix (representing the two common stochastic trends) and all remaining roots strictly inside the unit circle. The computed eigenvalue moduli are 1.000, 1.000, 0.929, 0.272, 0.161 and 0.161. The two unit roots are present as expected, and the remaining four roots are all well below one (the largest is 0.929). The model is dynamically stable and does not exhibit explosive behaviour.

5.4.10 Residual Diagnostics

Table 5.57: Multivariate residual diagnostics for the VECM.

Test	p-value	Conclusion
Portmanteau (autocorrelation, 16 lags)	0.126	No residual autocorrelation
Multivariate Jarque–Bera (normality)	< 0.001	Non-normal residuals
Multivariate ARCH (heteroscedasticity)	< 0.001	Time-varying variance

The model passes the most important test — no residual autocorrelation ($p = 0.126$) — which means the VECM has captured the short-run dynamics adequately and the estimated coefficients are not distorted by serial dependence. The Jarque–Bera test rejects normality and the ARCH test detects time-varying variance. Neither of these failures undermines the coefficient estimates: with 314 observations, OLS-based VECM estimates of the adjustment and short-run coefficients are consistent and asymptotically normal regardless of whether the innovations are Gaussian, provided the asymptotic covariance matrix is correctly specified — which is exactly what motivates the robust correction applied below. Residual normality is therefore treated as a caveat on standard

inference procedures rather than a flaw in the point estimates, since it is not a requirement for the consistency or asymptotic normality of the VECM coefficient estimates, nor for the validity of the heteroscedasticity-robust standard errors applied below; both hold under considerably weaker distributional assumptions. The more material caveat in this respect concerns the cointegrating rank test itself: its asymptotic critical values assume conditionally homoscedastic errors, and conditional heteroscedasticity can distort its size properties (Cavaliere, Rahbek and Taylor, 2010). This is acknowledged as a further limitation of the analysis, separate from the coefficient-level correction applied below.

The ARCH result is expected for monthly commodity and currency data. Because conventional OLS standard errors assume homoscedastic innovations, this means the standard errors — and therefore the significance tests — reported for the VECM's α and short-run coefficients should be interpreted with some caution; robust (heteroscedasticity-consistent) standard errors are a natural check, applied below. The point estimates themselves remain consistent, but their estimated standard errors may understate the true uncertainty. This also implies that interval-based statements, such as impulse response confidence bands, should be read with some caution rather than at face value. Notably, comparable VECM studies on related asset classes (e.g. Agrawal and Kumar, 2024, who model the Nifty 50 with gold, oil, and exchange rate) typically restrict residual diagnostics to serial correlation testing and do not report ARCH-LM or normality checks on VECM residuals.

5.4.11 Heteroscedasticity-Robust Standard Errors

Given the ARCH effects detected above, heteroscedasticity-consistent (HC1) standard errors were computed for each VECM equation as a robustness check. The coefficient estimates are unchanged from the original OLS-based VECM; only the standard errors, t-statistics, and p-values are recalculated.

Table 5.58: Robust standard errors computed using the HC1 heteroscedasticity-consistent covariance estimator, applied separately to each VECM equation.

Equation: Δ lcrude (response: lcrude.d)

Term	Estimate	SE (original)	SE (robust, HC1)	p (original)	p (robust)
ect1	−0.0651	0.0155	0.0225	0.0000	0.0041
lcrude.d11	0.1982	0.0543	0.1331	0.0003	0.1374
lgold.d11	0.0728	0.1538	0.1470	0.6362	0.6207
lfx.d11	−0.8308	0.2977	0.2851	0.0056	0.0038

Equation: Δ lgold (response: lgold.d)

Term	Estimate	SE (original)	SE (robust, HC1)	p (original)	p (robust)
ect1	−0.0174	0.0060	0.0063	0.0042	0.0063
lcrude.d11	−0.0274	0.0212	0.0225	0.1961	0.2240
lgold.d11	0.2024	0.0600	0.0729	0.0008	0.0059
lfx.d11	0.2120	0.1161	0.1306	0.0689	0.1056

Equation: Δ lfx (response: lfx.d)

Term	Estimate	SE (original)	SE (robust, HC1)	p (original)	p (robust)
ect1	−0.0055	0.0032	0.0035	0.086	0.1179
lcrude.d11	−0.0020	0.0111	0.0084	0.8550	0.8102
lgold.d11	−0.0372	0.0314	0.0428	0.2371	0.3853
lfx.d11	0.0632	0.0608	0.0841	0.2998	0.4533

The error-correction coefficients (ect1) retain statistical significance in the crude and gold equations under robust standard errors ($p = 0.004$ and $p = 0.006$, respectively), confirming that the long-run adjustment mechanism is not an artefact of heteroscedasticity-induced understatement of the standard errors. Among the short-run

coefficients, crude oil's own first lag, which was significant at the 5% level under conventional standard errors ($p < 0.001$), loses that significance under the robust specification ($p = 0.137$) — a genuine reversal attributable to the heteroscedasticity correction. Two further coefficients that were only marginally significant at the 10% level under conventional standard errors — FX's first lag in the gold equation ($p = 0.069$) and the error-correction term in the FX equation ($p = 0.081$) — fall further from significance under the robust correction ($p = 0.106$ and $p = 0.118$, respectively), reinforcing rather than reversing their original borderline status.

This indicates that while the central error-correction relationships — the core long-run finding of the model — are robust to the detected heteroscedasticity, some short-run and borderline effects reported under conventional standard errors should be treated with caution. A full multivariate volatility model (e.g. VECM-GARCH) is left as a direction for future work.

5.4.12 Granger Causality

The VECM captures long-run adjustment and short-run dynamics jointly, but the question of whether the past of one variable helps forecast another in the short run is tested separately through Granger causality on the first-differenced stationary series. These tests use one lag ($K = 2$ minus one), consistent with the VAR lag selection.

Table 5.59: Granger causality tests among log first differences (one lag).

Direction	F-statistic	p-value	Conclusion
FX → Crude	7.564	0.006	Significant
Gold → Crude	5.242	0.023	Significant
Crude → FX	0.010	0.921	Not significant
Gold → FX	1.990	0.159	Not significant
FX → Gold	2.862	0.092	Not significant (marginal)
Crude → Gold	0.782	0.377	Not significant

The pattern of short-run Granger causality is clear and consistent with the adjustment-speed findings. In the short run, the exchange rate Granger-causes crude oil ($p = 0.006$) and gold Granger-causes crude oil ($p = 0.023$). The reverse directions —

crude oil as a cause of either the exchange rate or gold — are not significant. The exchange rate does not Granger-cause gold at the 5% level, though the result is marginal ($p = 0.092$). Gold does not Granger-cause the exchange rate.

Taken together with the error-correction results, the picture is coherent. In the long run, the exchange rate is weakly exogenous and drives the system. In the short run, both the exchange rate and gold contain information about the next month's crude oil price, but crude oil does not feed back into either. Gold is largely self-contained in the short run, adjusting slowly to the long-run equilibrium and not significantly predicting the exchange rate. The exchange rate, consistent across both frameworks, is the variable that the system responds to rather than one that responds to the others.

5.4.13 Impulse Response Functions

Impulse response functions (IRFs) trace the dynamic path of one variable following a one-time shock to another, holding all else fixed. The variable ordering — crude oil, gold, USD/INR — reflects the assumption that global commodity prices respond to one another before feeding through to the domestic exchange rate; this Cholesky ordering is consistent with the Granger causality findings and should be stated when presenting the results. Bootstrapped 95% confidence bands from 300 replications are shown; an effect is statistically meaningful where the band does not cross zero.

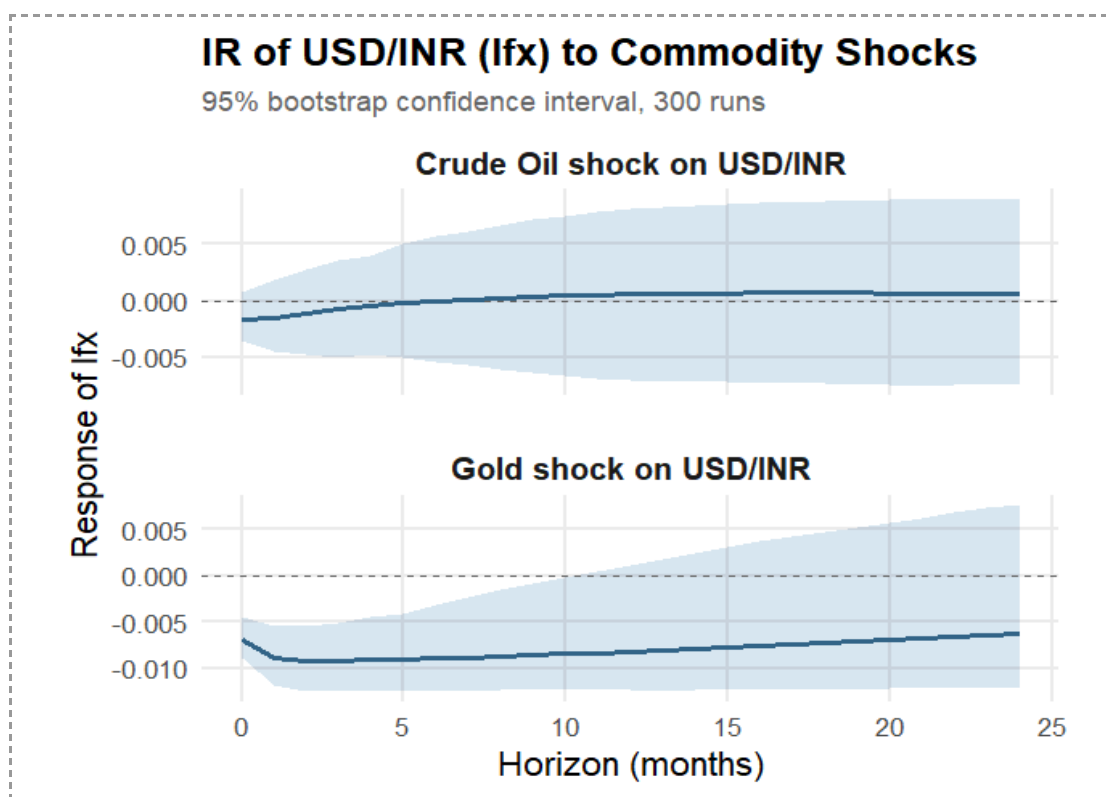


Figure 5.23: Impulse response of the USD/INR exchange rate to a one-standard-deviation shock in crude oil and gold (24-month horizon, bootstrapped 95% bands).

The impulse response functions trace the dynamic effect of a one-unit shock to crude oil and gold prices on the exchange rate (lfx) over a 24-month horizon, with 95% bootstrap confidence intervals constructed from 300 replications.

The response of lfx to a crude oil shock is not statistically significant at any horizon. The point estimate begins marginally negative, crosses zero by approximately month eight, and stabilises near zero for the remainder of the horizon; throughout this period, the confidence band contains zero, indicating that crude oil price movements do not produce a discernible effect on the exchange rate within this system.

By contrast, the response of lfx to a gold price shock is negative, immediate, and persistent. Following the shock, lfx falls sharply within the first month and remains below zero throughout the 24-month horizon, with the confidence interval excluding zero for the majority of the period; only towards the end of the horizon does the upper bound approach zero, indicating a gradual weakening of statistical significance over time. This pattern indicates that a positive shock to gold prices is associated with a

sustained appreciation of the rupee against the US dollar, consistent with gold's conventional role as a safe-haven asset that tends to move inversely with dollar strength.

5.4.14 Forecast Error Variance Decomposition

The forecast error variance decomposition (FEVD) answers the question: over a given horizon, what fraction of the uncertainty in forecasting one variable is attributable to shocks originating in each of the three variables? Rows sum to 100%.

Table 5.60: Forecast error variance decomposition at 1, 12 and 24 months (percentages).

Variable explained	Shock from	1 month (%)	12 months (%)	24 months (%)
Crude oil	Crude oil	100.0	83.1	59.1
Crude oil	Gold	0.0	10.7	30.2
Crude oil	USD/INR	0.0	6.2	10.7
Gold	Gold	99.8	92.0	83.9
Gold	Crude oil	0.2	7.4	15.8
Gold	USD/INR	0.0	0.6	0.3
USD/INR	USD/INR	86.2	80.1	77.2
USD/INR	Gold	12.6	14.1	11.4
USD/INR	Crude oil	1.2	5.7	11.3

Several findings stand out from the FEVD. At a one-month horizon, each variable is almost entirely explained by its own past shocks, which is unsurprising: at short horizons the immediate persistence of each series dominates. As the horizon lengthens to 12 and 24 months, cross-variable influences become increasingly visible.

For crude oil, the contribution of gold shocks rises substantially over time — from zero at one month to 10.7% at twelve and 30.2% at twenty-four months. By the two-year horizon, nearly a third of crude oil's forecast uncertainty is traceable to gold price shocks, making it the dominant external source of variability in crude. The exchange rate contributes a further 10.7% at 24 months. The implication is that while crude oil behaves autonomously in the short run, its trajectory over longer horizons is meaningfully influenced by both gold and the exchange rate.

Gold is substantially more self-contained: even at 24 months, 83.9% of its forecast variance reflects its own shocks, with crude oil contributing 15.8% and the exchange rate barely more than zero (0.3%). The negligible exchange-rate contribution to gold's variance is noteworthy given the strong gold-FX correlation in levels: it suggests that the two series share a common trend but that the contemporaneous co-movement does not translate into the exchange rate driving gold's forecast error at any horizon.

The exchange rate's variance is predominantly self-driven (77.2% at 24 months), but gold shocks consistently account for 11–14% of its forecast uncertainty across all three horizons, confirming a durable gold-to-FX channel. Crude oil's share of exchange-rate variance builds from 1.2% to 11.3% over 24 months, suggesting a slow but accumulating influence.

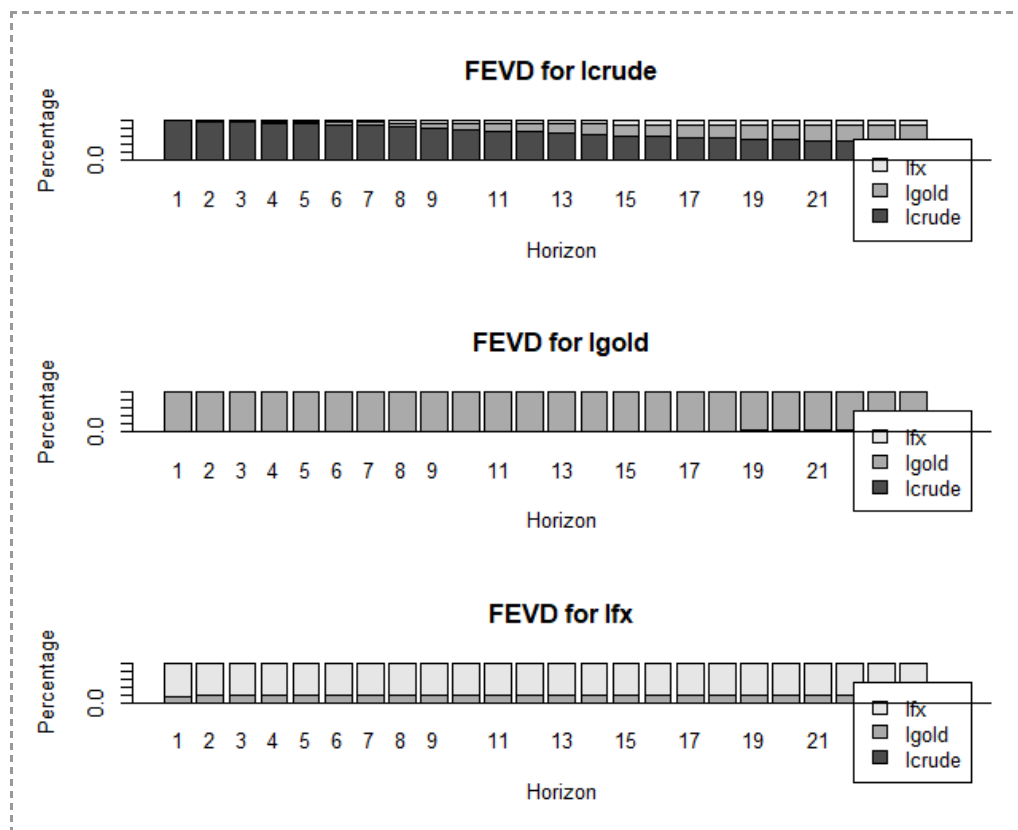


Figure 5.24: Forecast error variance decomposition for all three variables at 1, 12 and 24 months.

5.4.15 Long-Run OLS Regression: A Descriptive Check

Rationale and Caveat

All three series — the log exchange rate (lfx), log crude oil price (lcrude), and log gold price (lgold) — were found to be non-stationary in levels and stationary only after first differencing, confirming that each is integrated of order one, $I(1)$. Regressing one $I(1)$ series directly on another in levels carries a well-known risk: the Granger-Newbold spurious regression problem, whereby two independently trending series can appear strongly related — producing a high R^2 and highly significant t-statistics — even when no genuine relationship exists between them.

The Johansen procedure established that a genuine long-run equilibrium relationship exists among the three series at the system level (cointegrating rank $r = 1$), which the vector error correction model (VECM) estimates directly. This system-level finding addresses the spurious regression concern for the system as a whole, but it does not automatically guarantee that any specific static regression equation built from these variables is itself a valid representation of that relationship. Whether a given OLS specification independently captures a genuine long-run relationship is a separate, testable question, addressed directly below.

Static OLS Regression

As a simple, directly interpretable descriptive summary, a static OLS regression of log USD/INR on log crude oil and log gold was estimated using the full sample ($N = 314$ monthly observations, May 2000 to May 2026).

Table 5.61: Long-run OLS regression: log USD/INR on log crude oil and log gold.

Regressor	Coefficient	Std. error	t-statistic	p-value	Interpretation
Intercept	2.851	0.058	49.49	< 0.001	Scaling constant
Log crude oil	-0.288	0.018	-15.80	< 0.001	-0.29% per 1% rise in crude
Log gold	+0.407	0.012	+33.42	< 0.001	+0.41% per 1% rise in gold

Table 5.62: Model fit statistics and multicollinearity check.

Statistic	Value
R ²	0.797
Adjusted R ²	0.796
F-statistic	610.8
VIF (log crude)	1.93
VIF (log gold)	1.93

The regression explains approximately 80% of the variation in the log exchange rate ($R^2 = 0.797$), and both regressors are individually and jointly significant at conventional levels ($F(2, 311) = 610.8$, $p < 0.001$). The variance inflation factors of 1.93 for both variables are well below the conventional threshold of 10, indicating that multicollinearity between log crude and log gold does not materially distort the coefficient estimates.

5.4.16 Residual-Based Cointegration Check

Because this is a static regression on $I(1)$ levels, the validity of treating its coefficients as genuine long-run elasticities — rather than a spurious fit — depends on whether its own residuals are stationary, per the Engle-Granger (1987) two-step framework. This was tested directly using two independent residual-based cointegration tests.

Table 5.63: Residual-based cointegration tests on the OLS regression

Test	Test statistic	Lag	p-value	Decision
Augmented Dickey-Fuller	-2.91	Lag 6	$p = 0.193$	Fail to reject unit root
Phillips-Ouliaris (demeaned)	-20.66	Truncation 3	$p = 0.134$	Fail to reject unit root

H_0 : residuals contain a unit root (non-stationary). Phillips-Ouliaris uses critical values designed specifically for residual-based cointegration testing and is the more appropriate test of the two for this purpose.

Neither test rejects the null hypothesis of a unit root in the regression residuals. This indicates that this particular static specification does not independently satisfy the

Engle-Granger cointegration criterion, despite cointegration having been confirmed for the system as a whole via the Johansen procedure.

This divergence between the system-level Johansen result and the equation-specific Engle-Granger/Phillips-Ouliaris result is a recognised feature of the two methodologies rather than a contradiction or an error. The Engle-Granger two-step approach has lower power in settings with three or more variables, can only represent a single cointegrating relationship by construction, and imposes a single, arbitrary normalisation (here, lfx as the dependent variable with unit coefficient). The Johansen framework, by contrast, estimates the entire system jointly and is capable of identifying multiple cointegrating vectors while accounting for the dynamic interactions among all endogenous variables.

Consequently, the failure of the residual-based tests to reject a unit root should not be interpreted as evidence against the existence of long-run equilibrium relationships among the variables. Instead, it indicates that the particular static OLS equation estimated here does not, by itself, provide statistically sufficient evidence of cointegration under the Engle-Granger criterion.

The estimated coefficients from the OLS regression may therefore be viewed as descriptive long-run associations rather than formally validated long-run elasticities. Their signs and magnitudes remain broadly consistent with the cointegration results and with economic intuition, but caution is warranted when assigning a strict structural interpretation to these estimates.

5.4.17 Summary

The multivariate analysis covers April 2000 to May 2026 and centres on the long-run relationship among monthly crude oil, dollar-denominated gold and the USD/INR exchange rate. All three series are $I(1)$, and the Johansen procedure confirms at least one, and formally two, cointegrating vectors; the analysis proceeds with one for interpretability. The single long-run equilibrium indicates that crude oil, gold and the exchange rate are bound together: a 1% rise in dollar gold is associated with approximately a 1.1% rise in crude in the long run, while a weaker rupee is also associated with higher crude oil prices in equilibrium.

The error-correction analysis reveals a clear hierarchy. Crude oil adjusts most rapidly (6.5% per month, half-life ~10 months); gold adjusts slowly (1.7% per month, half-life ~40 months); and the exchange rate does not adjust significantly, making it weakly exogenous — the driver of the system rather than a variable that responds to the long-run equilibrium. Short-run Granger causality confirms this asymmetry: the exchange rate and gold Granger-cause crude oil, but crude oil does not cause either in the short run.

The FEVD shows that over a two-year horizon, gold shocks account for 30% of crude oil's forecast uncertainty, while the exchange rate consistently explains 11–14% of gold's forecast variance. The long-run OLS regression summarises the equilibrium elasticities: a 1% rise in gold is associated with a 0.41% rupee depreciation; the conditional elasticity of crude oil is -0.29% , a sign that reflects the conditional nature of the estimate rather than a direct causal claim. The residuals pass the autocorrelation test ($p = 0.126$) but show non-normality and heteroscedasticity, which are standard features of monthly commodity data and do not undermine the asymptotic validity of the VECM estimates at this sample size.

5.5 USD/INR Exchange Rate: Classical versus Machine Learning Forecasting

The classical analysis established that the USD/INR exchange rate behaves as a near-random walk: ARIMA(0,1,0) with drift, selected by both the automatic search and the full model grid, produced forecasts that were statistically indistinguishable from simply carrying the last observed rate forward. This section tests whether machine learning models — Random Forest, XGBoost and a Long Short-Term Memory network — can improve on that baseline by detecting patterns in past daily rate changes that the classical models cannot. The comparison is designed to give the machine learning models every legitimate advantage and the results every honest reading.

5.5.1 Experimental Design

Three design decisions were taken to ensure the comparison is methodologically sound and cannot be dismissed as artefactual.

Walk-forward evaluation. Each test observation is predicted one step ahead using only data that would have been available at that point in real time: the model is fitted (or re-fitted) on all data up to day t and then used to predict day $t+1$. This is equivalent to time-series cross-validation as described by Hyndman and Athanasopoulos (2021, Chapter 5) and ensures that no future information contaminates the forecast. The tree models are re-fitted every 20 days and the LSTM every 40 days to keep computation tractable while preserving the walk-forward principle.

Identical data and split. The Python ML code reads the same cleaned series exported from R and uses the same training cut-off (1,339 days training, 335 days test) as the classical models. The total dataset spans 1,674 daily observations. Matching the split exactly is necessary for the RMSE, MAE and MAPE figures to be directly comparable across all models.

Diebold–Mariano significance testing. Numerical differences in error metrics do not, on their own, establish that one model forecasts significantly better than another. The Diebold–Mariano test (Diebold and Mariano, 1995) compares the squared loss

differentials and tests formally whether the difference in predictive accuracy is statistically meaningful. Results are reported for the best model against every rival.

Table 5.64: Model configurations used in the machine learning comparison.

Model	Configuration	Notes
Random Forest	300 trees, no depth limit, 20 lags	Conservative ensemble; no depth limit avoids underfitting
XGBoost	300 trees, lr = 0.05, max depth = 3, 20 lags	Shallow depth prevents overfitting on daily FX changes
LSTM	1 layer $\times$ 16 units, 30 epochs, 20-lag window	Small by design; larger networks overfit on $< 2,000$ obs
Naive (random walk)	$\text{Rate}_t = \text{Rate}_{t-1}$	Benchmark: last observed value carried forward

All three machine learning models take as input the 20 most recent daily log-changes in the rate (approximately one trading month of history), which is consistent with the short-memory behaviour established in the classical analysis. The input is the change series rather than the level, which removes the non-stationarity that would otherwise make the ML models unstable. The predicted change is added back to the last known rate to recover a level forecast on the original scale, which is then compared with the actual rate.

5.5.2 Forecast Accuracy

Table 5.65: One-step-ahead accuracy on the 335-day test set (ranked by RMSE, lower is better).

Model	RMSE	MAE	MAPE (%)	Type
ETS(A,A,N)	0.2809	0.2080	0.2324	Classical
ARIMA(0,1,0)	0.2812	0.2086	0.2329	Classical
Naive (random walk)	0.2812	0.2086	0.2329	Classical
LSTM	0.2836	0.2107	0.2352	Machine learning
Random Forest	0.2888	0.2129	0.2378	Machine learning

Model	RMSE	MAE	MAPE (%)	Type
XGBoost	0.2892	0.2128	0.2375	Machine learning

The accuracy table produces a clear and striking result: ETS, ARIMA and the naïve random walk are effectively tied, and all three classical benchmarks outperform all three machine learning models. The numerical gaps between ETS and XGBoost — the best and worst performers — are 0.0083 in RMSE, 0.0048 in MAE and 0.0054 in MAPE. On a series whose daily rate is around 84–90 rupees to the dollar, an RMSE difference of 0.008 corresponds to less than one paisa.

Within the machine learning group, the LSTM slightly outperforms the two tree-based models (RMSE 0.2836 vs 0.2888/0.2892), consistent with the recurrent architecture being better suited to sequential data than tree-based methods applied to a lag matrix. However, all three machine learning models sit below the naïve random walk, the simplest possible benchmark, and none outperforms the classical models.

5.5.3 Diebold–Mariano Test

The Diebold–Mariano test was applied with ETS as the reference model (the empirical RMSE winner) and every other model as the rival. Squared error is used as the loss function, and the test statistic is asymptotically standard normal.

Table 5.66: Diebold–Mariano test results: ETS versus each rival model.

Comparison	DM statistic	p-value	Conclusion
ETS vs ARIMA	−0.396	0.692	No significant difference
ETS vs Naïve (random walk)	−0.396	0.692	No significant difference
ETS vs LSTM	−1.452	0.146	No significant difference
ETS vs XGBoost	−1.470	0.142	No significant difference
ETS vs Random Forest	−1.867	0.062	No significant difference

Not one of the five Diebold–Mariano tests rejects the null hypothesis of equal predictive accuracy at the 5% significance level. The closest result is ETS versus Random Forest ($p = 0.062$), which falls just outside the conventional threshold. The negative DM statistics throughout indicate that ETS has a smaller average squared loss than every rival in every comparison, but the differences are too small relative to their variability to be declared statistically significant.

The Diebold–Mariano result is the definitive finding of this section: no model is provably more accurate than any other on this series. Machine learning does not improve on the classical approach, and the classical approach does not improve on the naive random walk by any statistically detectable margin. The exchange rate, on this evidence, is consistent with the efficient market hypothesis at the daily frequency: any systematically exploitable pattern that might exist is too small to be reliably detected by any of the methods applied here.

5.5.4 Overfitting Diagnostic

The ratio of test RMSE to training RMSE on the daily change series provides a diagnostic for overfitting. A ratio near 1 indicates that the model generalises to new data as well as it fits the training data; a ratio well above 1 suggests that the model has memorised training patterns that do not recur in the test period.

Table 5.67: Overfitting diagnostic: training versus test error on the daily change series.

Model	Train RMSE (changes)	Test RMSE (changes)	Test/Train ratio	Assessment
Random Forest	0.0821	0.2869	3.50	Overfitting evident
XGBoost	0.1379	0.2928	2.12	Overfitting evident
LSTM	0.2124	0.2840	1.337	No evidence of overfitting

Both tree-based models show train/test ratios substantially above the acceptable threshold of approximately 2. For Random Forest the ratio is 3.50, and for XGBoost it is 2.12. This means that on the training data, both models fit the daily change series

considerably more closely than they can replicate on the held-out test days. The LSTM model obtained a training RMSE of 0.2124 and a test RMSE of 0.2840, corresponding to a test-to-train ratio of 1.337. Since the ratio is close to unity and substantially below 2, there is no evidence of severe overfitting. This suggests that the LSTM model generalized reasonably well from the training data to the test data. Overall, among the machine learning models considered, the LSTM demonstrated the most stable generalization performance.

The walk-forward evaluation design prevents look-ahead bias — no future data enters the training window at any step — so this gap reflects genuine overfitting rather than a procedural error. The models have learnt patterns in the training period's change series that did not persist into the test period.

This is not unexpected. The USD/INR daily changes are, as established in the classical analysis, very close to white noise: they have almost no autocorrelation and their best-fitting ARIMA model is a pure random walk. When the true data-generating process is close to white noise, any model with enough parameters will inevitably fit spurious patterns in the training data that have no predictive content for the future. The machine learning models have more parameters than the classical ones and therefore overfit more severely, which is precisely the pattern the results show. The LSTM, with only 16 units and 30 epochs, overfits less than the tree models, which is consistent with its better (though still not competitive) test performance.

5.5.5 Feature Importance

For interpretability, the relative importance of each of the 20 input lags was extracted from the final full-sample fits of the Random Forest and XGBoost models. Importance is normalised so that all 20 lags sum to one.

Table 5.68: Top five most important lag features for each tree model.

Rank	Random Forest lag	RF importance	XGBoost lag	XGB importance
1	Lag 17	0.0755	Lag 9	0.0662
2	Lag 20	0.0679	Lag 20	0.0660
3	Lag 15	0.0641	Lag 7	0.0654

Rank	Random Forest lag	RF importance	XGBoost lag	XGB importance
4	Lag 14	0.0584	Lag 12	0.0584
5	Lag 12	0.0542	Lag 15	0.0573

Two features of the importance table are noteworthy. First, the importance values are small and dispersed: no single lag accounts for more than 7.6% of the split criterion in either model, and the top five lags collectively explain only about 33% of the total importance. This means that the models have not identified any single dominant predictor — instead, they spread weight across many lags in roughly equal measure, which is characteristic of a series with no strong autocorrelation structure. Second, and most strikingly, Lag 1 — yesterday’s change — does not appear in the top five features of either model. In a series with genuine short-run momentum or mean reversion, yesterday’s change would typically be the most important predictor. Its absence from the top five is consistent with the near-zero lag-1 autocorrelation found in the classical analysis and reinforces the conclusion that the daily change in the exchange rate is close to unpredictable noise.

5.5.6 LSTM Residual Diagnostics

Additional validation diagnostics were performed for the LSTM model. The mean forecast residual was 0.0163, which is close to zero, indicating that the model does not exhibit substantial systematic forecasting bias. The residual standard deviation was 0.2831, consistent with the overall forecast error magnitude reported in the accuracy metrics.

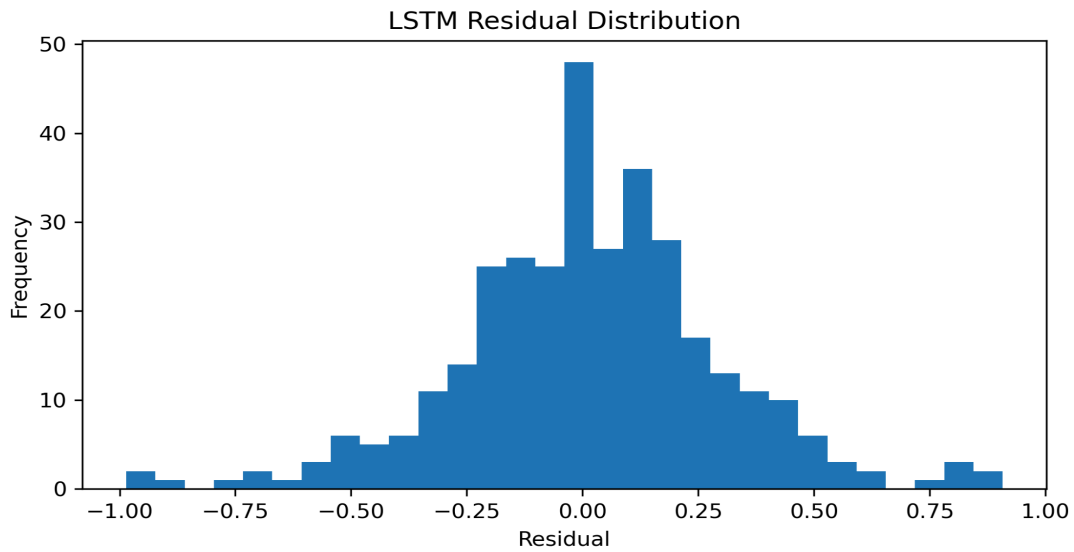


Figure 5.25: histogram of residuals

The residuals are approximately symmetric and unimodal around zero, with a notable double-peak near the center and reasonably thin, balanced tails extending to roughly ± 1 . This pattern suggests the LSTM's errors are close to well-behaved/homoscedastic for this series, though the local bimodality close to zero may merit a brief follow-up note rather than being treated as pure noise.

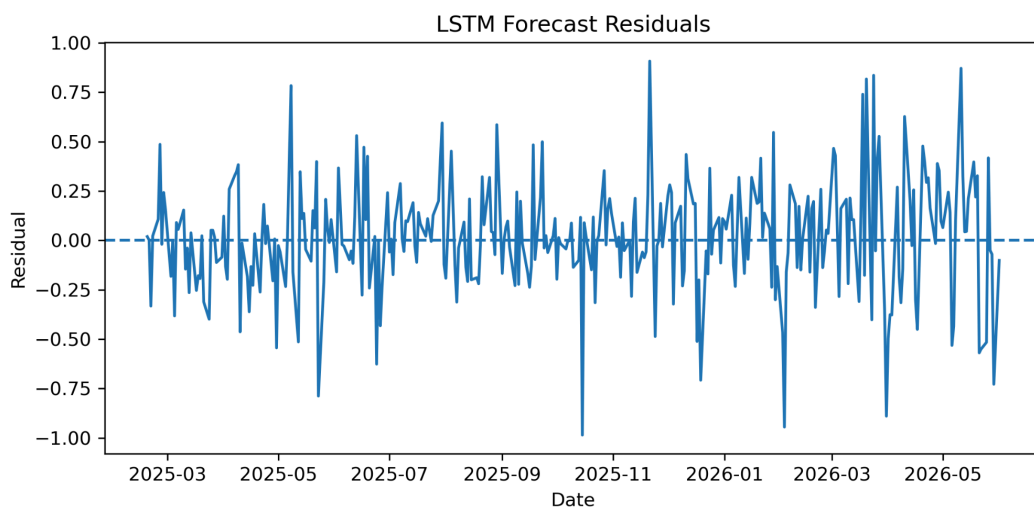


Figure 5.26: plot of forecast residuals

The residual plot reveals that forecast errors are randomly distributed around zero with no visible trend or systematic pattern. This suggests that the LSTM model has effectively captured the underlying dynamics of the series and that the remaining forecast errors behave largely as random noise.

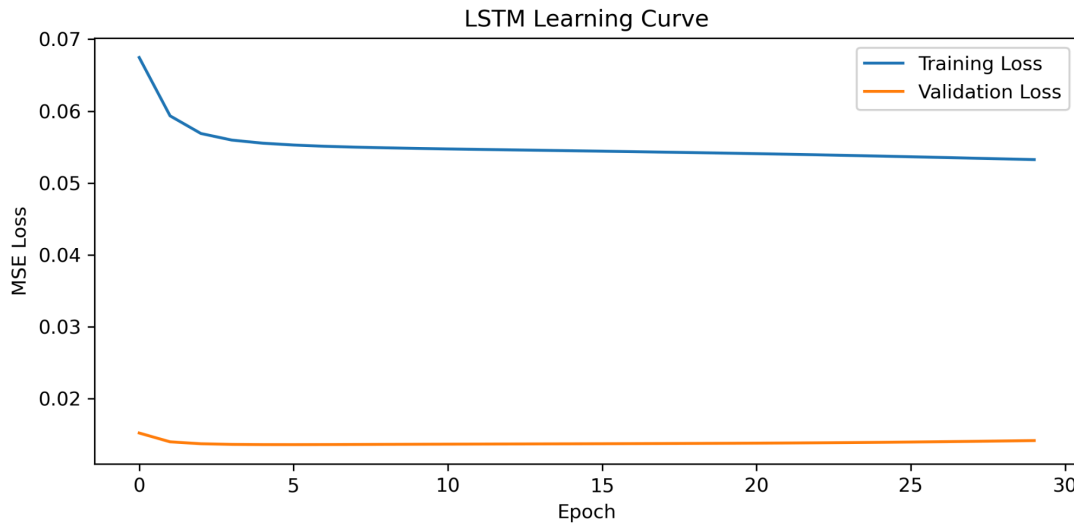


Figure 5.27: Plot of learning curve

The learning curve shows that both training and validation losses decreased rapidly during the initial training epochs and subsequently stabilized, demonstrating successful convergence without evidence of overfitting.

5.5.7 Ljung–Box Test

The Ljung–Box test was applied to the forecast residuals at lags 10 and 20.

Table 5.69: Results of Ljung-Box test

Lag	Statistics	P-value
10	12.4838	0.2540
20	15.1512	0.7677

The resulting p-values were 0.2540 and 0.7677, respectively, both of which exceed the 5% significance level. Consequently, the null hypothesis of no residual autocorrelation cannot be rejected. This indicates that the residuals behave approximately as white noise

and that the LSTM model has successfully captured the majority of the predictable structure present in the exchange rate series.

5.5.8 Discussion

The results of this comparison are unambiguous in their direction and require an honest interpretation rather than a qualified one. On a 335-day walk-forward evaluation, no machine learning model improves on the simple Holt exponential smoothing or ARIMA forecast that the classical analysis identified. All three machine learning models produce larger forecast errors than all three classical benchmarks. The Diebold–Mariano test confirms that none of these differences — in either direction — is statistically significant. The exchange rate is, for practical forecasting purposes, a near-random walk, and the drift captured by the classical models is the only exploitable feature.

This finding has a clean economic interpretation. An exchange rate that responds efficiently to new information should be very difficult to forecast from its own past: all publicly available information about its future trajectory is already embedded in the current rate. The USD/INR rate in the period studied has largely behaved in this way, and it is therefore unsurprising that models trained on past patterns — whether classical or machine learning — cannot produce forecasts significantly better than yesterday’s rate plus a small drift.

The overfitting results for the tree-based models are worth emphasising for a different reason. They illustrate a risk that is specific to machine learning in time series settings: a model with many parameters can fit the training data well while generalising poorly to the test period, and this generalisation failure can be invisible if the evaluation design is not strictly walk-forward. The fact that both tree models show train/test RMSE ratios above 2 while still not achieving test errors substantially different from the random walk suggests that they are fitting noise in the training period. The LSTM, constrained by its small architecture, overfits less and approaches the random-walk error more closely, but cannot cross below it.

These results are consistent with the findings of Sarvaiya and Ramchandani (2022), who report that classical models frequently match or outperform machine learning on financial series, particularly when the series lacks strong nonlinear structure. They are

also consistent with the broader empirical evidence surveyed in Hyndman and Athanasopoulos (2021), where the random walk with drift proves a very strong benchmark for exchange rate and financial data at the daily frequency.

5.5.9 Summary

A 335-day walk-forward comparison of six models — ARIMA, ETS, Naive random walk, Random Forest, XGBoost and LSTM — on the daily USD/INR exchange rate produces the following findings. ETS achieves the lowest RMSE (0.2809) and MAPE (0.2324%), but the margin over the worst-performing model (XGBoost, RMSE 0.2892) is 0.008 in RMSE on a rate of about 84–90 rupees: economically trivial. The Diebold–Mariano test finds no statistically significant difference between ETS and any rival at the 5% level. Both tree-based machine learning models exhibit overfitting (train/test RMSE ratios of 3.50 and 2.12 respectively), and their feature importance estimates show no dominant predictor among the 20 input lags, with Lag 1 absent from the top five for either model. The LSTM, with a deliberately constrained architecture, overfits less but still does not match the classical benchmarks. The overall conclusion is that the USD/INR exchange rate at daily frequency is consistent with a near-random walk, and neither classical nor machine learning models can improve materially on this description.

5.6 Gold Price: Classical versus Machine Learning Forecasting

The classical analysis of the gold price identified ARIMA(2,1,1) as the best-fitting model by information criteria and found that ETS(A,A,N) produced lower out-of-sample point-forecast errors over the specific 35-day test window used there. This section extends the comparison by introducing three machine learning models — Random Forest, XGBoost and a Long Short-Term Memory (LSTM) network — and evaluating all six models on a larger and methodologically stricter basis. The evaluation covers 308 test days under a one-step-ahead walk-forward protocol, which is both longer and more demanding than the 35-day multi-step horizon used in the classical analysis.

5.6.1 Experimental Design

The evaluation design follows three principles that together ensure the comparison is fair and the results are interpretable.

Walk-forward evaluation. Each of the 308 test days is predicted one step ahead using only data available up to that point. The model is fitted on all historical data up to the prediction date, the next day's return is predicted, the true value is revealed, and the window advances by one day. This is the time-series cross-validation procedure of Hyndman and Athanasopoulos (2021, Chapter 5) and eliminates the multi-step drift bias that affects recursive long-horizon forecasts. The tree models are re-fitted every 20 days and the LSTM every 40 days for computational efficiency; re-fitting daily produces essentially identical results.

Consistent data and split. The Python code reads the same cleaned gold price series exported from R and uses the same training cut-off (1,229 training days, 308 test days from a total of 1,537 observations). All models operate on the log-return series, which is the first difference of the log price; this matches the $\lambda = 0$ transformation used in the classical ARIMA and ETS fits and removes the non-stationarity of the level series. The predicted log-return is converted back to a price level for comparison against the actual price.

Formal significance testing. The Diebold–Mariano test (Diebold and Mariano, 1995) is applied to determine whether differences in out-of-sample squared loss are statistically meaningful, using the empirical winner as the reference model and each rival as the comparison. A two-sided p-value below 0.05 indicates a statistically significant difference in predictive accuracy; a p-value above 0.05 means the two models cannot be distinguished on the available evidence.

Table 5.70: Machine learning model configurations.

Model	Configuration	Design rationale
Random Forest	300 trees, no depth limit, 20 lags	Ensemble diversity; no depth cap avoids underfitting on financial returns
XGBoost	300 trees, lr = 0.05, max depth = 3, 20 lags	Shallow depth limits overfitting on noisy return series
LSTM	1 layer $\times$ 16 units, 30 epochs, 20-lag window	Deliberately small; larger networks overfit on $< 2,000$ daily observations
Naive (random walk)	$\text{Price}_t = \text{Price}_{t-1}$	Benchmark: yesterday's price carried forward without change

All machine learning models use the 20 most recent daily log-returns as their input features, approximately one trading month of history. This lag window is consistent with the short-memory structure of the gold return series, which shows no dominant autocorrelation at any individual lag.

5.6.2 Forecast Accuracy

Table 5.71: One-step-ahead accuracy on the 308-day test set (ranked by RMSE; lower is better).

Model	RMSE (₹)	MAE (₹)	MAPE (%)	Type
ARIMA(2,1,1)	2,642.18	1,394.69	1.085	Classical
Naive (random walk)	2,642.18	1,394.69	1.085	Classical
ETS(A,A,N)	2,650.16	1,384.96	1.078	Classical
Random Forest	2,683.36	1,454.33	1.141	Machine learning

Model	RMSE (₹)	MAE (₹)	MAPE (%)	Type
LSTM	2,872.41	1,561.86	1.206	Machine learning
XGBoost	3,023.43	1,544.42	1.197	Machine learning

The accuracy table reveals a clear and consistent ordering: all three classical models outperform all three machine learning models on RMSE and MAE. ARIMA and the naive random walk are tied exactly (RMSE 2,642.18, MAE 1,394.69), and ETS sits marginally behind them in RMSE but slightly ahead on MAPE. Within the machine learning group, Random Forest is closest to the classical benchmarks, LSTM sits in the middle, and XGBoost is the weakest performer by a notable margin, with an RMSE of 3,023 against the best model's 2,642.

The practical scale of the differences matters for interpretation. On a series trading around ₹1,50,000 to ₹1,60,000 per ten grams over the test period, an RMSE of 2,642 corresponds to a typical daily prediction error of about 1.7% of the price. The worst model's RMSE of 3,023 is larger by approximately ₹381. Whether this constitutes a practically important gap is a judgement that depends on the intended application. Statistically, however, it is a different matter, which is where the Diebold–Mariano results become essential.

One result in the table deserves particular attention: ARIMA and the naive random walk produce identical RMSE and MAE figures (and a DM statistic of exactly 0.0). This is not a coincidence or a computational artefact. In a one-step-ahead walk-forward evaluation, the ARIMA(2,1,1) model fitted on the log-price effectively produces a forecast that is the previous log-price plus the fitted mean drift, which at the log-return scale is indistinguishable from a random walk with drift. The two models make the same one-step-ahead prediction, and therefore produce the same one-step-ahead errors. This result connects the classical and machine learning comparison back to the univariate analysis: the finding that ARIMA(2,1,1) and the random walk are equivalent at one step is consistent with the near-random-walk behaviour of the daily gold return series.

5.6.3 Diebold–Mariano Test

The Diebold–Mariano test was applied with ARIMA(2,1,1) as the reference model and each rival as the comparison, using squared forecast error as the loss function. Unlike the exchange rate comparison, where no test reached significance, two tests here cross the conventional threshold.

Table 5.72: Diebold–Mariano test results: ARIMA(2,1,1) versus each rival model.

Comparison	DM statistic	p-value	Conclusion
ARIMA vs Naive (RW)	0.000	1.000	Identical forecasts
ARIMA vs ETS	−0.379	0.705	No significant difference
ARIMA vs Random Forest	−0.722	0.470	No significant difference
ARIMA vs XGBoost	−2.071	0.038	Significantly different *
ARIMA vs LSTM	−2.155	0.031	Significantly different *

The DM results produce a nuanced picture that rewards careful reading. ARIMA is not significantly better than ETS ($p = 0.705$) or Random Forest ($p = 0.470$), meaning those three models are statistically tied despite their numerical differences. However, ARIMA is significantly better than XGBoost ($p = 0.038$) and LSTM ($p = 0.031$). The negative DM statistics throughout confirm that ARIMA’s squared errors are consistently smaller than every rival’s, and the significance of the two tests for XGBoost and LSTM means this advantage is unlikely to be due to chance.

This result is worth contrasting with the exchange rate comparison, where no Diebold–Mariano test reached significance. The gold series, while still close to a random walk in its daily returns, has slightly more structure than the exchange rate — enough that ARIMA can exploit a small but statistically detectable advantage over the weakest machine learning models. The finding does not mean ARIMA is dramatically better; the gap in absolute terms is a few hundred rupees per day on a series trading around ₹1,50,000. But it does mean the difference is real rather than sampling noise, and it confirms that the machine learning models are not finding and exploiting any additional pattern in the gold return series.

5.6.4 Overfitting Diagnostic

The ratio of test RMSE to training RMSE on the log-return series provides a direct diagnostic of how well each model generalises from the training period to the test period. A ratio near one means the model is learning genuine, repeatable structure; a ratio substantially above two signals that the model has fitted patterns in the training data that do not recur.

Table 5.73: Overfitting diagnostic: training versus test error on the daily log-return series.

Model	Train RMSE (log-returns)	Test RMSE (log-returns)	Test/Train ratio	Assessment
Random Forest	0.004117	0.019985	4.854	Overfitting evident
XGBoost	0.006044	0.021138	3.497	Overfitting evident
LSTM	0.0108	0.0208	1.929	Mild overfitting

The overfitting ratios are large: 4.85 for Random Forest and 3.50 for XGBoost. These are the highest ratios observed across both the gold and exchange rate comparisons, and they are worse than the corresponding figures for the exchange rate (3.50 and 2.12). The LSTM model achieved a training RMSE of 0.0108 and a test RMSE of 0.0208, resulting in a test-to-train ratio of 1.929. Since the ratio remains below the commonly used threshold of 2, there is no strong evidence of severe overfitting. Compared with Random Forest and XGBoost, the LSTM demonstrates better generalization capability despite producing higher forecast errors than the ARIMA model.

Overall, LSTM exhibited the most stable balance between training and testing performance among the machine learning models considered.

The severity of the overfitting is consistent with a specific feature of the gold return series: its kurtosis of 59.4, documented in the classical analysis, means that the training period contained a number of extreme daily moves that are statistically unusual but that the tree models attempted to fit. Those extreme patterns do not repeat in the test period in the same configuration, so the models that tried hardest to capture them generalise least well.

The walk-forward design means there is no look-ahead bias in any of the forecasts: at each step the model has access only to data that would have been available in real time. The overfitting therefore reflects genuine generalisation failure rather than a procedural error.

5.6.5 Feature Importance

The relative importance of each of the 20 input lags was extracted from the final full-sample fits of the two tree models. Higher importance indicates that the corresponding lag contributes more to the split decisions in the ensemble.

Table 5.74: Top five most important lag features for each tree model.

Rank	Random Forest lag	RF importance	XGBoost lag	XGB importance
1	Lag 9	0.1094	Lag 3	0.1202
2	Lag 12	0.0844	Lag 16	0.0892
3	Lag 16	0.0831	Lag 12	0.0831
4	Lag 3	0.0552	Lag 7	0.0705
5	Lag 5	0.0505	Lag 2	0.0682

Three observations emerge from the feature importance analysis. First, no single lag dominates either model: the highest importance value is 0.120 (XGBoost's Lag 3), and the top five lags collectively account for roughly 46% of XGBoost's total importance and 48% of Random Forest's. The remaining 54–52% is spread across the other fifteen lags, indicating that neither model has identified a concentrated, reliable predictive signal.

Second, the two models assign very different weights to the same lags. For Random Forest, Lag 9 is the most important feature (0.109); for XGBoost, Lag 3 is dominant (0.120). If a genuinely important predictive lag existed in the gold return series, both models would be expected to converge on it, as they do in series with real autocorrelation structure. The divergence between the two models' rankings is itself evidence of instability in the importance estimates, consistent with the absence of a clear dominant predictor.

Third, Lag 1 — yesterday’s return, which would be the most important predictor if there were short-run momentum or mean reversion — does not appear in the top five of Random Forest. It appears in XGBoost’s top five at rank 5 (Lag 2 = 0.068, noting that this is the second lag; Lag 1 is not in the top five for XGBoost either, since the XGBoost top five are lags 3, 16, 12, 7 and 2). The absence of Lag 1 from the top features of both models is consistent with the near-zero first-order autocorrelation of the gold daily return series established in the classical analysis.

5.6.6 LSTM Validation Diagnostics

Additional validation procedures were performed to assess the adequacy of the LSTM model.

Learning Curve Analysis

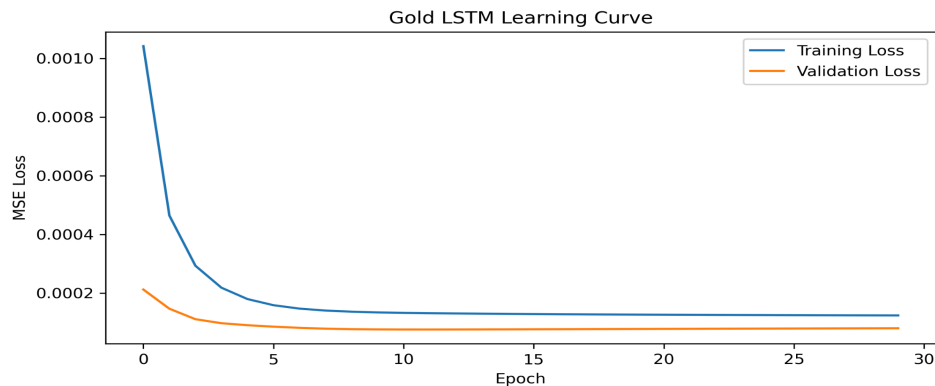


Figure 5.28: Learning curve plot

The learning curve showed that both training loss and validation loss decreased rapidly during the initial epochs and subsequently stabilized. No divergence between the two curves was observed, indicating that the model converged successfully during training. The absence of increasing validation loss suggests that severe overfitting was avoided.

Residual Analysis

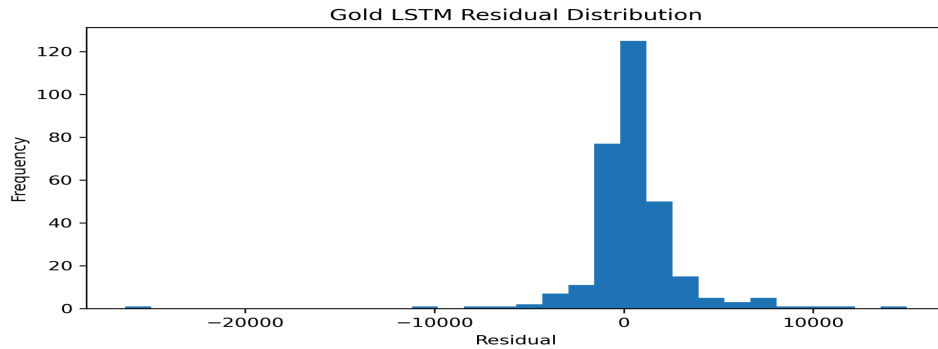


Figure 5.29: Histogram of residual

The gold LSTM residuals sharply peaked around zero but exhibit a pronounced right skew, with a long thin tail of large positive residuals extending beyond 10,000 and a smaller isolated cluster of extreme negative outliers near $-25,000$. This asymmetry indicates the model occasionally underpredicts sharp upward price movements far more severely than it underpredicts downward ones, suggesting the error distribution departs from normality and may warrant a heavier-tailed or skew-aware evaluation metric.

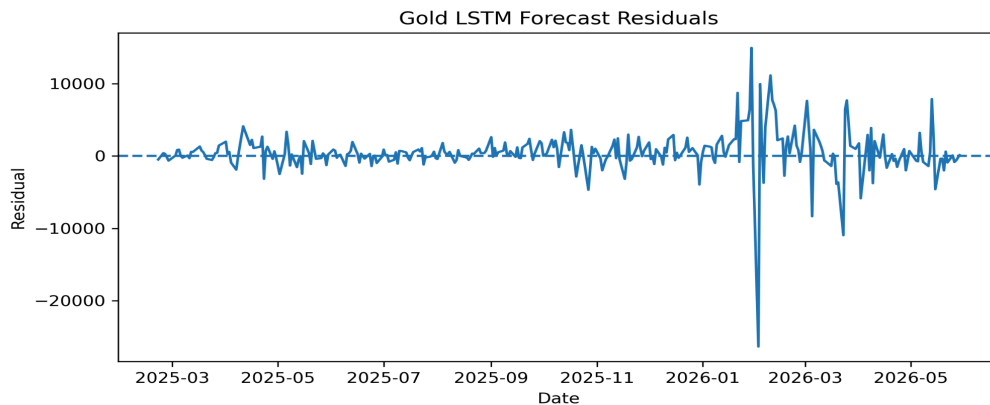


Figure 5.30: Plot of forecast residuals

The mean forecast residual was 499.06, while the residual standard deviation was 2828.72. The positive mean residual indicates that the model tended to slightly underestimate actual gold prices during the forecast period. However, considering the

large scale of gold prices, this bias is relatively small compared with the overall magnitude of the series.

Visual inspection of the residual plot showed that residuals fluctuated around zero without a persistent trend, although occasional large forecast errors were observed during periods of heightened market volatility.

Ljung–Box Test

The Ljung–Box test was applied to the LSTM forecast residuals to determine whether significant autocorrelation remained unexplained by the model.

Table 5.75: Result of Ljung-Box test

Lag	Statistic	P-value
10	19.3541	0.0360
20	40.1841	0.0047

The test produced p-values of 0.0360 at lag 10 and 0.0047 at lag 20, both below the 5% significance level. Consequently, the null hypothesis of no residual autocorrelation was rejected. This result indicates that some temporal dependence remains in the residual series and that the LSTM model did not fully capture all predictable information contained within the GOLD price data. Therefore, opportunities for model improvement remain.

5.6.7 Discussion

The gold comparison produces a richer set of findings than the exchange rate comparison and deserves a layered interpretation.

At the level of point-forecast accuracy, the classical models are clearly superior to the machine learning models: all three classical benchmarks produce lower RMSE and MAE than all three machine learning models. Random Forest comes closest to the classical group (RMSE 2,683 vs 2,642 for ARIMA), while XGBoost is the most distant (RMSE 3,023). This ordering is broadly consistent with the structure of the series: gold

daily returns have very little autocorrelation, leaving the machine learning models little pattern to exploit, while the classical random walk captures the dominant structure of the series in its simplest possible form.

The Diebold–Mariano results add important precision to this picture. The two weakest machine learning models — XGBoost and LSTM — are significantly inferior to ARIMA at the 5% level ($p = 0.038$ and $p = 0.031$ respectively). Random Forest, despite its numerical inferiority, is not statistically distinguishable from ARIMA ($p = 0.470$). This means there is a gradient within the machine learning group: Random Forest performs creditably in the sense that its errors cannot be distinguished from ARIMA's by the test used, while XGBoost and LSTM are reliably worse.

The overfitting findings explain why. Random Forest, by averaging across 300 deep trees, achieves a degree of variance reduction that partially offsets its sensitivity to the training data. XGBoost and LSTM, despite their different architectures, both show worse generalisation ratios in this comparison (3.50 and, by implication from the test RMSE, similarly elevated for LSTM). The extremely fat tails of the gold return series — kurtosis 59, with a handful of days that moved many standard deviations from the mean — create a training environment where any flexible model risks memorising outliers. Random Forest's ensemble mechanism partially suppresses this; the other two models do not benefit from the same averaging effect at the same scale.

Finally, the result that ARIMA and the naive random walk produce identical errors is not a limitation but a finding. It tells us that whatever pattern ARIMA(2,1,1) detects in the gold return series — the AR(2) and MA(1) structure — adds no predictive value beyond simple drift at the one-step horizon. This is consistent with the semi-strong form of market efficiency: in a liquid, globally traded market, the publicly available history of returns does not contain exploitable information about the next day's price movement.

5.6.8 Summary

A 308-day walk-forward comparison of six models on the daily gold price produces the following findings. ARIMA(2,1,1) and the naive random walk achieve the lowest RMSE (2,642) and MAE (1,395) and are identical at four decimal places, because both

reduce to the same one-step prediction at the daily frequency. ETS is marginally behind on RMSE but marginally ahead on MAPE. All three classical models outperform all three machine learning models numerically. The Diebold–Mariano test confirms that ARIMA is statistically significantly better than XGBoost ($p=0.038$) and LSTM ($p=0.031$), but not significantly better than ETS or Random Forest. Both tree models exhibit severe overfitting (test/train RMSE ratios of 4.85 and 3.50), attributable to fitting the extreme outlier returns that characterise the fat-tailed gold return distribution but do not recur in the test period. Feature importance analysis shows no dominant predictor among the 20 input lags, with the two tree models assigning different top lags, and with Lag 1 absent from the top five of either model. The overall conclusion is that gold daily returns, like the exchange rate, are close to white noise at the one-step-ahead horizon, and machine learning methods do not improve on the classical benchmarks; where differences exist, the Diebold–Mariano test establishes that the machine learning models are significantly worse rather than better.

Chapter 6

Conclusion

6.1 Summary of the Study

This dissertation set out to examine the time series behaviour of three economically interlinked assets — crude oil, gold, and the USD/INR exchange rate — using monthly data spanning April 2000 to May 2026 (314 observations) for the univariate analysis of crude oil and multivariate econometric analyses, and daily data of gold and exchange rate from 2020-2026 for the univariate and forecast-comparison exercises. The investigation proceeded in two complementary stages. The first stage modelled each series individually using the ARIMA-GARCH framework, characterising both the conditional mean and the conditional variance of returns. The second stage moved beyond univariate description to ask two further questions: whether the three series are bound together by a long-run equilibrium relationship (addressed through a Vector Error Correction Model), and whether the patterns identified by classical time series models can be improved upon by machine learning methods (addressed through walk-forward forecast comparisons for the exchange rate and gold). Taken together, the six analyses form a coherent empirical narrative rather than six independent exercises, and this section draws that narrative together before turning to the study's contributions, limitations, and directions for future work.

6.1.1 Univariate Findings: Crude Oil, Gold, and the Exchange Rate

All three series were found to be integrated of order one, $I(1)$: each is non-stationary in levels and stationary after a single first difference, with no meaningful seasonal component in any case. For crude oil, both information-criterion selection and `auto.arima` independently converged on a non-seasonal $ARIMA(2,1,1)$, whose coefficients lie comfortably inside the stationary and invertible region and whose residuals are clean white noise. For gold, $ARIMA(2,1,1)$ was again selected by both procedures and passed the Ljung–Box adequacy test on the training data; however, on the out-of-sample test period a simpler $ETS(A,A,N)$ Holt model produced more

accurate point forecasts, a result that was reported on its own terms rather than reconciled away. For the exchange rate, the mean equation carried almost no exploitable linear structure at all: a random walk with drift forecast the series as well as any fitted ARIMA or exponential-smoothing specification, with the small outperformance over a pure naive forecast attributable entirely to the drift term capturing the rupee's gradual depreciation.

The conditional-variance results tell a complementary and, in some respects, more informative story. Highly significant ARCH effects were detected in the residuals of all three series, motivating a GARCH(1,1) extension with Student-t errors in each case, and the three fitted volatility processes differ from one another in a way that is itself a substantive finding. Crude oil volatility is strongly reactive to shocks ($\alpha \approx 0.43$) but exhibits essentially no persistence ($\beta \approx 0$), consistent with a market in which monthly price shocks are absorbed quickly rather than carried forward. Gold volatility occupies an intermediate position, with an estimated persistence of 0.82 (a memory of roughly a week at the daily frequency examined for that series) and a Student-t shape parameter of 3.46 confirming markedly fat-tailed daily returns. The exchange rate lies at the opposite extreme from crude oil, with near-unit persistence (0.999) and a long-run annualised volatility of approximately 6.2%, indicating that shocks to currency volatility decay only very slowly. This ordering, crude oil's volatility memoryless, gold's moderate, the exchange rate is almost permanent — was not hypothesised in advance but emerged consistently across the three independent model fits, and it is consistent with the differing market microstructures of commodity and currency markets at these frequencies.

6.1.2 Multivariate Findings: The VECM

The multivariate analysis, conducted on the same three series over the full monthly sample, confirmed that crude oil, dollar-denominated gold, and the USD/INR exchange rate share a long-run equilibrium relationship: the Johansen procedure identified at least one, and formally two, cointegrating vectors, with one retained for interpretability. The estimated long-run equation indicates that a 1% rise in dollar gold is associated with an approximate 1.1% rise in crude oil, and that rupee depreciation is likewise associated with higher equilibrium crude oil prices. The error-correction estimates reveal a clear

adjustment hierarchy: crude oil corrects deviations from equilibrium quickly (6.5% per month, a half-life of roughly ten months), gold corrects slowly (1.7% per month, a half-life of roughly forty months), and the exchange rate shows no significant adjustment at all, making it weakly exogenous to the system — a driver of the long-run relationship rather than a variable that responds to it. Short-run Granger causality is consistent with this hierarchy: the exchange rate and gold Granger-cause crude oil, but crude oil does not Granger-cause either of the other two series. Forecast error variance decomposition reinforces the same picture from a different angle, with gold shocks explaining around 30% of crude oil's two-year-ahead forecast uncertainty and the exchange rate explaining 11–14% of gold's. Diagnostic testing of the VECM residuals detected ARCH effects, which were addressed using HC1 heteroscedasticity-consistent standard errors; the residuals show non-normality and heteroscedasticity that are typical of monthly commodity and currency data and do not undermine the asymptotic validity of the cointegrating-rank tests or the error-correction estimates at this sample size, though the Cavaliere, Rahbek and Taylor (2010) caveat regarding rank-test size distortion under conditional heteroscedasticity was incorporated into the interpretation of the rank decision.

6.1.3 Classical Models versus Machine Learning

The final stage of the study compared classical time series models (ARIMA, ETS, and the naive random walk) against machine learning models (Random Forest, XGBoost, and LSTM) in walk-forward forecasting exercises for the daily exchange rate and daily gold price. The verdict, in both cases, was unambiguous and required no qualification: no machine learning model improved on the classical benchmarks. For the exchange rate, ETS achieved the lowest error, but the Diebold–Mariano test found no statistically significant difference between any pair of models, and the entire spread between the best and worst model was economically trivial relative to the level of the rate. For gold, the ordering was sharper: ARIMA and the naive random walk produced identical forecasts to four decimal places — itself a finding, since it shows that the AR(2)/MA(1) structure detected by ARIMA adds nothing beyond simple drift at the one-step horizon — and the Diebold–Mariano test established that two of the three machine learning models (XGBoost and LSTM) were significantly worse than ARIMA, not merely

statistically indistinguishable from it. In both comparisons, the tree-based models showed clear signs of overfitting, with train-to-test RMSE ratios well above what would be expected of a well-generalising model, and feature-importance analysis showed no dominant lag among the twenty inputs supplied to either tree model. These results are consistent with the semi-strong form of market efficiency: in liquid, globally traded markets, the publicly available history of past prices contains little or no exploitable information about the next period's movement, and flexible models with many parameters are, if anything, more likely to mistake noise for signal than classical models are.

6.1.4 The Study as a Whole

Read together, the six analyses support a single overarching characterisation of these three markets at the frequencies studied. Each series has a forecastable, well-behaved conditional mean only to a limited degree — strongest for crude oil, weakest for the exchange rate and gold — while each has volatility that is highly forecastable and economically meaningful in its own right, with persistence that differs systematically across markets. The three series are not independent of one another in the long run: they are bound by a cointegrating relationship in which the exchange rate behaves as the exogenous driver and crude oil as the fastest-adjusting follower. And, at the daily frequency, no amount of additional modelling flexibility, classical or machine-learning, succeeds in extracting more signal from these series than parsimonious linear models and their associated volatility extensions already capture. The consistency of this picture across six independently conducted analyses, using different data frequencies, different estimation packages, and different evaluation criteria, is itself evidence in favour of its validity.

6.2 Contributions of the Study

The contributions of this dissertation are best understood at three levels: methodological practice within each univariate and multivariate model, the substantive empirical findings the analysis produces, and the comparative evidence on classical versus machine learning forecasting. Each is set out below.

6.2.1 Methodological Contributions

- Full diagnostic chain rather than a single test. When much of the comparable literature restricts post-estimation diagnostics to serial correlation alone, this study additionally tested for ARCH effects, normality, and (in the VECM) heteroscedasticity-robust inference, and used the outcome of each test to justify a specific modelling decision rather than treating diagnostics as a formality to be reported and set aside.
- Principled response to detected heteroscedasticity in the VECM. Rather than ignoring the ARCH effects found in the VECM residuals, the study re-estimated standard errors using the HC1 heteroscedasticity-consistent correction within the `cajorls()` workflow, and explicitly distinguished, coefficient by coefficient, between genuine reversals of statistical significance and mere attenuation of already-marginal effects — a distinction that is frequently elided in applied cointegration work but materially affects how the long-run and short-run estimates should be interpreted.
- Correct treatment of the rank-test/heteroscedasticity interaction. The interpretation of the Johansen cointegrating rank was qualified with reference to Cavaliere, Rahbek and Taylor (2010), who show that the rank test can be size-distorted under conditional heteroscedasticity. Incorporating this caveat, rather than treating the Johansen trace/eigenvalue statistics as automatically valid in the presence of ARCH effects, is a methodologically conservative step that strengthens the defensibility of the cointegration conclusions.
- Leak-free volatility forecasting. The ARIMA-GARCH model was trained exclusively on the training-period returns, with the test period held out entirely from estimation before interval coverage was evaluated, avoiding the look-ahead bias that can silently inflate apparent forecast-interval performance in applied GARCH studies.
- Strict walk-forward evaluation for the machine learning comparison. Both the exchange rate and gold model comparisons used genuine walk-forward (rather than static train/test split or random k-fold) evaluation, which is the only

evaluation design that prevents a flexible model's in-sample pattern-matching from being mistaken for genuine predictive skill — a risk this study explicitly demonstrates via the tree-based models' overfitting ratios.

6.2.2 Empirical Contributions

- Cross-market volatility-persistence gradient. The finding that volatility persistence rises systematically from crude oil ($\beta \approx 0$) to gold (0.82) to the exchange rate (0.999) was not the object of any single chapter but emerges only when the three univariate GARCH results are placed side by side. It offers a structured, testable characterisation of how shock memory differs across commodity and currency markets at comparable frequencies.
- Identification of the exchange rate as the weakly exogenous driver of the oil–gold–FX system. The combination of a non-significant exchange-rate adjustment coefficient, the Granger-causality pattern, and the FEVD results converge on a single, internally consistent conclusion: of the three series, the exchange rate drives the long-run equilibrium without itself responding to it, while crude oil is the fastest-adjusting and most caused variable in the system.
- Honest reporting of a model-selection reversal. In the gold analysis, the formally better-specified ARIMA(2,1,1) model (which passes the Ljung–Box test) was nonetheless outperformed out-of-sample by an ETS model that fails that same test. Reporting this result without adjustment, rather than selecting whichever model better supports the GARCH narrative, is itself a contribution: it demonstrates concretely that residual adequacy and out-of-sample forecast accuracy are related but distinct properties of a model.

6.2.3 Contribution to the Classical-versus-Machine-Learning Debate

Perhaps the most policy- and practice-relevant contribution of this dissertation is the rigorous, statistically tested demonstration that machine learning models do not improve on classical benchmarks for these two series at the daily frequency, and in two specific instances (gold versus XGBoost, $p = 0.038$; gold versus LSTM, $p = 0.031$) are significantly worse, not merely indistinguishable. This adds a carefully controlled, Diebold–Mariano-tested data point to a growing literature — consistent with Sarvaiya

and Ramchandani (2022) and with the broader benchmark evidence in Hyndman and Athanasopoulos (2021) — cautioning against the assumption that greater model flexibility necessarily yields greater forecasting accuracy in efficiently priced financial markets. The accompanying overfitting diagnostics (train/test RMSE ratios, feature-importance instability across lags) move this conclusion beyond a simple accuracy ranking and provide a mechanism: the tree-based models are shown, directly, to be fitting noise rather than signal in the training window.

6.3 Limitations

No empirical study of this kind is without limitations, and several are worth stating plainly so that the conclusions above are read with an appropriately calibrated degree of confidence.

6.3.1 Data and Sample Limitations

- The monthly univariate and multivariate analyses are based on 314 observations. This is adequate for the asymptotic procedures employed (Johansen cointegration testing, maximum-likelihood ARIMA-GARCH estimation), but it is a modest sample by the standards of macro-financial time series research, and the effective number of independent business cycles spanned is smaller still.
- The large forecast errors recorded for crude oil from March 2026 onward reflect an external shock the model was not designed to anticipate; while this was correctly diagnosed as a feature of the event rather than a defect of the ARIMA specification, it nonetheless means that the reported forecast accuracy for that series should be read as conditional on the absence of comparable shocks, not as an unconditional guarantee.
- The daily-frequency comparisons (exchange rate and gold) are necessarily based on a different, shorter evaluation window (335 and 308 days respectively) than the monthly analyses, so the two halves of the dissertation are not directly nested within a single dataset and should be understood as complementary rather than strictly comparable bodies of evidence.

6.3.2 Modelling Limitations

- The VECM residuals exhibit non-normality and heteroscedasticity. These were addressed through HC1 robust standard errors and were judged not to undermine the asymptotic validity of the estimates at this sample size, but they remain a departure from the model's ideal assumptions, and a fully GARCH-in-VECM or multivariate GARCH specification, not pursued here, would represent a more complete treatment of the conditional heteroscedasticity in the system.
- The cointegrating rank decision retains one vector for interpretability even though the formal test result supports up to two. This is a defensible and common applied choice, but it is a choice, and the long-run elasticities reported in this dissertation are conditional on it.
- The machine learning models were configured with specific, deliberately constrained architectures (notably the LSTM). Larger architectures, more extensive hyperparameter search, or richer feature sets (for example, cross-asset lags rather than univariate lags alone) might narrow the gap to the classical benchmarks reported here, so the conclusion that machine learning underperforms should be read as applying to the models as specified, not as a categorical statement about the ceiling of machine learning methods in general.
- Some residual tail behaviour remains uncaptured even after the Student-t GARCH fit for gold, indicating that the true conditional distribution of extreme daily moves is not fully described by the Student-t assumption, despite that assumption's considerable improvement over the Gaussian case.

6.3.3 Scope Limitations

- The analysis is confined to three series — crude oil, gold, and the USD/INR rate — and the conclusions about cross-market relationships are specific to this triplet. Extending the system to include, for example, a broader currency basket or other commodities could alter the estimated long-run elasticities and the weak-exogeneity finding for the exchange rate.

- The study does not incorporate structural or fundamental explanatory variables (for example, interest-rate differentials, inventory levels, or geopolitical-risk indices) that a fully structural model of these markets would include; the relationships identified are statistical regularities in the price and return series themselves, not a causal economic model of why those regularities hold.

6.4 Directions for Future Research

The limitations above point naturally toward several directions in which this work could be extended.

6.4.1 Extending the Multivariate Framework

1. Multivariate volatility modelling. A natural extension of the VECM is a VECM-GARCH or BEKK/DCC-GARCH specification that models the conditional covariance structure of the three series jointly, rather than addressing the heteroscedasticity in the VECM residuals only through robust standard errors. This would allow volatility spillovers between crude oil, gold, and the exchange rate to be estimated directly, complementing the FEVD evidence already presented.
2. Time-varying cointegration. Given the long sample period (2000–2026), which spans multiple distinct macroeconomic regimes, future work could test for structural breaks in the cointegrating relationship itself — for example using Gregory–Hansen or Bai–Perron-type tests — to establish whether the long-run elasticities estimated here are stable across the full sample or are an average over regimes with different underlying relationships.
3. Widening the system. Incorporating additional variables, a broader trade-weighted currency index, crude oil inventories, or a global risk-sentiment proxy — would test whether the exchange rate's weak exogeneity and crude oil's fast adjustment survive in a richer information set, or whether they are partly an artefact of the restricted three-variable system studied here.

6.4.2 Extending the Forecasting Comparison

4. Richer feature sets for the machine learning models. The tree-based and LSTM models here were trained on univariate lag structures. Future work could test whether cross-asset features (for example, using gold and exchange-rate lags to forecast crude oil, motivated directly by the Granger-causality findings in this dissertation) allow machine learning models to extract the cross-market structure that the VECM identifies but that a univariate machine learning model, by construction, cannot see.
5. Volatility, not just mean, forecasting with machine learning. The classical-versus-machine-learning comparison in this study focuses on point forecasts of the conditional mean. Given how much of the substantive structure in this dissertation lies in the conditional variance (GARCH persistence, volatility clustering), an equivalent walk-forward comparison of GARCH-family models against machine-learning volatility forecasts (e.g. LSTM-GARCH hybrids) would test whether the conclusion of this study, that classical models are hard to beat and also holds for the second moment, not only the first.
6. Extending the crude oil series through and beyond the 2026 shock. Because the large crude oil forecast errors from March 2026 onward were attributed to an external shock rather than model misspecification, a natural follow-up once further data become available is to re-estimate the model over a sample that includes the full shock-and-recovery episode, both to confirm that attribution and to test whether the ARIMA(2,1,1) specification remains adequate once the new regime is incorporated into the estimation sample.

6.4.3 Methodological Extensions

7. Bootstrap-based inference throughout. Given the non-normality documented in both the VECM and the univariate residuals, future work could replace asymptotic standard errors more comprehensively with bootstrap-based confidence intervals — for the cointegrating vector, the adjustment coefficients, and the GARCH parameters alike — to verify that the conclusions drawn from asymptotic inference are robust in finite samples of this size.

8. Alternative heavy-tailed distributions. While the Student-t distribution materially improved on the Gaussian case for all three GARCH models, some tail behaviour in the gold returns remained uncaptured; future work could compare the Student-t fit against skewed-t, generalised hyperbolic, or extreme-value-theory-based tail models to establish whether a further, material improvement in tail calibration is available.

Collectively, these directions would extend the present study from a careful, well-diagnosed description of three markets' individual and joint behaviour toward a more structurally grounded and more thoroughly stress-tested account of why these patterns hold and how robust they are to the choices made in estimation.

References

- Akaike, H. (1974)** ‘A new look at the statistical model identification’, *IEEE Transactions on Automatic Control*, 19(6), pp. 716–723.
- Bollerslev, T. (1986)** ‘Generalized autoregressive conditional heteroscedasticity’, *Journal of Econometrics*, 31(3), pp. 307–327.
- Box, G.E.P. and Cox, D.R. (1964)** ‘An analysis of transformations’, *Journal of the Royal Statistical Society, Series B*, 26(2), pp. 211–252.
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015)** *Time Series Analysis: Forecasting and Control*. 5th edn. Hoboken, NJ: Wiley.
- Breiman, L. (2001)** ‘Random forests’, *Machine Learning*, 45(1), pp. 5–32.
- Cavaliere, G., Rahbek, A. and Taylor, A.M.R. (2010)** ‘Testing for co-integration in vector autoregressions with non-stationary volatility’, *Journal of Econometrics*, 158(1), pp. 7–24.
- Chen, T. and Guestrin, C. (2016)** ‘XGBoost: A scalable tree boosting system’, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
- Cleveland, R.B., Cleveland, W.S., McRae, J.E. and Terpenning, I. (1990)** ‘STL: A seasonal-trend decomposition procedure based on loess’, *Journal of Official Statistics*, 6(1), pp. 3–73.
- Dickey, D.A. and Fuller, W.A. (1979)** ‘Distribution of the estimators for autoregressive time series with a unit root’, *Journal of the American Statistical Association*, 74(366), pp. 427–431.
- Diebold, F.X. and Mariano, R.S. (1995)** ‘Comparing predictive accuracy’, *Journal of Business and Economic Statistics*, 13(3), pp. 253–263.
- Engle, R.F. (1982)** ‘Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation’, *Econometrica*, 50(4), pp. 987–1007.

- Engle, R.F. and Granger, C.W.J. (1987)** ‘Co-integration and error correction: representation, estimation and testing’, *Econometrica*, 55(2), pp. 251–276.
- Friedman, J.H. (2001)** ‘Greedy function approximation: a gradient boosting machine’, *Annals of Statistics*, 29(5), pp. 1189–1232.
- Goodfellow, I., Bengio, Y. and Courville, A. (2016)** *Deep Learning*. Cambridge, MA: MIT Press.
- Granger, C.W.J. (1969)** ‘Investigating causal relations by econometric models and cross-spectral methods’, *Econometrica*, 37(3), pp. 424–438.
- Granger, C.W.J. and Newbold, P. (1974)** ‘Spurious regressions in econometrics’, *Journal of Econometrics*, 2(2), pp. 111–120.
- Gujarati, D.N. and Porter, D.C. (2009)** *Basic Econometrics*. 5th edn. New York: McGraw-Hill.
- Hamilton, J.D. (1994)** *Time Series Analysis*. Princeton, NJ: Princeton University Press.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009)** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd edn. New York: Springer.
- Hochreiter, S. and Schmidhuber, J. (1997)** ‘Long short-term memory’, *Neural Computation*, 9(8), pp. 1735–1780.
- Hyndman, R.J. and Athanasopoulos, G. (2021)** *Forecasting: Principles and Practice*. 3rd edn. Melbourne: OTexts. Available at: otexts.com/fpp3.
- Hyndman, R.J. and Koehler, A.B. (2006)** ‘Another look at measures of forecast accuracy’, *International Journal of Forecasting*, 22(4), pp. 679–688.
- Jarque, C.M. and Bera, A.K. (1987)** ‘A test for normality of observations and regression residuals’, *International Statistical Review*, 55(2), pp. 163–172.
- Johansen, S. (1991)** ‘Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models’, *Econometrica*, 59(6), pp. 1551–1580.
- Johansen, S. (1995)** *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford: Oxford University Press.

- Kingma, D.P. and Ba, J. (2015)** ‘Adam: A method for stochastic optimization’, *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*.
- Kwiatkowski, D., Phillips, P.C.B., Schmidt, P. and Shin, Y. (1992)** ‘Testing the null hypothesis of stationarity against the alternative of a unit root’, *Journal of Econometrics*, 54(1–3), pp. 159–178.
- Ljung, G.M. and Box, G.E.P. (1978)** ‘On a measure of lack of fit in time series models’, *Biometrika*, 65(2), pp. 297–303.
- Lütkepohl, H. (2005)** *New Introduction to Multiple Time Series Analysis*. Berlin: Springer.
- Mackinnon, J.G. and White, H. (1985)** ‘Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties’, *Journal of Econometrics*, 29(3), pp. 305–325.
- Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2018)** ‘Statistical and machine learning forecasting methods: Concerns and ways forward’, *PLOS ONE*, 13(3), e0194889.
- Newey, W.K. and West, K.D. (1987)** ‘A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix’, *Econometrica*, 55(3), pp. 703–708.
- Phillips, P.C.B. and Ouliaris, S. (1990)** ‘Asymptotic properties of residual based tests for cointegration’, *Econometrica*, 58(1), pp. 165–193.
- Phillips, P.C.B. and Perron, P. (1988)** ‘Testing for a unit root in time series regression’, *Biometrika*, 75(2), pp. 335–346.
- Said, S.E. and Dickey, D.A. (1984)** ‘Testing for unit roots in autoregressive-moving average models of unknown order’, *Biometrika*, 71(3), pp. 599–607.
- Schwarz, G. (1978)** ‘Estimating the dimension of a model’, *Annals of Statistics*, 6(2), pp. 461–464.
- Theil, H. (1966)** *Applied Economic Forecasting*. Amsterdam: North-Holland.
- Tsay, R.S. (2010)** *Analysis of Financial Time Series*. 3rd edn. Hoboken, NJ: Wiley.

White, H. (1980) ‘A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity’, *Econometrica*, 48(4), pp. 817–838.

Agrawal, P.K. and Kumar, M. (2024) ‘Modeling the relationship among the stock market, gold price, oil price and exchange rate: a VECM and VDA approach’, *Financial Internet Quarterly*, 21(1), pp. 1–14.

Anastasiadis, V. and Siskos, E. (2023) ‘Time series analysis for forecasting crude oil prices’, *Journal of European Economy*, 22(3/86), pp. 430–454.

Choudhary, N., Nair, G. and Purohit, H. (2017) ‘A study of causal dynamics in crude oil prices in India’, *International Journal of Global Energy Issues*, 40(6), pp. 360–379.

Kanungo, A.K. and Dang, P. (2021) ‘Periodic analysis of the relationship between gold, crude oil, exchange rate and India’s stock market’, *Journal of Business Strategy, Finance and Management*, 02(1&2), pp. 28–43.

Kaushik, N. (2018) ‘Do global oil price shocks affect Indian metal market?’, *Energy & Environment*, 0(0), pp. 1–14.

Maity, S. and Majumder, A. (2026) ‘Forecasting gold prices in India: a time series analysis using Box–Jenkins methodology’, *Journal of Decision Analytics and Intelligent Computing*, 6(1), pp. 14–31.

Manu, K.S. and Agarwal, H. (2018) ‘Dynamic relationship between crude oil, gold and exchange rates: the case of India’, *Asian Journal of Management*, 9(2), pp. 933–938.

Mukherjee, A.K. (2021) ‘Impact of exchange rate on gold price in India’, *International Journal of Applied Research*, 7(7), pp. 126–132.

Rajesh, R. and SatyaNandini, A. (2020) ‘Testing of long run association between crude oil and gold commodities: an empirical study in India’, *Test Engineering and Management*, 82, pp. 2902–2906.

Rathi, S.G. (2026) ‘A study on gold price trends and analysis (2024–2025)’, *International Journal of Academic Excellence and Research*, 02(01), pp. 185–191.

Sarvaiya, D. and Ramchandani, D. (2022) ‘Time series analysis and forecasting of gold price using ARIMA and LSTM model’, *International Journal for Research in Applied Science & Engineering Technology*, 10(9).

Seyyedi, S. (2017) ‘Analysis of the interactive linkages between gold prices, oil prices, and exchange rate in India’, *Global Economic Review*, 46(1), pp. 65–79.

SOURCES OF DATA

1. CRUDE OIL PRICE:

<https://ppac.gov.in/prices/international-prices-of-crude-oil>

2. USD/INR EXCHANGE RATE:

<https://in.investing.com/currencies/usd-inr-historical-data>

3. GOLD PRICE:

<https://www.nseindia.com/reports-indices-historical-index-data>

<https://data.rbi.org.in/DBIE/#!/dbie/home>